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Growing the Efficient Frontier on Panel Trees

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Lin William Cong / Guanhao Feng / Jingyu He / Xin He

NBER Working Paper 30805
2022年12月发布；2023年12月修订

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阅读导航

原PDF第 01 页 | 封面与致谢

原PDF第 02 页 | 摘要与作者信息

原PDF第 03 页 | 1 引言

原PDF第 10 页 | 2 用于资产定价的面板树

原PDF第 12 页 | 2.2 生长一棵P-Tree

原PDF第 16 页 | 2.3 提升与多因子

原PDF第 17 页 | 2.4 模拟概述

原PDF第 18 页 | 3 实施与首棵P-Tree

原PDF第 25 页 | 4 提升P-Tree与多个因子

原PDF第 32 页 | 4.3 资产定价表现

原PDF第 34 页 | 4.4 因子投资

原PDF第 37 页 | 5 可解释性、特征与宏观状态

原PDF第 39 页 | 5.2 评价单项特征

原PDF第 42 页 | 5.3 宏观经济状态

原PDF第 44 页 | 6 结论

原PDF第 45 页 | 参考文献（原文保留）

原PDF第 50 页 | 附录：单棵P-Tree算法

原PDF第 51 页 | 图A.1 子样本树

原PDF第 52 页 | 表A.1 61项公司特征

原PDF第 53 页 | 表A.2 宏观变量

原PDF第 54 页 | 表A.3 子样本检验

原PDF第 55 页 | 网络附录 I 模拟

原PDF第 61 页 | 网络附录 II 基准调整

原PDF第 68 页 | 网络附录 III 替代切分准则

原PDF第 70 页 | III.C 实证结果

原PDF第 75 页 | III.D 时间序列切分

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NBER 工作论文系列，第30805号

Lin William Cong / Guanhao Feng / Jingyu He / Xin He

2022年12月；2023年12月修订

美国国家经济研究局（NBER），1050 Massachusetts Avenue，Cambridge，MA 02138。

论文链接：http://www.nber.org/papers/w30805

作者感谢INQUIRE Europe和香港研究资助局（项目编号CityU 11507022）的资助。Will Cong同时担任BlackRock，Inc.的顾问。作者感谢Fahiz Baba-Yara、Svetlana Bryzgalova、John Campbell、Andrei Goncalves、Bing Han、Kewei Hou、Bob Jarrow、Ron Kaniel、Chris Malloy、Alberto Rossi、Gideon Saar、Shrihari Santosh、Peixuan Yuan和Guofu Zhou提出的详细意见与建议，其中Fahiz Baba-Yara、Svetlana Bryzgalova、Alberto Rossi、Shrihari Santosh及Peixuan Yuan为讨论人。

作者还感谢Doron Avramov、Jie Cao、Zhanhui Chen、Eric Ghysels、P. Richard Hahn、Dashan Huang、Sophia Zhengzi Li、Asaf Manela、Stefan Nagel、Adam Reed、Artem Streltsov、Yinan Su、Fred Sun、Yaki Tsaig、Mao Ye、Dacheng Xiu和Bobby Yu，以及ArrowStreet Capital、Baylor University、BUAA、Ca' Foscari University of Venice、Cambridge University Algorithmic Trading Society Quant Conference、CFTRC 2022、CityUHK、CKGSB、第二届杭州金融科技创新与发展会议、Cornell、Goethe University Frankfurt、GSU-RFS FinTech Conference 2022、HKAIFT-Columbia joint seminar、Hunan University、INFORMS 2021、第4届RUC金融科技国际论坛、2023 Mid-South DATA Conference、KAIST、NFA 2022、Oxford SML-Fin、OSU、PKU Guanghua、PKU-NSD、PKU-NUS金融计量年会、Qube Research and Technology、Reichman University、Schroders Investments、第三届Shanghai Financial Forefront Symposium、SHUFE、SMU、SUSTech、TAMU、Tsinghua SEM、UNC、University of Bath、University of Hawai'i、USC、USTC、University of Macau、World Online Seminars on Machine Learning in Finance、WFA 2022及2023 XJTLU AI and Big Data in Accounting and Finance Conference的研讨会和会议参与者，感谢他们的建设性讨论与意见。

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NBER WORKING PAPER SERIES

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Working Paper 30805
http://www.nber.org/papers/w30805

NATIONAL BUREAU OF ECONOMIC RESEARCH
1050 Massachusetts Avenue
Cambridge, MA 02138
December 2022, revised December 2023

The authors acknowledge financial support from INQUIRE Europe and the Hong Kong Research Grants Council (Project No. CityU 11507022). Will Cong is also an advisor to BlackRock, Inc. We are grateful to Fahiz Baba-Yara (discussant), Svetlana Bryzgalova (discussant), John Campbell, Andrei Goncalves, Bing Han, Kewei Hou, Bob Jarrow, Ron Kaniel, Chris Malloy, Alberto Rossi (discussant), Gideon Saar, Shrihari Santosh (discussant), Peixuan Yuan (discussant), and Guofu Zhou for detailed comments and suggestions. We also thank Doron Avramov, Jie Cao, Zhanhui Chen, Eric Ghysels, P. Richard Hahn, Dashan Huang, Sophia Zhengzi Li, Asaf Manela, Stefan Nagel, Adam Reed, Artem Streltsov, Yinan Su, Fred Sun, Yaki Tsaig, Mao Ye, Dacheng Xiu, Bobby Yu, and seminar and conference participants at ArrowStreet Capital, Baylor University, BUAA, Ca' Foscari University of Venice, Cambridge University Algorithmic Trading Society Quant Conference, CFTRC 2022, CityUHK, CKGSB, Conference on FinTech, Innovation and Development at Hangzhou (2nd), Cornell University, Goethe University Frankfurt, GSU-RFS FinTech Conference 2022, HKAIFT-Columbia joint seminar, Hunan University, INFORMS Annual Meeting 2021, 4th International FinTech Research Forum (RUC), 2023 Mid-South DATA Conference (Memphis), KAIST Digital Finance Conference, NFA 2022, University of Oxford SML-Fin Seminar, OSU, PKU Guanghua, PKU-NSD, PKU-NUS Annual International Conference on Quantitative Finance and Economics, Qube Research and Technology, Reichman University (IDC Herzliya), Schroders Investments, Shanghai Financial Forefront Symposium (3rd), SHUFE, SMU, SUSTech, TAMU, Tsinghua SEM, UNC, University of Bath, University of Hawai'i, USC, USTC, University of Macau, World Online Seminars on Machine Learning in Finance, WFA 2022, and 2023 XJTLU AI and Big Data in Accounting and Finance Conference for constructive discussions and feedback. The views expressed herein are those of the authors and do not necessarily reflect the views of the National Bureau of Economic Research.

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NBER工作论文第30805号；2022年12月，2023年12月修订。JEL分类：C1、G11、G12。

摘要

我们提出一类新的树模型——面板树（P-Trees），用于分析非平衡面板等面板数据。模型使用经济目标引导的全局切分准则，而不是局部准则，在保持稀疏性和可解释性的同时防范过拟合。为推广证券排序并更好地估计有效前沿，我们自上而下生长P-Tree，划分资产收益横截面以构造测试资产，并在均值—方差有效框架下恢复随机贴现因子，同时将公司特征之间的非线性及非对称交互可视化。

应用于美国股票后，提升式多因子P-Trees相较于由成熟因子和常见测试资产构成的前沿，显著拓展了有效前沿。P-Tree测试资产具有分散化特征，且相对基准模型存在显著的未解释alpha。统一生成的P-Tree因子，在横截面收益定价方面优于大多数已知的可观测和潜在因子模型，并提供透明、有效的交易策略。除资产定价外，本文框架还为通过目标导向的高维聚类分析面板数据提供了一种替代方法，相比近期机器学习与人工智能模型具有更高的可解释性和计算效率。

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P-Tree的GitHub软件包：https://github.com/Quantactix/TreeFactor

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Lin William Cong, Guanhao Feng, Jingyu He, and Xin He
NBER Working Paper No. 30805
December 2022, revised December 2023
JEL No. C1,G11,G12

ABSTRACT

We develop a new class of tree-based models (P-Trees) for analyzing (unbalanced) panel data using economically guided, global (instead of local) split criteria that guard against overfitting while preserving sparsity and interpretability. To generalize security sorting and better estimate the efficient frontier, we grow a P-Tree top-down to split the cross section of asset returns to construct test assets and re-cover the stochastic discount factor under the mean-variance efficient framework, visualizing (asymmetric) nonlinear interactions among firm characteristics. When applied to U.S. equities, boosted (multi-factor) P-Trees significantly advance the efficient frontier relative to those constructed with established factors and common test assets. P-Tree test assets are diversified and exhibit significant unexplained alphas against benchmark models. The unified P-Tree factors outperform most known observable and latent factor models in pricing cross-sectional returns, delivering transparent and effective trading strategies. Beyond asset pricing, our framework offers a more interpretable and computationally efficient alternative to recent machine learning and AI models for analyzing panel data through goal-oriented, high-dimensional clustering.

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A P-Tree Github Package is available at https://github.com/Quantactix/TreeFactor

1 引言

估计均值—方差有效（MVE）前沿，对资产定价和投资管理仍然至关重要。然而，试图直接从个体资产收益的高维非平衡面板估计切点组合，以达到有效前沿，并不现实（Cochrane, 2009）。实证资产定价研究通常考虑一组“分散化”的资产作为基础组合，并识别少数共同因子，以实现多因子组合有效性（Fama, 1996）。从张成有效前沿的角度看，使用回归左端测试资产与右端风险因子的两类方法在统计上等价（Gibbons、Ross和Shanken, 1989），因为在alpha为零的MVE框架下，测试资产的联合夏普比率与风险因子的联合夏普比率相等。

但流行因子模型甚至难以张成常见测试资产集合，更不用说个股收益所构成的最终有效前沿（Kozak et al., 2018；Daniel et al., 2020；Lopez-Lira和Roussanov, 2020）。实证文献大多采用人为指定的特征排序组合作为测试资产，例如Fama和French（1993、1996）。由于高维排序困难（Cochrane, 2011），这些方法通常局限于单变量和双变量排序。即便采用条件排序（Daniel et al., 1997），研究者也往往忽略非线性和非对称交互，只关注对全部资产统一适用的交互。无法捕捉有效前沿，会削弱模型估计和评价的有效性（Lewellen et al., 2010；Ang et al., 2020）。缩小真实有效前沿与资产组合或因子组合的切点组合之间差距的关键，是系统利用高维公司特征；这些特征包含资产收益动态联合分布的丰富信息。

为此，我们提出面板树（P-Tree）框架。这是一类新型、用途广泛的模型，通过对面板观测进行聚类来实现给定的经济目标。

脚注1：为改进估计，研究者使用大量特征构造特征管理因子（Kozak和Nagel, 2022）或机器学习潜在因子（Kelly et al., 2019；Lettau和Pelger, 2020b；Chen et al., 2023），尽管这些方法存在解释困难。

脚注2：Cochrane（2011）认为，预期收益、方差和协方差是特征的稳定函数；另见Kelly et al.（2019）、Kozak et al.（2020）。Feng et al.（2023）基于高维特征，使用深度学习估计切点组合。

1 Introduction

Estimating the mean-variance efficient (MVE) frontier remains crucial in asset pricing and investment management. However, attempting to reach the efficient frontier by estimating the tangency portfolio from the high-dimensional, unbalanced panel of individual asset returns proves impracticable (Cochrane, 2009). Empirical asset pricing studies address the challenge by considering a "diversified" set of assets as basis portfolios and identifying a few common factors to achieve multi-factor portfolio efficiency (Fama, 1996). These approaches for using left-hand-side test assets and right-hand-side risk factors are statistically equivalent in terms of spanning the efficient frontier (Gibbons, Ross, and Shanken, 1989), because under the MVE framework with zero alphas, the collective Sharpe ratios of test assets equal those of risk factors.

However, popular factor models hardly span the common test asset universe (e.g., Kozak et al., 2018; Daniel et al., 2020; Lopez-Lira and Roussanov, 2020), let alone the ultimate efficient frontier constructed by individual asset returns.[1] The empirical literature mostly uses ad hoc characteristics-sorted portfolios as test assets (e.g., Fama and French, 1993, 1996), which are typically limited to univariate- and bivariate-sorted approaches, due to the difficulties of high-dimensional sorting (Cochrane, 2011). Even with dependent sorting (e.g., Daniel et al., 1997), researchers leave out nonlinearity and asymmetric interactions, instead focus on interactions uniformly applicable to all assets. The inability to capture the efficient frontier hampers the effectiveness of model estimations and evaluations (Lewellen et al., 2010; Ang et al., 2020). The key to bridging the gap between the true efficient frontier and tangency portfolios of assets or factor portfolios lies in systematically utilizing the high-dimensional firm characteristics that contain rich information on the joint distribution of asset returns dynamics.[2]

To this end, we introduce the Panel Tree (P-Tree) framework, a new and versatile model family that clusters panel observations to achieve given economic objec-

[1] To improve the estimation, numerous characteristics are considered in constructing characteristics-managed factors (Kozak and Nagel, 2022) or machine-learning-based latent factors (Kelly et al., 2019; Lettau and Pelger, 2020b; Chen et al., 2023), despite interpretation challenges.

[2] Cochrane (2011) asserts that expected returns, variances, and covariances are stable functions of characteristics (also see, e.g., Kelly et al., 2019; Kozak et al., 2020). Based on high-dimensional characteristics, Feng et al. (2023) estimate the tangency portfolio via deep learning.

P-Tree擅长分析非平衡面板等面板数据，继承了树模型的优势，同时通过经济目标引导的全局切分准则，实现可解释的目标导向搜索。应用于个体资产收益面板时，P-Tree在MVE框架下利用高维公司特征，联合生成分散化测试资产和潜在因子，以恢复随机贴现因子（SDF）。因此，它将树模型的用途拓展到纯预测之外，特别是拓展实证有效前沿和构建分散化基础组合，对资产定价和投资具有丰富启示。

具体来说，P-Tree采用自上而下的方法划分数千只个体资产的横截面，把特征值相近的资产归入少量群组，并形成市值加权组合。在资产定价目标指导下，模型依据全局切分准则生长，构造分散化测试资产与潜在因子模型，并估计有效前沿。根据Hansen和Jagannathan（1991），通过张成有效前沿而最大化夏普比率所对应的SDF，也会最小化用于解释横截面的HJ距离。在统一框架下生成的叶基础组合，为实证研究者提供了用于模型评价的一组分散化测试资产。此外，决策树的布局直观，可以将经济含义可视化，并识别一小组有用的公司特征；这些特征通过交互，共同生成测试资产和潜在因子。

从本质上说，P-Tree是一种受经济目标指导的贪心算法，在庞大的聚类空间中，依据公司特征值寻找较优的资产分组。资产定价中的基础组合通常按国家、行业等分类构造，或根据收益相关性、公司特征进行无监督聚类。相比之下，P-Tree生成由特征管理的叶基础组合，以最大化所形成MVE组合的联合夏普比率。

脚注3：树模型善于在低信噪比和小样本等环境中，预测具有高维、非线性和变量交互的复杂数据，相关研究包括Sorensen et al.（2000）、Rossi和Timmermann（2015）、Gu et al.（2020）、Bianchi et al.（2021）、Bryzgalova et al.（2023）。决策树易于可视化和解释，但当时尚无为证券排序与测试资产构造专门定制的树生长算法。多数现成机器学习方法针对独立同分布观测的预测任务设计，既非经济原则引导，也未针对金融面板数据定制。

tives. It is adept at analyzing (unbalanced) panel data and inherits the advantages of tree-based models while allowing interpretability under economics-guided global split criteria for goal-oriented search. When applied to the panel of individual asset returns, P-Trees utilize high-dimensional firm characteristics under the MVE framework to jointly generate diversified test assets and latent factors for recovering the stochastic discount factor (SDF). Therefore, they extend the scope of tree models beyond pure prediction tasks, particularly expanding the empirical efficient frontier and constructing diversified basis portfolios, with rich asset pricing and investment implications.[3]

Specifically, P-Trees employ a top-down approach, splitting the cross section of thousands of individual assets and grouping them into a small number of clusters with similar characteristic values to form value-weighted portfolios. Guided by asset pricing considerations, we grow P-Trees following global split criteria to construct diversified test assets and latent factor models, estimating the efficient frontier. According to Hansen and Jagannathan (1991), the resulting SDF that maximizes the Sharpe ratio by spanning the efficient frontier also minimizes the HJ-distance for explaining the cross section. Generated under the unified framework, the resulting leaf basis portfolios provide empirical researchers with a diversified set of test assets for model evaluation. Furthermore, the decision trees have an intuitive layout, allowing visualized economic interpretability and identifying a sparse set of useful firm characteristics that interact to jointly generate test assets and latent factors.

Fundamentally, P-Tree is a greedy algorithm guided by economic goals for optimal clustering based on firm characteristic values within a large clustering space. In asset pricing, basis portfolios are created by grouping assets based on classifications such as country or industry, or through unsupervised clustering based on the correlation of returns or firm characteristics. By contrast, P-Trees generate characteristics-managed leaf basis portfolios to maximize the collective Sharpe ratio of the resulting

[3]Tree-based models excel in predicting complex data with high dimensionality, nonlinearity, and variable interactions, even in low signal-to-noise environments and small sample sizes (e.g., Sorensen et al., 2000; Rossi and Timmermann, 2015; Gu et al., 2020; Bianchi et al., 2021; Bryzgalova et al., 2023). Decision trees are easy to visualize and interpret, but no algorithm currently customizes tree generation for security sorting and constructs test assets. Most off-the-shelf machine learning (ML) methods, including tree-based models, are designed for predicting tasks on independent and identically distributed (i.i.d.) observations and are not guided by economic principles or tailored for financial panel data.

这一方法既不属于通常意义上的监督学习，也不属于无监督学习。由于目标可以灵活指定，P-Tree适用于广泛的面板数据分析。

方法创新

资产定价应用清楚展示了本文的核心方法创新。

首先，现成机器学习方法，包括著名的分类回归树CART（Breiman et al., 1984），通常假设观测独立同分布，并非为从面板数据提取异质信息而设计。直接把CART、提升树或随机森林拟合到特征—收益数据上，虽然取得了一定效果（Gu et al., 2020），却忽视了面板结构。相比之下，P-Tree采用适用于多期观测的向量化叶参数，充分利用面板结构，并使全部时期共享不随时间变化的树结构。这种时间不变结构增强了面板机器学习模型的可解释性。随着树生长，节点对应的是基础组合收益向量，P-Tree因子则为这些组合的MVE组合。

其次，包括CART在内的标准树模型主要关注预测。CART递归生长，优化预测的二次损失；为提高计算效率，它通常在每个节点局部优化，不考虑兄弟节点。这种“短视”策略常导致过拟合，因为树越深，节点内观测越少。P-Tree将用途扩展到测试资产和潜在因子的生成，以及横截面收益的解释。借助经济目标，其迭代生长过程使用整个横截面的数据，以防范过拟合。它以非线性机器学习方法生成测试资产，并通过全局切分准则最大化联合夏普比率。

脚注4：作者开发并公开了R语言P-Tree统计软件包TreeFactor（https://github.com/Quantactix/TreeFactor）。研究者可通过修改切分准则与潜在因子生成方式，采用其他投资目标，例如由个股收益构建最小方差组合；也可用最小化总体定价误差的全局准则为个体资产定价。详见附录III。

MVE portfolios, without being either supervised or unsupervised. The objective can be flexibly specified, making the P-Tree framework applicable for a wide range of panel data analyses.[4]

Methodological innovations. Our emphasis on the asset pricing application effectively demonstrates the key methodological innovations. First, off-the-shelf ML methods, including the famous Classification and Regression Trees (CART, Breiman et al., 1984), typically assume that observations are independent and identically distributed (i.i.d.), and not designed for extracting heterogeneous information from panel data. Though naively fitting CART, boosted trees, or random forests on characteristics-return data shows positive performance (e.g., Gu et al., 2020), they ignore the panel data structure. By contrast, P-Tree fully embraces it and incorporates vectorized leaf parameters tailored for multi-period observations, enabling a time-invariant tree structure for all periods. The time-invariant P-Tree allows for interpretability when building ML models on panel data. As P-Tree grows, its nodes evolve into vectorized values of basis portfolio returns, with the P-Tree factor being their MVE portfolio.

Second, the standard tree-based models in ML, including CART, focus purely on prediction. Furthermore, CART-based models grow recursively, optimizing the quadratic loss of prediction. These models optimize locally at each node without considering sibling nodes, mainly for computational efficiency. However, this "myopic" strategy often leads to overfitting, because it operates on fewer observations in each node as the tree grows. In contrast, P-Trees broaden the applications beyond prediction, encompassing the generation of test assets and latent factors and explaining cross-sectional returns. With an economics-guided optimization objective, the iterative growth of P-Trees is designed to utilize data from the entire cross section to guard against overfitting. P-Tree uses a nonlinear ML approach to generate test assets in asset pricing, maximizing their collective Sharpe ratio with global split criteria.

[4]We have developed and shared the P-Tree statistical package, `TreeFactor`, in a public repository in `R` for other researchers to explore (see https://github.com/Quantactix/TreeFactor). For instance, we can adopt different investment objectives, such as constructing the minimum variance portfolio using individual asset returns, which can be achieved by modifying the split criteria and latent factor generation. We can also consider the global split criteria to minimize aggregated pricing errors for pricing individual assets, with further details provided in Appendix III.

这种方法将经济原则与统计性质相结合，同时保持单棵树的透明性和可解释性。作者认为，P-Tree提供了在MVE框架下构造分散化基础组合的首个连贯的一般性方法。

第三，P-Tree能够自然结合机器学习中的boosting策略，使多个因子展现出优良的资产定价与投资表现。如§2.3所述，提升式多因子P-Trees在MVE框架下利用尚未被张成的有效前沿，扩充资产与因子集合。通过先将个股分组为叶测试资产，多因子组合构造获得自然的分块结构，无须估计数千测试资产的大规模协方差矩阵。P-Tree还能评价相对于Fama－French等基准因子模型的增量信息，对额外特征进行交互与非线性建模。总体而言，提升生成多个因子时，P-Tree是PCA方法的一种替代方案，在捕捉非对称交互的同时，具有更高的可解释性和稀疏性。

实证发现

第一，我们将P-Tree应用于1981—2020年的美国个股月度收益与61项公司特征。P-Tree显著拓展有效前沿：单棵树含10项测试资产，年化夏普比率为6.37；20棵提升树含200项测试资产，达到15.63。这些数值显著高于传统基础组合，即数百个单变量或双变量排序组合，以及常用因子模型。这为现有实证文献所刻画的前沿与潜在有效前沿极限之间存在显著差距提供了证据。在统一MVE框架中，提升式P-Trees生成的多因子模型能有效解释横截面收益，最近20年的样本外年化夏普比率超过3。

第二，这些分散化P-Tree测试资产张成有效前沿，对其他因子模型提出显著挑战。许多测试资产存在经济意义和统计意义上都未被解释的收益。

脚注5：例如，P-Tree可用ME切分低波动股票，用BM切分高波动股票。相比对所有资产统一应用相同特征交互的传统对称排序，这提供了更细致的分析。

The approach combines economic principles and statistical properties while ensuring transparency and interpretability of individual trees. P-Tree offers the first coherent general approach for creating diversified basis portfolios under the MVE framework.

Third, P-Tree seamlessly integrates with ML's boosting strategy, enabling multiple factors to demonstrate exceptional asset pricing and investment performance. As detailed in section 2.3, boosted and multi-factor P-Trees expand the set of assets and factors by exploiting the unspanned efficient frontier under the MVE framework. The multi-factor efficiency of our boosted P-Trees enjoys a natural block-diagonal covariance structure in portfolio construction, achieved by grouping individual assets into P-Tree leaf test assets, without estimating a large covariance matrix of thousands of test assets. Furthermore, P-Trees can assess incremental information on benchmark factor models, such as the Fama-French factors, for interactive and nonlinear modeling of additional characteristics. Overall, when boosted to generate multiple factors, P-Trees offer an alternative to PCA methods, with greater interpretability and sparsity, while capturing (asymmetric) interactions.[5]

Empirical findings. First, we apply P-Tree to monthly U.S. individual stock returns from 1981 to 2020 using 61 firm characteristics. P-Trees significantly advance the efficient frontier with high annualized Sharpe ratios, ranging from 6.37 for a single P-Tree (with 10 test assets) to 15.63 for 20 boosted P-Trees (with 200 test assets). These numbers are substantially higher than those constructed by conventional basis portfolios (hundreds of univariate- or bivariate-sorted portfolios) or commonly used factor models. These findings provide strong evidence of a significant gap between the current empirical literature and the potential limits of the efficient frontier. Moreover, generated under the unified MVE framework, boosted P-Trees produce multi-factor models that effectively price the cross-sectional returns and offer annualized Sharpe ratios over 3 out-of-sample in the recent 20 years.

Second, these diversified P-Tree test assets span the efficient frontier and pose a significant challenge for alternative factor models. We identify many economically

[5]For instance, P-Trees might choose `ME` to split low-volatility stocks and `BM` to split high-volatility stocks, providing a more nuanced analysis than conventional symmetric sorting, which applies the same characteristic interactions to all.

多数月度alpha超过1%；首棵P-Tree相对于Fama－French五因子（FF5）模型的GRS统计量，即加权总体定价误差，达到141.27。这些由特征管理、包含非对称交互和非线性的叶基础组合，也难以由其他知名可观测因子模型（Hou et al., 2021）和机器学习潜在因子模型（Kelly et al., 2019；Lettau和Pelger, 2020b）解释。鉴于单变量与双变量排序组合对MVE前沿的张成不足、模型检验门槛较低，作者建议未来模型评价纳入这些多特征、系统排序的P-Tree测试资产。

最后，我们通过bagging设计随机P-Forest，评价特征重要性、考虑变量选择风险，并验证透明且可解释的P-Tree模型。结果确认，P-Tree算法选择的同一小组特征，例如SUE、DOLVOL和BM_IA，更可能代理SDF中真实的基本面风险，而人工选取因子的线性模型可能忽略这些信息。此外，在控制规模和价值等基准特征后，P-Tree提供了评价新增特征增量信息的新方法，发现投资、盈利、摩擦和无形资产相关的一些特征包含增量信号。P-Tree还展示全部切分规则，帮助研究者以图形方式理解公司特征之间的交互。

相关文献

本文首先与用于资产定价模型评价的测试资产选择和构造文献相关。实证文献包括正则化线性模型（Ao et al., 2019；Kozak et al., 2020；Bryzgalova et al., 2023）和PCA方法（Kozak et al., 2018；Lettau和Pelger, 2020b；Kim et al., 2021），大多将预先指定或特征管理的组合作为既定输入。有些研究分别评价不同测试资产集合以检验稳健性，有些选择测试资产，还有些组合测试资产。

脚注6：P-Tree允许多空组合的多头和空头分别与不同特征交互，从而使组合加载于不同叶基础组合。这种非对称交互排序不同于传统上把多空组合当成单一资产的处理，与Jarrow et al.（2021）分别建模异常收益组合两条腿的开创性工作互为补充。

and statistically unexplained test assets, indicated by most monthly alphas larger than 1%, and an extremely high GRS test statistic of 141.27 (aggregated weighted pricing error) for the first P-Tree against the Fama-French five-factor (FF5) model. These asymmetric, interactive, and nonlinear characteristics-managed leaf basis portfolios are difficult to explain using other well-known observable factor models (e.g., Hou et al., 2021) and ML-based latent factor models (e.g., Kelly et al., 2019; Lettau and Pelger, 2020b). Given the insufficient MVE spanning and low hurdles of univariate- or bivariate-sorted portfolios for model testing, we recommend including these multi-characteristic, systematically sorted P-Tree test assets in any future model evaluation.

Finally, we design the random P-Forest with bagging to evaluate characteristic importance, account for variable selection risk, and validate transparent and interpretable P-Tree models. We confirm the same small set of characteristics (e.g., `SUE`, `DOLVOL`, and `BM_IA`) selected by the P-Tree algorithm is more likely to proxy for the true fundamental risk inherent in the SDF, which may be overlooked in a linear factor model with ad hoc selected factors. In addition, P-Tree offers a novel approach to assess the incremental information of newly added characteristics by controlling for baseline characteristics of size and value, uncovering some characteristics associated with investment, profitability, frictions, and intangibles that contain incremental signals. Last but not least, P-Trees display all split rules, aiding researchers in understanding the interactions among firm characteristics graphically.[6]

Literature. Foremost, our paper relates to the selection and construction of test assets used for asset pricing model evaluations. The empirical literature, including regularized linear models (e.g., Ao et al., 2019; Kozak et al., 2020; Bryzgalova et al., 2023) and the PCA approaches (e.g., Kozak et al., 2018; Lettau and Pelger, 2020b; Kim et al., 2021), largely takes pre-specified or characteristics-managed portfolios as given. Some evaluate different sets of test assets separately for robustness checks (e.g., Fama and French, 1996, 2016), select test assets (e.g., Daniel et al., 2020; Giglio et al., 2022; Bryz-

[6]P-Trees allow the long and short legs of a long-short portfolio to interact with different characteristics, thereby loading the portfolio on different leaf basis portfolios. This asymmetric interactive sorting contrasts with the traditional treatment of a long-short portfolio as a single asset and complements the pioneering work of Jarrow et al. (2021) in modeling the two legs of anomaly portfolios separately.

相关研究包括Fama和French（1996、2016）的多集合检验，Daniel et al.（2020）、Giglio et al.（2022）、Bryzgalova et al.（2023）的测试资产选择，以及Feng et al.（2020）、Giglio和Xiu（2021）的测试资产组合。

作者将P-Tree定位为在MVE框架下，对个体证券进行目标导向的系统聚类并生成叶基础组合的首个方法。Hoberg和Welch（2009）提出通过优化目标函数而非排序来构建因子和测试组合。无监督聚类方面，Ahn et al.（2009）通过建模组内、组间个股收益相关性生成基础组合；Chen et al.（2017）通过聚类对冲基金alpha评价业绩；Patton和Weller（2022）则按资产在风险因子上的异质风险溢价进行横截面分组。本文与Cong et al.（2022）属于较早通过最大化张成有效前沿等特定经济目标、迭代聚类面板数据的研究。

纳入大量测试资产会带来高维组合估计或弱因子等实证挑战（Giglio et al., 2022），促使研究者开发新的测试资产选择方法，如Daniel et al.（2020）、Zhu et al.（2020）。特别地，Bryzgalova、Pelger和Zhu（2023）在大量基础组合上收缩无用资产以估计最优组合，并将证券排序与一组决策树联系起来。他们侧重自下而上的剪枝或正则化，没有指定树生长的切分准则，并人为设置少量切分候选及较浅的初始树深度，因为枚举全部树结构以供剪枝在计算上不可行。P-Tree则自上而下生长，以目标导向聚类高效搜索高维特征上的广义顺序排序空间，并允许非对称交互。

本文也对潜在因子模型文献作出贡献。早期研究用纯统计准则的主成分分析提取潜在因子，例如Chamberlain和Rothschild（1983）、Connor和Korajczyk（1986、1988）。较新的投影PCA（Kim et al., 2021）及工具变量PCA（Kelly et al., 2019、2023）使用公司特征作为工具，建模时变因子载荷并估计主成分。Lettau和Pelger（2020a、b）提出风险溢价PCA及风险溢价的正则化估计。

galova et al., 2023), or combine test assets (e.g., Feng et al., 2020; Giglio and Xiu, 2021).

P-Trees constitute the first goal-oriented, systematic clustering of individual securities and generation of leaf basis portfolios (test assets) under the MVE framework. Hoberg and Welch (2009) propose constructing factors and test portfolios by optimizing objective functions instead of sorting. For unsupervised clustering, Ahn et al. (2009) generates basis portfolios by modeling within- and between-group return correlations of individual stocks, Chen et al. (2017) groups hedge fund alpha values for performance evaluation, and Patton and Weller (2022) groups assets in the cross section based on their heterogeneous risk premia on risk factors. We join Cong et al. (2022) as the earliest studies that iteratively cluster panel data by maximizing specific economic objectives such as spanning the efficient frontier.

Including a large set of test assets presents empirical challenges, such as large portfolio estimation or weak factors (Giglio et al., 2022), leading to the development of new methods for selecting test assets in multiple studies (e.g., Daniel et al., 2020; Zhu et al., 2020). Notably, Bryzgalova, Pelger, and Zhu (2023) shrinks useless assets for estimating the optimal portfolio on a large number of basis portfolios and links security sorting to a collection of decision trees. The authors focus on pruning or regularizing the tree bottom-up without specifying a split criterion for tree growth and manually specify a small set of split candidates and a shallow depth for initial trees (since enumerating all possible tree configurations to prune from is computationally infeasible). P-Trees differ by growing top-down to provide a goal-oriented clustering approach and efficiently scanning the large space of generalized sequential sorting on high-dimensional characteristics, allowing for asymmetric interactions.

Moreover, our study contributes to the growing literature on latent factor models in asset pricing. Early works apply principal component analysis (PCA) to extract latent factors with a purely statistical criterion, such as Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986, 1988). For recent developments, the projected PCA (Kim et al., 2021) and the instrumental PCA (Kelly et al., 2019, 2023) use firm characteristics as instruments for modeling the time-varying factor loadings and estimating principal components. Lettau and Pelger (2020a,b) develop the risk premia

近期研究还开发了用于非线性潜在因子建模的定制深度学习模型，包括自编码器（Gu et al., 2021）、生成对抗网络（Chen et al., 2023）、特征排序因子近似（Feng et al., 2023）和结构神经网络（Fan et al., 2022）。在这些进展基础上，P-Tree提供了特征非线性和非对称交互的图形表示，而PCA或深度学习方法通常不提供这种表示。

最后，由于具有利用高维特征进行非线性建模的能力，机器学习和树方法已广泛用于金融，参见Gu et al.（2020）、Bianchi et al.（2021）、van Binsbergen et al.（2023）、Leippold et al.（2022）、Bali et al.（2023）。但多数研究要么直接应用现成机器学习进行预测，而未考虑面板结构或经济引导；要么集中于监督或无监督学习。近期例外包括Cong et al.（2022）将贝叶斯尖峰—厚尾先验与树模型用于横截面资产定价，Cong et al.（2021）用强化学习开发金融组合管理的首个“大”模型，以及Creal和Kim（2023）将贝叶斯树用于货币收益。本文也补充了在估计和评价机器学习或因子模型时引入经济约束的研究，如Gagliardini et al.（2016）、Avramov et al.（2023）。

本文结构如下：第2节介绍经济目标引导的P-Tree切分框架；第3节讨论实施、有效前沿估计和分散化测试资产生成；第4节介绍提升式多因子P-Trees并比较表现；第5节解释特征重要性、展示另一项应用，并评价不同宏观状态下的稳健性；第6节总结。网络附录I提供模拟证据；网络附录II、III通过基准因子调整和其他经济目标展示框架的灵活性。

PCA and provide a regularized estimation for risk premia. Recent studies, such as the auto-encoder (Gu et al., 2021), generative adversarial network (Chen et al., 2023), characteristics-sorted factor approximation (Feng et al., 2023), and structural neural network (Fan et al., 2022), have also developed customized deep learning models for nonlinear latent factor modeling. Following these developments, P-Trees provide a graphical representation for variable nonlinearity and asymmetric interactions, which PCA or deep learning methods do not offer.

Finally, ML and tree-based methods are widely adopted in finance due to their powerful nonlinear modeling capabilities using high-dimensional characteristics (e.g., Gu et al., 2020; Bianchi et al., 2021; van Binsbergen et al., 2023; Leippold et al., 2022; Bali et al., 2023). However, most studies either apply off-the-shelf ML models for prediction without considering the panel data structure or incorporating economic guidance, or focus on supervised or un-supervised learning. Recent exceptions include Cong et al. (2022), who combine Bayesian spike-and-slab priors and tree models for cross-sectional asset pricing, Cong et al. (2021), who develop the first "large" model in finance for portfolio management using reinforcement learning, and Creal and Kim (2023), who adapt Bayesian trees to currency returns. We also contribute to recent studies advocating imposing economic restrictions when estimating and evaluating ML or factor models (e.g., Gagliardini et al., 2016; Avramov et al., 2023).

The remainder of the paper follows: Section 2 introduces the P-Tree framework for the economic-guided split criteria. Section 3 discusses the implementation of P-Tree, estimation performance on the efficient frontier, and generation of diversified test assets. Section 4 introduces the boosted and multi-factor P-Trees and compares performance. Section 5 interprets P-Trees for character importance, demonstrates another application, and evaluates robust performance across macro states. Section 6 concludes. Simulation exercises demonstrating P-Trees' superiority can be found in Internet Appendix I. Internet Appendices II and III illustrate P-Tree's flexibility by allowing adjustments to benchmark factors and using alternative economic objectives.

2 用于资产定价的面板树

P-Tree针对经济面板数据设计，将树模型的用途扩展到预测之外。§2.1回顾CART并介绍P-Tree的动机；§2.2详细说明树的生长；§2.3介绍多因子提升式P-Trees；§2.4简述模拟设置与发现。

2.1 CART与P-Tree的创新

标准CART及其变体采用“分而治之”的预测策略（Breiman et al., 1984）。CART把预测变量空间划分为互不重叠的区域，并为每个区域赋予一个常数叶参数（Hastie et al., 2009）。它使用二元切分规则的决策树搜索并记录这种划分。假设有K个预测变量，第i个观测为z_i=(z_i,1,⋯,z_i,K)。第j条切分规则记为 $\tilde{c}(j)=(z_{\cdot},k,c_j)$，通过比较第k个预测变量与阈值c_j来划分数据。CART只考虑二元切分，因为任意多路切分都可以用多次二元切分表示。最优规则通过最小化预测误差选出。

经过J次切分后，CART把预测变量空间划分为J+1个叶区域，并为各节点赋予常数参数 μ_n。包含切分规则和叶参数的回归树可写成高维阶梯函数，见公式(1)（完整数学排版保留于对应原文页）。

指示函数对观测所属叶节点取1，对其他节点取0。叶参数代表阶梯函数的高度，通过对叶内训练数据求平均估计。这种非参数方法使CART适用于低信噪比和小样本环境。训练完成后，树找到新观测所在的叶，并用该叶参数作为预测值。

脚注7：CART是二元决策树，也是随机森林（Breiman, 2001）和提升树（Freund和Schapire, 1997）的基础。其他著名贝叶斯树模型包括BART（Chipman et al., 2010）和XBART（He et al., 2019；He和Hahn, 2023）。

2 Panel Tree for Asset Pricing

P-Tree is designed for economic panel data and expands tree models beyond predictions. Section 2.1 reviews CART and motivates the P-Tree framework. Section 2.2 delves into the growth of P-Trees, before section 2.3 introduces boosted P-Trees for a multi-factor model. Section 2.4 briefly presents the simulation settings and findings.

2.1 CART and P-Tree Innovations

The standard CART (Breiman et al., 1984) model and its variants use "divide-and-conquer" strategies for making predictions.[7] CART partitions the predictor space into distinct, non-overlapping regions of leaf nodes and assigns a constant leaf parameter to each region (Hastie et al., 2009). It searches and records the partition using a decision tree structure with binary split rules. Suppose the data constitute K predictors, the i-th observation is denoted by $\mathbf{z}_i = (z_{i,1}, \cdots, z_{i,K})$. The j-th split rule of the decision tree is denoted by $\tilde{\mathbf{c}}^{(j)} = (z_{\cdot,k}, c_j)$, which partitions data by checking whether the value of the k-th predictor $z_{\cdot,k}$ is greater or smaller than cutoff c_j. CART considers only binary splits, since any multiway-split tree can be represented as one with multiple binary splits. The optimal split rule is chosen to minimize the prediction error.

After J splits, CART partitions the predictor space into $J+1$ leaf nodes denoted by $\{\mathcal{R}_n\}_{n=1}^{J+1}$ and assigns a constant leaf parameter μ_n to each node. The regression tree $\mathcal{T}$, with parameters $\Theta_J = \{\{\tilde{\mathbf{c}}^{(j)}\}_{j=1}^{J}, \{\mu_n\}_{n=1}^{J+1}\}$, can be expressed as high-dimensional step functions:

$$\mathcal{T}(\mathbf{z}_i \mid \Theta_J) = \sum_{n=1}^{J+1} \mu_n \mathbb{I}\left\{\mathbf{z}_i \in \mathcal{R}_j\right\}. \tag{1}$$

The indicator $\mathbb{I}\{\mathbf{z}_i \in \mathcal{R}_j\}$ takes value 1 for one leaf node and 0 for all others. The leaf parameters, which represent the height of the step function, are estimated by averaging the training data within each leaf node. This nonparametric approach adapts CART to low signal-to-noise environments and small sample sizes. After training, the tree model predicts new observations by locating the corresponding leaf and using its

[7] CART is a binary decision tree model and serves as the foundation for ensemble methods such as random forest (Breiman, 2001) and boosting trees (Freund and Schapire, 1997). Other notable Bayesian tree models include BART (Chipman et al., 2010) and XBART (He et al., 2019; He and Hahn, 2023).

（承上页：训练好的树以新观测所在叶节点的参数作为预测。）

图1：决策树示例

左图为具有两次切分、三个叶节点和三个叶参数的决策树；右图为规模与价值两个预测变量空间中的相应划分。size为规模，value为价值，Yes/No表示满足/不满足切分条件，μ为叶参数。原始图形保留在随后原文页。

图1以一般性例子说明CART如何使用公司特征预测股票收益。通过递归划分特征空间，该特征预测模型形成小市值、大市值价值型及大市值成长型三类聚类结果。底部节点通常称为叶节点，与常数叶参数对应，因此同一节点内资产共享相同的预期收益估计。

CART等传统树模型存在多项限制，使其不适合经济或金融面板数据，也不适合生成测试资产或定价核。第一，它们把横截面或时间序列中的资产收益当作独立同分布观测。第二，它们侧重预测，使用统计性的局部切分准则，例如最小化某次切分两个子节点中的误差平方和，因此没有纳入面板结构或经济约束。随着树生长，节点中的观测减少，个体特有噪声导致过拟合。随机森林等集成方法虽然减少过拟合，却不如单棵树容易解释。

P-Tree通过两点处理这些问题：（i）用叶基础组合的收益时间序列向量作为叶参数；（ii）用考虑全部观测、而不只考虑某一父节点观测的经济目标指导生长。前者使我们能充分利用面板信息估计有效前沿；后者以包含经济原则的全局准则迭代建树，以防止过拟合，并将树的用途由纯预测扩展至测试资产生成。

脚注8：Gu et al.（2020）的图1也使用类似图示说明树预测模型。

译注：本页“防止过拟合”为作者的原文表述，不应理解为一般性的理论保证。

parameter as the prediction.

Figure 1: **Example of a Decision Tree**

Left: A decision tree has two splits, three leaf nodes, and three leaf parameters. Right: The corresponding partition plot is generated for the sample of predictor space on value and size.



For a general example, Figure 1 illustrates how CART uses firm characteristics to predict stock returns.[8] The characteristics-based prediction model is achieved by recursively partitioning the characteristic space, providing the clustering outcome (small-cap, large-cap value, and large-cap growth). The bottom nodes, often called leaf nodes, are associated with constant leaf parameters, meaning assets within the same node share the same expected return estimate.

Conventional tree-based models such as CART have multiple limitations, rendering them unsuitable for economic or financial panel data and for generating test assets or pricing kernels. First, they treat asset returns as i.i.d in the cross section or the time series. Second, they focus on predictions using statistical local split criteria, such as minimizing the sum of squared error objectives in the child nodes of a particular split. Therefore, they do not incorporate panel structure or economic restrictions. As the tree grows, the number of observations in each node decreases, and idiosyncratic noise causes overfitting. Although ensemble methods such as random forests reduce overfitting, they are less interpretable than single trees.

P-Tree solves both problems by (i) using a vectorized time series of leaf basis portfolio and (ii) using economic objectives that consider all observations instead of just those in a parent node to guide the tree growth. (i) allows us to extract panel data to estimate the efficient frontier thoroughly. (ii) iteratively builds P-Trees using global split criteria to include economic principles and prevent overfitting, extending the scope of trees from pure prediction to generating test assets.

[8] Gu et al. (2020) has a similar Figure 1 for demonstrating tree-based predictive models.

2.2 生长一棵P-Tree

P-Tree根据排名后特征的数值，将个体资产集合划为互不重叠的叶节点；在T期中共享不随时间变化的树结构，每期有I_t个资产。叶基础组合是各叶内股票的市值加权组合，其收益时间序列保存在向量化叶参数中。P-Tree从根节点顺序切分，每增加一次切分，就增加一个叶基础组合。因此，第j次切分后有j+1个叶组合，将数千个股降维为j+1个组合。这种特征管理的叶基础组合推广了常见的特征排序组合，允许多个特征的非对称交互。

以R_t^(j)表示第j次切分后全部叶基础组合的收益向量，以f_t^(j)表示由这些收益张成的因子。树的所有叶节点共有j+1个基础组合，记为[R_1,t^(j),…,R_n,t^(j),…,R_j+1,t^(j)]；其中原文将第n个叶节点在T期的收益表示为长度T的向量。第一次切分从包含全部资产的组合R_t^(0)开始，它就是根节点。树通过迭代更新叶收益与因子而扩展：先随着树生长增加叶基础组合，再利用扩展后的组合重新估计P-Tree因子。为了加快计算，我们采用顺序生长的贪心算法，而不枚举所有树结构。

图2：第一次切分示意

为了确定最优特征和阈值，逐项评价候选切分并计算准则值。图示为SUE≤0.4，将根节点切为两个叶组合。原文图注将SUE写成“特质波动率”，但后文与表A.1将其定义为盈利意外，二者不一致，此处保留缩写并提示。

第一次切分：根节点包含全部资产，对应市值加权市场因子。每一期的公司特征均在横截面上均匀标准化到[0,1]。

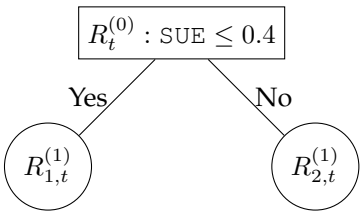
2.2 Growing a P-Tree

P-Tree partitions the universe of individual assets into non-overlapping leaf nodes based on the values of ranked characteristics with a time-invariant structure over T periods, with I_t assets in each period.The leaf basis portfolios are (value-weighted) portfolios of stocks within each leaf node, and their time series of returns are saved in a vectorized leaf parameter. P-Tree splits sequentially from the root node and generates one extra leaf basis portfolio after each additional split. Specifically, it produces $j+1$ leaf basis portfolios after the j-th split, reducing the dimension from thousands of individual stocks to $j+1$ portfolios. This concept of characteristics-managed leaf basis portfolios generalizes the popular characteristics-sorted portfolios to allow for asymmetric interactions of multiple characteristics.

Let $\mathbf{R}_t^{(j)}$ denote all return vectors of leaf basis portfolios after the j-th split of the tree, and let $f_t^{(j)}$ denote the factor spanned by $\mathbf{R}_t^{(j)}$. The tree has $j+1$ basis portfolios in all leaf nodes, denoted by $\mathbf{R}_t^{(j)} = [R_{1,t}^{(j)}, \cdots, R_{n,t}^{(j)}, \cdots, R_{j+1,t}^{(j)}]$, where $R_{n,t}^{(j)}$ represents a length T vector of returns over T periods for the n-th leaf node. To find the first split, we begin with a portfolio of all assets, denoted as $\mathbf{R}_t^{(0)} = \left[R_t^{(0)}\right]$, which serves as the root node for P-Tree. The tree is expanded through iterative updates of $\left\{\mathbf{R}_t^{(j)}, f_t^{(j)}\right\}$. First, the leaf basis portfolios $\mathbf{R}_t^{(j)}$ are expanded as the tree grows and splits into additional leaves. Second, using the expanded leaf basis portfolios, we re-estimate the P-Tree factor $f_t^{(j)}$. We use a greedy algorithm to grow the tree sequentially to expedite the process instead of enumerating all possible configurations.

Figure 2: **Illustration of the First Split**

To determine the optimal characteristic and cutpoint value, we evaluate a list of candidate splits (i.e., idiosyncratic volatility ($\texttt{SUE}$) ≤ 0.4) for calculating the split criterion value.



First split. The root node contains all assets corresponding to the value-weighted market factor. Firm characteristics are uniformly normalized cross-sectionally to the range

我们为特征z_·,k考察不同阈值c_m，例如五分位切分点[0.2,0.4,0.6,0.8]。给定候选规则 $\tilde{c}_{k,m}=(z_{\cdot,k},c_m)$，依据特征是否小于或等于阈值，把根节点分为左右子节点（图2）。

每个候选子叶中的股票收益构成一个叶基础组合，得到收益向量R_1,t^(1)和R_2,t^(1)。因为它们是收益向量，所以保留了资产收益的面板结构。P-Tree因子估计为所有叶基础组合的MVE组合，见公式(2)：因子是叶收益按权重的线性组合，权重与协方差矩阵的逆乘平均超额收益成正比。Σ_1和μ_1分别表示第一次切分后叶组合的协方差矩阵与平均超额收益。潜在因子的生成与切分准则一致，因为P-Tree因子正是全部叶基础组合的MVE组合。

切分准则最大化MVE组合的夏普比率平方，见公式(4)：$L(\tilde{c})=\mu_F'\,\Sigma_F^{-1}\mu_F$。其中F是生成的潜在因子，μ_F与Σ_F为其均值和协方差。该准则使模型在估计有效前沿这一全局目标下，同时构造和估计基础组合、潜在因子及有效组合权重。

每个候选切分都产生不同的数据划分、叶基础组合和P-Tree因子，因而得到不同准则值。我们遍历全部候选，并选择使准则最大的规则作为第一次切分。

脚注9：五分位排序是常用而高效的缩小搜索空间方法。作者的结果对不同设置，包括更密集的切分点，具有稳健性。

脚注10：参照Kozak et al.（2020）和Bryzgalova et al.（2023）的正则化方法，使用小收缩参数γ_Σ=10^(-4)稳健估计组合权重。公式(3)在叶组合收益协方差矩阵上加入γ_Σ乘单位矩阵，再求逆并乘平均收益。单位矩阵记为I_k+1。收缩参数使权重估计更稳定并防止过度杠杆，可调节正则化程度；此外，将权重归一化，使绝对权重之和为1。公式(2)—(4)原貌保留于随后原文页。

of $[0, 1]$ within each period. We assess different split threshold options c_m for characteristics $z_{\cdot,k}$, such as quintile splits $c_m \in [0.2, 0.4, 0.6, 0.8]$.[9] Consider a candidate split $\tilde{\mathbf{c}}_{k,m} = (z_{\cdot,k}, c_m)$ (Figure 2). The candidate partitions the root node into left and right child nodes based on whether $z_{\cdot,k}$ is less than or equal to c_m.

The observations of stock returns within each potential child leaf form a leaf basis portfolio, with vectors of return $R^{(1)}_{1,t}$ and $R^{(1)}_{2,t}$, respectively. Note they are vectors of returns, thus maintaining the panel structure of asset returns. The P-Tree factor is estimated as the MVE portfolio of all leaf basis portfolios,

$$f^{(1)}_t = \mathbf{w}^{(1)\intercal}\mathbf{R}^{(1)}_t, \quad \mathbf{w}^{(1)} \propto \mathbf{\Sigma}^{-1}_1 \boldsymbol{\mu}_1, \tag{2}$$

where $\mathbf{\Sigma}_1$ and $\boldsymbol{\mu}_1$ are the covariance matrix and average excess returns, respectively, for leaf basis portfolios $\mathbf{R}^{(1)}_t = \left[R^{(1)}_{1,t}, R^{(1)}_{2,t}\right]$ after the first split.[10] Note that generating latent factors is consistent with the P-Tree split criteria, because the P-Tree factor represents the MVE portfolio of all leaf basis portfolios.

Our split criterion aims to maximize the squared Sharpe ratio of the MVE portfolio as follows:

$$\mathcal{L}(\tilde{\mathbf{c}}_{k,m}) = \boldsymbol{\mu}'_{\mathbf{F}} \mathbf{\Sigma}^{-1}_{\mathbf{F}} \boldsymbol{\mu}_{\mathbf{F}}, \tag{4}$$

where $\mathbf{F} = f^{(1)}_t$ is the generated latent factor, with mean and covariance $\boldsymbol{\mu}_F$ and $\mathbf{\Sigma}_{\mathbf{F}}$, respectively. The criterion allows P-Tree to construct and estimate basis portfolios, latent factors, and efficient portfolio weights simultaneously, for the global objective of estimating the efficient frontier.

Each candidate split $\tilde{\mathbf{c}}_{k,m}$ generates a different partition of the data, resulting in unique leaf basis portfolios, corresponding P-Tree factors, and ultimately, varying valuations of the split criteria in (4). Consequently, we loop over all candidate splits and

[9] Quintile sorting is a popular and efficient method that reduces search space. Our results are robust to different specifications, including denser breakpoints.

[10] Following the regularization approaches in Kozak et al. (2020) and Bryzgalova et al. (2023), we utilize small shrinkage parameters, $\gamma_\Sigma = 10^{-4}$, for robustly estimating efficient portfolio weights:

$$\mathbf{w}^{(j)} = \left(\text{Cov}(\mathbf{R}^{(j)}_t) + \gamma_\Sigma \mathbf{I}_{k+1}\right)^{-1} \text{E}(\mathbf{R}^{(j)}_t), \tag{3}$$

where $\mathbf{I}_{k+1}$ is the identity matrix. Shrinkage parameters stabilize portfolio weight estimation and prevent over-leveraging, which can be adjusted to control the degree of regularization. Also, portfolio weights are normalized such that the sum of absolute weights equals one.

第二次切分可以发生在根节点左子节点，也可以发生在右子节点。我们评价两叶上的全部候选。原文此句写“选择使公式(4)准则最小的切分”；但前后公式与算法均要求最大化，存在文字不一致。图3展示两种第二次切分候选：任一情况下，一个叶变成内部节点，并产生两个新叶。

此时P-Tree因子根据全部三个叶基础组合构建，见公式(5)。Σ_2和μ_2分别是第二次切分后这三个组合的协方差矩阵和平均超额收益。三组合的具体构成取决于候选切分哪个节点。更新后的因子代入准则，最大化夏普比率平方，其中F=f_tˆ(2)。

图3：第二次切分示意

两图展示第二次迭代中切分一个节点的两种候选。两种情况下都得到三个叶基础组合，用于生成潜在因子。（a）在第一个叶节点按DOLVOL切分；（b）在第二个叶节点按DOLVOL切分。图中SUE、DOLVOL与阈值原样保留；Yes/No为条件成立/不成立。

我们提出的是全局切分准则，因为构造F，即全局MVE组合时，要考虑全部叶基础组合。CART侧重某一个叶节点；本模型则搜索全部叶上的全部候选，找出投资表现改善最大的一项。

生长终止：后续切分按同样方式决定。算法1列出树生长过程。选择叶节点数是树生长的自然停止方式。

select the one that maximizes the split criteria as our initial split rule.

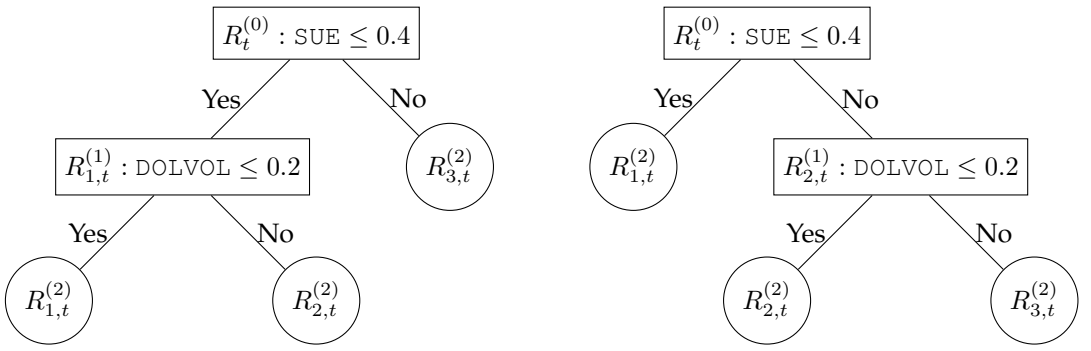
Second split. The second split can occur at the root's left or right child node. We assess the split criteria for all candidate splits for *both* leaf nodes and choose the split that minimizes the criteria as in (4). Figure 3 depicts the tree of the candidates for the second split. In either scenario, one leaf node splits, becoming an internal node and generating two new leaf nodes. The P-Tree factor is then constructed based on *all three* leaf basis portfolios:

$$f_t^{(2)} = \mathbf{w}^{(2)\intercal}\mathbf{R}_t^{(2)}, \quad \mathbf{w}^{(2)} \propto \boldsymbol{\Sigma}_2^{-1}\boldsymbol{\mu}_2, \tag{5}$$

where $\boldsymbol{\Sigma}_2$ and $\boldsymbol{\mu}_2$ are the covariance matrix and average excess returns for leaf basis portfolios $\mathbf{R}_t^{(2)} = \left[R_{1,t}^{(2)}, R_{2,t}^{(2)}, R_{3,t}^{(2)}\right]$ after the second split. The construction of the three basis portfolios depends on which node the candidate splits, as shown in Figure 3. The updated P-Tree factor is plugged in to maximize the squared Sharpe ratio, $\mathcal{L}(\tilde{\mathbf{c}}_{k,m}) = \boldsymbol{\mu}_{\mathbf{F}}'\boldsymbol{\Sigma}_{\mathbf{F}}^{-1}\boldsymbol{\mu}_{\mathbf{F}}$, where $\mathbf{F} = f_t^{(2)}$ is the updated P-Tree factor.

Figure 3: **Illustration of the Second Split**

These two figures illustrate two potential candidates for the second split, where one of the nodes splits. During the second iteration of the tree model, three leaf basis portfolios remain available to produce the latent factor.



(a) Splitting node $R_{1,t}^{(1)}$ at DOLVOL.

(b) Splitting node $R_{2,t}^{(1)}$ at DOLVOL.

Notably, our proposed objective is a *global* split criterion, because it considers all leaf basis portfolios when constructing $\mathbf{F}$ (the global MVE portfolio). Unlike CART, which focuses on a specific leaf node, our model explores all candidate splits in all leaf nodes to find the one with the largest investment improvement.

Growth termination. All subsequent splits are determined in the same manner. The tree-growing procedure is outlined in Algorithm 1. Determining the number of leaves

作者将叶节点数视作P-Tree所需的唯一调节参数。基准设置为J+1=10个叶节点。由于叶节点需要充当基础组合，生长时还要求最小叶规模为20；不满足此要求的叶不再切分。生长结束后输出切分序列、全部叶基础组合收益以及P-Tree因子。每个叶组合都可通过顺序切分揭示的经济基本面来解释。

其他经济目标引导准则：P-Tree能够灵活容纳公式(4)之外的目标，用个体资产收益估计MVE组合。例如，目标可以是最小方差组合，也可以是构造最能解释个股或基础组合等测试资产的潜在因子。关键在于明确经济目标，并用树中的贪心生长算法搜索庞大的观测聚类空间，以有效优化该目标。

在网络附录III，我们用估计带时变beta的潜在因子模型之后的已实现资产定价误差作为切分准则，展示这种灵活性。最小化定价误差的目标，与近期用深度学习生成潜在因子的文献精神一致（Gu et al., 2021; Chen et al., 2023; Feng et al., 2023）。我们对树结构运行因子回归，为叶参数提供条件期望值。这类P-Tree同样产生优良的投资与定价表现。

利用宏观变量识别状态：面板数据有多个维度，因此除横截面切分之外，也可以切分时间序列，这提供了计量经济学结构突变建模的替代方法。作为灵活扩展，P-Tree可以考虑不同宏观状态，在递归优化资产定价目标时识别这些状态，并分别拟合状态特定的树。附录III.D展示以宏观变量为工具识别状态、用于横截面资产定价的数据驱动时间切分。

is a natural turning point in tree growth, which is also the only tuning parameter needed for P-Tree. We consider P-Trees with $J + 1 = 10$ leaf nodes in the baseline specification. Furthermore, we mandate a minimum leaf size of 20 for growing the tree because the leaf needs to serve as a basis portfolio, and any leaves that do not meet this criterion are not subjected to further splitting. Once the tree growing process terminates, it outputs the P-Tree split sequence, leaf basis portfolios $\mathbf{R}_t^{(J)} = [R_{1,t}^{(J)}, \cdots, R_{J+1,t}^{(J)}]$, and the P-Tree factor $f_t^{(J)}$. Note each leaf portfolio can be interpreted through economic fundamentals revealed by the sequential splits.

Other economically guided criteria The P-Tree framework is flexible to accommodate objectives alternative to the global split criteria in (4) for estimating the MVE portfolio using individual asset returns. For instance, one might seek to create the minimum-variance portfolio or construct latent factors that best explain test assets, such as individual assets or basis portfolio returns. The key lies in specifying a clear economic goal and using the greedy growth algorithm embedded in trees to search the large observation clustering space to optimize the objective effectively.

In Internet Appendix III, we illustrate P-Tree's flexibility using a split criterion based on realized asset pricing errors after estimating a latent factor model with time-varying beta. The asset pricing goal of minimizing pricing errors shares the same spirit in the recent literature for latent factor generation via deep learning (e.g., Gu et al., 2021; Chen et al., 2023; Feng et al., 2023). We run a regression (factor model) for the tree structure and provide a conditional expected value for the leaf parameter. Such P-Trees again produce superior investment and pricing performances.

Regime detection by macro variables The panel data has multiple dimensions, so besides splitting the cross section, one can also split the time series, which is an alternative to structural break modeling in econometrics. For future flexible extensions, P-Tree can be extended to consider different macroeconomic regimes and detect those regimes when recursively optimizing the asset pricing objective, fitting regime-specific P-Trees. The data-driven time-series splits of P-Tree are shown by detecting macro-instrumented regimes for cross-sectional asset pricing in Appendix III.D.

2.3 提升式P-Trees

Boosting是一种把弱学习器结合为强学习器的机器学习技术（Freund和Schapire, 1997；Rossi和Timmermann, 2015）。提升式P-Trees在MVE框架下依次生成一系列增补已有树的加性树，通过新增基础组合进一步张成有效前沿，并生成多因子模型。考虑已有因子后，新增P-Tree因子以最大化切点组合的联合夏普比率，使boosting自然融入框架。

步骤1：初始因子f_1,t可以由第一棵P-Tree生成，也可以是研究者选择的基准因子，例如市场因子。

步骤2：第二棵树生成f_2,t，以补充第一个因子并最大化夏普比率平方。第二棵树采用类似的提升切分准则，见公式(6)，此时F=[f_1,t,f_2,t]，直接由全部因子形成MVE组合。

步骤3：重复第二步，依次生成P个因子，直到触发生长终止条件，例如夏普比率平方的改善已不显著（Barillas和Shanken, 2017）。

提升式P-Tree通过两步把多组叶基础组合用于构建MVE组合或SDF。先在每棵树内部，将其全部叶组合形成树特定的MVE组合，得到当前P-Tree因子；再将该因子与之前全部P-Tree因子或基准因子结合，形成多因子模型的全树MVE组合。顺序树的分块结构，间接对全树MVE权重施加正则化。虽然其夏普比率未必超过直接合并所有树全部基础组合的结果，但这种方法避免了高维估计问题，从而获得稳健估计与样本外表现。

2.3 Boosted P-Trees

Boosting is an ML technique that combines weak learners to form a strong learner (Freund and Schapire, 1997; Rossi and Timmermann, 2015). Boosted P-Trees sequentially grow a list of additive trees that augment previous trees under the MVE framework, which helps further span the efficient frontier with additional basis portfolios and generate multi-factor models. Considering existing factors, the boosting procedure seamlessly integrates into the P-Tree framework by creating additional P-Tree factors to maximize the collective Sharpe ratio of the tangency portfolio. The boosting procedure is outlined below.

1. The initial factor, denoted as $f_{1,t}$, can be either generated by the first single P-Tree or benchmark factors selected by researchers (e.g., the market factor).

2. The second factor $f_{2,t}$ is generated by the second P-Tree to complement the first factor and maximize the squared Sharpe ratio. The second tree growth follows a similar boosting split criterion:

$$\mathcal{L}(\tilde{\mathbf{c}}_{k,m}) = \boldsymbol{\mu}_{\mathbf{F}}' \boldsymbol{\Sigma}_{\mathbf{F}}^{-1} \boldsymbol{\mu}_{\mathbf{F}}, \tag{6}$$

   where $\mathbf{F} = [f_{1,t}, f_{2,t}]$. The MVE portfolio is generated from all the factors directly.

3. Repeat Step 2 to sequentially generate P factors $\mathbf{f}_t = [f_{1,t}, \cdots, f_{P,t}]$ until the stopping criteria are triggered (e.g., one finds insignificant squared Sharpe ratio improvement Barillas and Shanken, 2017).

The boosted P-Tree utilizes a two-step process to produce either the MVE portfolio or the SDF by incorporating multiple sets of leaf basis portfolios. Initially, the tree-specific MVE portfolio of all leaf basis portfolios within each P-Tree is computed to determine the current P-Tree factor. Subsequently, this factor is merged with all previous P-Tree factors or benchmarks to create an all-tree MVE portfolio for the multi-factor P-Tree model. Boosted P-Trees indirectly impose regularization on the all-tree MVE portfolio weights through the sequential tree block structure, even though the

2.4 模拟证据

本节模拟展示P-Tree选择特征的能力，以及测试资产捕捉特征非线性交互的有效性。附录I用同一实证样本校准收益生成过程，设定三个真正起作用的特征及其交互项和非线性项，同时加入大量与真实特征相关、但不驱动收益的冗余特征。

首先，我们比较P-Tree测试资产与无法建模这些非线性交互的传统特征排序组合的夏普比率和alpha。在不同信噪比下，无论样本内还是样本外，单棵与提升式P-Tree均能产生夏普比率更高、未解释定价误差更大的叶基础组合。即使传统排序直接使用真实特征，单变量和双变量排序仍不如利用高维特征生成的P-Tree测试资产。这与§4.2中P-Tree测试资产具有更好均值—方差有效性的实证发现一致。

其次，我们展示P-Tree在大量冗余或无用特征中选择稀疏真实特征的能力。单棵树的可解释性依赖变量选择能力。重复模拟表明，P-Tree在前几次切分中以较高频率选出真实特征，且信噪比越高，正确选择率越高。只有模拟能识别这一性质，因为真实世界中的稀疏真实特征集合未知。这些稳定的选择性质与§5.1的特征重要性评价相关。

最后，我们考察无法观测全部真实特征、但仍可观测大量冗余或无用特征时的效率损失。这模拟了因各种原因出现的信息不完整或无用信息环境。

resulting Sharpe ratio may not surpass the one generated directly from all basis portfolios in all P-Trees. This avoids high-dimensional estimation issues and, in turn, leads to robust estimation and out-of-sample performance.

2.4 Simulation Evidence

Simulation studies in this section demonstrate the ability of P-Tree models to select characteristics and the effectiveness of P-Tree test assets in capturing non-linear interactions among characteristics. In Appendix I, we calibrate the return-generating process on the same empirical data sample, considering three true characteristics, their interactions, and nonlinear terms. Additionally, we include a large set of redundant characteristics that are correlated with the true characteristics but do not drive returns.

First, we compare the Sharpe ratios and alphas of P-Tree test assets with conventional characteristics-sorted portfolios, which cannot model the nonlinear characteristics interactions. The single P-Tree and boosted P-Trees consistently produce leaf basis portfolios with higher Sharpe ratios and larger unexplained pricing errors under various signal-to-noise-ratio levels, both in-sample and out-of-sample. Even though sorting is performed on the true characteristics directly, univariate- and bivariate-sorted portfolios still underperform P-Tree test assets generated on high-dimensional characteristics. This result confirms our empirical findings that P-Tree test assets achieve better mean-variance efficiency than sorted portfolios as in Section 4.2.

Second, we demonstrate that P-Tree helps select the sparse true characteristics and a large number of redundant or useless ones. The interpretability of the single P-Tree relies on its ability to select variables. With repeated simulations, we show high frequencies for P-Tree for selecting the true characteristics in the first few splits. These correct selection rates are higher for cases with a high signal-to-noise ratio. This finding can only be identified from simulations, because we remain agnostic regarding the true sparse characteristics in the real-world data. The consistent selection properties are related to evaluating characteristics' importance in Section 5.1.

Finally, we evaluate the efficiency loss without observing all true characteristics but still observe a large set of redundant or useless characteristics. This exercise evaluates the model's performance with incomplete or useless information, which may

在所校准的收益生成过程中，缺失真实特征后夏普比率与CAPM alpha明显下降，P-Tree更频繁地选择冗余或无用特征。因此，当已有大量特征时，实证研究应全部纳入，而非人为先行选择。详见附录I。

3 划分横截面

P-Tree可用于美国上市股票或公司债，本文集中研究前者。在大型模型兴起的背景下，P-Tree提供了一种无需过多计算资源、且易解释的替代方法。

3.1 美国上市股票数据

股票数据与特征：我们对美国股票集合采用标准筛选，与Fama－French因子构造等做法相同。仅纳入在NYSE、AMEX或NASDAQ上市超过一年、CRSP股票代码为10或11的观测；排除账面权益或滞后市值为负的股票。每只股票每月使用61项公司特征，覆盖动量、价值、投资、盈利能力、摩擦（或规模）和无形资产六类。特征在横截面标准化至[0,1]，清单见表A.1。

月度样本为1981—2020年。前20年每月股票数的平均数和中位数分别为5265与4925，后20年分别为4110与3837。个股收益在横截面1%和99%分位处缩尾，以减轻异常值影响。全部40年用于基准分析；前20年（1981—2000）与后20年（2001—2020）用于子样本稳健性检查，以及训练、测试样本划分。

脚注11：在含61项特征、220万观测的美国数据上，基准单树在Intel Xeon Gold 6230 CPU服务器上训练约需20分钟。

脚注12：例如2020年12月市值标准化为[0,1]，最低者为0，最高者为1，其余均匀分布其间。缺失特征填为0.5，即在证券排序中视为中性。

occur for various reasons. Under our calibrated return-generating process, the Sharpe ratios and CAPM alphas decline substantially, and P-Tree selects redundant or useless characteristics more frequently. Therefore, when a large set of characteristics is provided, we should include all of them rather than performing ad hoc variable selection in empirical studies. For more details, please refer to Appendix I.

3 Splitting the Cross Section

The P-Tree framework easily applies to U.S. public equities or corporate bonds, focusing on the former in this paper. In an era of emergent large models, P-Tree offers an interpretable alternative that does not require excessive computation resources.[11]

3.1 Data on U.S. Public Equities

Equity data and characteristics. The standard filters (e.g., same as in Fama-French factor construction) are applied to the universe of U.S. equities. This universe includes only stocks listed on NYSE, AMEX, or NASDAQ for more than one year and uses those observations for firms with a CRSP share code of 10 or 11. We exclude Stocks with negative book equity or lag market equity. We use 61 firm characteristics with monthly observations for each stock, covering six major categories: momentum, value, investment, profitability, frictions (or size), and intangibles. Characteristics are standardized cross-sectionally to the range $[0, 1]$. Table A.1 lists these input variables. [12]

The monthly data ranges from 1981 to 2020. The average and median monthly stock observations are 5,265 and 4,925, respectively, for the first 20 years, and 4,110 and 3,837, respectively, for the latter 20 years. We apply cross-sectional winsorization at 1% and 99% to mitigate the impact of outliers on individual stock returns. The entire 40-year sample is used for benchmark analysis. The sample is split into two parts—the first 20 years from 1981 to 2000 and the recent 20 years from 2001 to 2020—for a subsample robustness check and as training and test samples.

[11] The baseline single P-Tree model trained on U.S. data runs about 20 minutes on a server with an Intel Xeon Gold 6230 CPU, for a training dataset with 61 characteristics and 2.2 million observations.

[12] For example, market equity values in Dec. 2020 are uniformly standardized into $[0, 1]$. The firm with the lowest value is 0, and the highest is 1. All others are distributed uniformly in between. Missing values of characteristics are imputed as 0.5, which implies the firm is neutral in the security sorting.

宏观经济变量：我们另外使用10个宏观变量捕捉潜在状态转换。表A.2汇总了市场择时预测变量、债券市场预测变量和标普500整体特征。各变量用过去10年的历史分位数标准化；这种滚动窗口标准化便于比较预测变量水平并识别宏观状态。

3.2 单棵P-Tree的可视化

图4绘制P-Tree。S#代表切分顺序，N#代表节点编号，#后的数字是节点内每月股票数的中位数；终端叶中报告的就是这一组合规模指标。切分规则包括选中的特征和横截面五分位阈值[0.2,0.4,0.6,0.8]。第一次切分前，树从根节点N1开始，根节点基础组合代表市值加权市场组合。

数据驱动的树首先按季度盈利意外SUE（Rendleman Jr et al., 1982），在0.4即40%分位处切分。40%的股票进入左叶N2，60%进入右叶N3。第二次在N3按交易金额DOLVOL（Chordia et al., 2001）的0.2阈值切分，第三次在N2按同样DOLVOL阈值切分。后续特征包括行业调整账面市值比BM_IA、ROA、零交易指标ZEROTRADE和市值ME。九次切分后停止，得到十个叶基础组合。

聚类模式与非对称交互：模型依据特征，在经济目标引导准则下聚类相似资产。图4展示用于划分横截面的非对称特征交互。同一路径上的特征共同定义一个叶区域，从而形成交互。例如，流动性指标ZEROTRADE适合继续划分低SUE、低DOLVOL股票。

脚注13：通胀值高于0.7，意味着当前通胀高于过去十年70%的观测。

脚注14：相较于用五分位保障每组观测数量的传统单变量排序，作者R包允许更灵活、更细的网格搜索。

译注：原文把ROA文字展开为return-on-equity，而表A.1为return on assets，存在不一致；保留ROA缩写。

Macroeconomic variables. In addition, we use 10 macro variables to capture potential regime switches. Table A.2 summarizes the macro variables, which include market timing macro predictors, bond market predictors, and aggregate characteristics for S&P 500. We standardize these macro predictor data by the historical percentile numbers for the past 10 years.[13] This rolling-window data standardization is useful when comparing the predictor level to detect different macroeconomic regimes.

3.2 Visualizing a Single P-Tree

Figure 4 plots the P-Tree diagram. In each leaf node, S# represents the order of sequential splits, N# is the node index, and the # numbers are the monthly median number of stocks within the node. We provide the numbers in the terminal leaves for the portfolio size: the monthly median number of stock observations. The split rules are the selected splitting characteristics and cross-sectional quintile cutpoints $[0.2, 0.4, 0.6, 0.8]$.[14] Before the first split, the tree grows from the root node (N1), whose leaf basis portfolio represents the value-weighted market portfolio.

The data-driven P-Tree first splits along the unexpected quarterly earnings (`SUE`, Rendleman Jr et al., 1982) at 0.4 (40% quantile). After this split, 40% of the stocks go to the left leaf (labeled N2), and 60% go to the right (N3). Then, the second split is on the dollar trading volume (`DOLVOL`, Chordia et al., 2001) at 0.2 of the right leaf (N3), and the third split is also on `DOLVOL` at 0.2 of the left leaf (N2). Furthermore, subsequent splits include the industry-adjusted book-to-market ratio (`BM_IA`), return-on-equity (`ROA`), zero trade (`ZEROTRADE`), and market equity (`ME`). After nine splits, we stop the P-Tree growth and obtain 10 leaf basis portfolios.

Clustering patterns and asymmetric interactions. P-Tree helps cluster similar assets under the economics-guided split criterion based on underlying characteristics. Figure 4 illustrates the (asymmetric) characteristics interactions for splitting the cross section. By jointly defining the partition corresponding to the leaf node, P-Tree learns the interaction of characteristics appearing in the same path. For instance, `ZEROTRADE`

[13] For example, inflation greater than 0.7 implies the current inflation level is higher than 70% of observations during the past decade.

[14] Our `R` package offers more flexibility and finer grid searches than conventional univariate security sorting, which uses quintiles to ensure adequate observations in each portfolio.

而对低SUE、非低DOLVOL股票，在MVE框架下，规模ME可能是更好的后续切分指标。

图5的区域划分图提供另一种展示聚类和非对称模式的方式，报告每个叶的平均月度超额收益和年化夏普比率。第一步按SUE的0.4切分，低SUE组含40%股票，高SUE组含60%。N2与N3的月度预期收益差为0.46%，夏普比率分别为0.25和0.64。

第二步在N3按DOLVOL的0.2切分为N6、N7，低DOLVOL组N6的预期收益和夏普比率高于N7。第三步在N2按同一阈值切分为N4、N5，低DOLVOL组N4的预期收益和夏普比率却低于N5。这表明DOLVOL在N2与N3中的作用不同。

图4：1981—2020年的面板树

图中列出各父节点的切分特征、阈值、节点编号及切分编号。例如第一次切分S1在N1按SUE的0.4进行；第二次S2在N3按DOLVOL的0.2进行。终端叶同时列出月度股票数中位数，例如N8为134。股票特征定义见表A.1，子样本树见图A.1。Y/N分别表示切分条件成立/不成立。树图与全部节点数值在对应原文页完整保留。

of liquidity is a valuable indicator for further splitting low-SUE low-DOLVOL stocks. However, for low-SUE non-low-DOLVOL stocks, ME of size might be a better indicator for a further split under the MVE framework.

The partition plot in Figure 5 is an alternative way to visualize the clustering and asymmetric patterns. We report each leaf's monthly average excess returns and annualized Sharpe ratios. First, the P-tree splits on SUE at 0.4, which yields a low-SUE portfolio including 40% of the stocks and a high-SUE portfolio constituted by 60% of the stocks in the cross section. The spread of monthly expected returns between N2 and N3 is 0.46%, and the Sharpe ratios range from 0.25 to 0.64.

Second, we split on DOLVOL at 0.2 on N3 to harvest N6 and N7, where the low-DOLVOL portfolio N6 has *higher* expected returns and Sharpe ratio than N7. Third, we split on DOLVOL at 0.2 on N2 to harvest N4 and N5, where the low-DOLVOL portfolio N4 has *lower* expected returns and Sharpe ratio than N5. We find DOLVOL has dif-

Figure 4: **Panel Tree from 1981 to 2020**

We provide splitting characteristics, cutpoint values for each parent node, and their respective node and splitting indexes. For example, node N1 is split by SUE at 0.4 as the first split S1, and the second split S2 is on node N3 by DOLVOL at 0.2. The median monthly number of assets in the terminal leaf basis portfolios is also included. For example, node N8 has 134 stocks by monthly median. Table A.1 describes equity characteristics. Figure A.1 reports the P-Tree diagrams for subsamples.

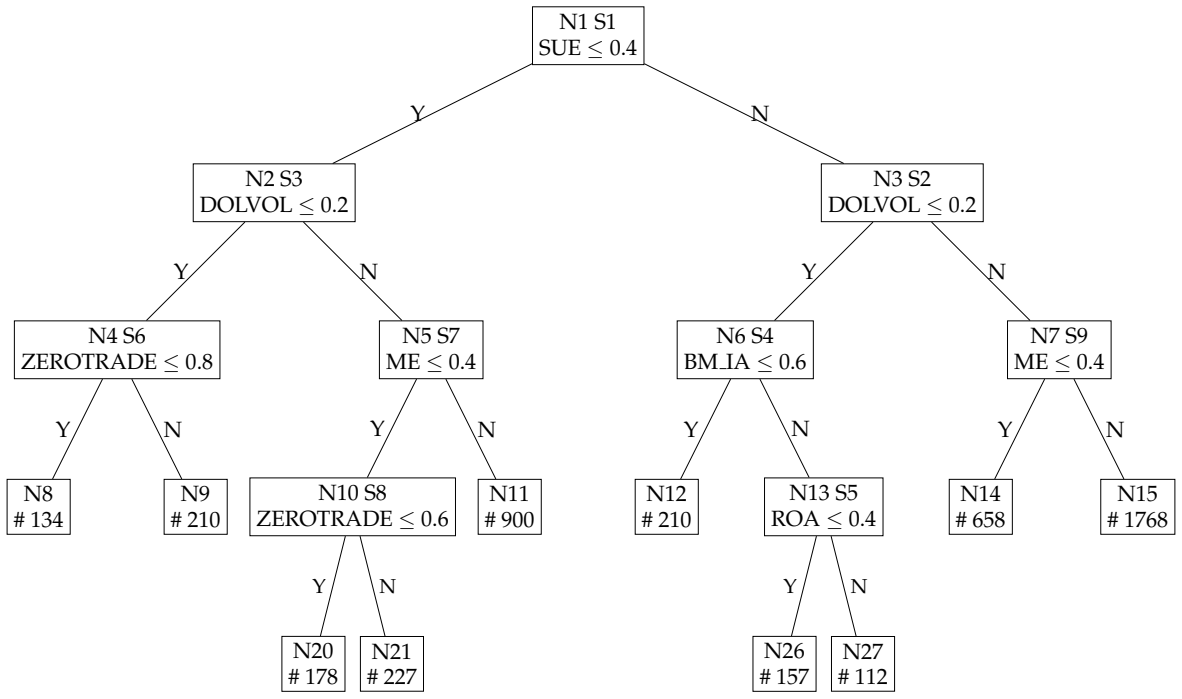


图5：用区域划分图展示非线性交互

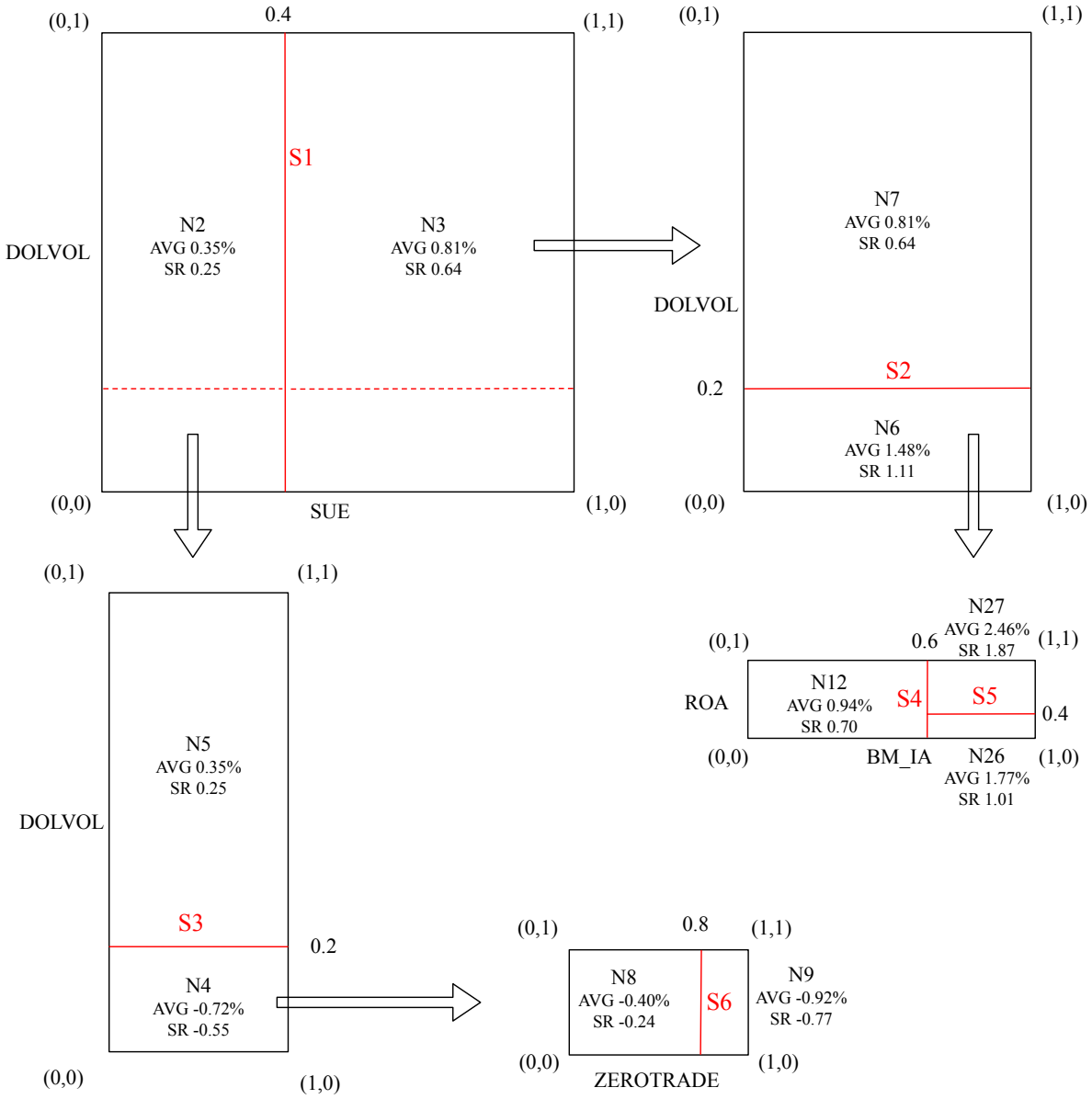
图示对应图4树结构前几次切分。第一次S1在全体股票SUE的40%分位处；图注原文将高SUE区域的第二次DOLVOL切分写为80%，但图中和正文显示0.2，二者不一致。每个区域报告月平均超额收益AVG与年化夏普比率SR。箭头表示下一次切分发生在上一步已划分的区域内。变量、区域坐标和数值完整保留在原图。

（承上页）原文随后称：在高SUE一侧，DOLVOL与资产收益正相关；在低SUE一侧，DOLVOL与资产收益负相关。这是SUE与DOLVOL非对称交互的例子。做空N4，即低SUE、低DOLVOL组合，同时做多N6，即高SUE、低DOLVOL组合的简单策略，月度预期收益超过2%。各叶之间的收益差异体现了利用特征非对称交互划分横截面的价值。

译注：上述“正相关/负相关”方向与前页N6/N7、N4/N5数值对比相反；译文保留原文主张并标注不一致，不自行改写数据。

Figure 5: **Visualizing Nonlinear Interactions with Partition Plots**

This diagram illustrates the partitions for the first few splits of the tree structure in Figure 4. The first split (S1) occurs at 40% of SUE on the entire stock universe, and the second split occurs at 80% of DOLVOL on the high SUE partition. The portfolio results for each partition are provided (monthly average excess returns (AVG) and annualized Sharpe ratios (SR)). The arrows indicate the next split is implemented on the partitioned area from the previous one.



ferent impacts on N2 and N3. On the high-SUE side, DOLVOL positively correlates to asset returns. However, on the low-SUE side, DOLVOL negatively correlates with asset returns. This is an example of asymmetric interaction between SUE and DOLVOL. A simple trading strategy that shorts N4 (Low-SUE -Low-DOLVOL portfolio) and longs N6 (High-SUE - Low-DOLVOL portfolio) makes over 2% monthly expected return. The return gaps among the leaves show the usefulness of splitting the cross section via the

3.3 P-Tree叶基础组合

资产聚类：P-Tree生成的叶就是叶基础组合。不同于CART为纯预测输出标量，P-Tree的一个叶代表时间不变树结构下的一条组合收益时间序列。这些组合由高维特征管理，包含非线性和交互。表1A总结图4第一棵树的叶组合，包括节点编号、叶内股票数中位数、平均收益、CAPM的alpha、beta及时间序列回归R^2，还有相对FF5、Q5、RP-PCA与IPCA五因子模型的alpha（Fama和French, 2015；Hou et al., 2015；Kelly et al., 2019；Lettau和Pelger, 2020b）。

两个较大叶为N11和N15，每月股票数中位数分别为900和超过1700。其CAPM R^2超过90%，beta接近1，表明收益与市场收益高度相关。作者指出，对这两个叶继续切分未带来更高夏普比率，而其他叶在MVE框架下有更大的投资改进。例如，N21为低SUE、高DOLVOL、低ME、高ZEROTRADE组，平均收益-1.14%、CAPM alpha为-1.83%；N27为高SUE、低DOLVOL、高BM_IA、高ROA组，平均收益2.46%、CAPM alpha为1.97%。原文称，每月再平衡、买入N27并卖空N21的简单策略，可获得约3.5%的平均收益。

这些叶组合通过61项特征上的非线性交互切分生成，预计难以被流行线性因子模型定价。十个叶中九个具有显著CAPM alpha。原文称，在10%水平上，FF5、Q5、RP-PCA5和IPCA5的显著非零alpha数量分别为10、9、9和7。

P-Tree以夏普比率最大化为目标进行聚类，因此预计这些测试资产相对不足以解释它们的基准模型，具有经济上较大、统计上显著的alpha。由于基础组合彼此不同，我们预计它们张成的前沿无法被基准因子张成，可能实现更高的均值—方差有效性。

asymmetric interaction of characteristics.

3.3 P-Tree Leaf Basis Portfolios

Asset clustering. P-Tree generates leaves, that is, leaf basis portfolios. Unlike the scalar output in CART for pure prediction, a leaf of P-Tree represents a time series of portfolio returns from a time-invariant P-Tree structure. These leaf basis portfolios are nonlinear, interactive, and high-dimensional characteristics-managed portfolios. Table 1, panel A, summarizes leaf basis portfolios generated by the first P-Tree in Figure 4: the index of nodes, median number of stocks in the leaf basis portfolio, average returns, CAPM α (%), β, time series regression R^2, and alphas with respect to FF5, Q5, RP-PCA, and IPCA five-factor models (Fama and French, 2015; Hou et al., 2015; Kelly et al., 2019; Lettau and Pelger, 2020b).

We observe two large leaves: N11 containing 900 stocks and N15 containing over 1,700 stocks by monthly median. Their CAPM R^2 are over 90%, and the β are close to one, meaning their return time series are highly correlated with the market returns. No further splitting results in higher Sharpe ratios on these two leaves, yet other leaves offer higher investment improvement under the MVE framework. For instance, N21 represents the low-SUE, high-DOLVOL, low-ME, and high-ZEROTRADE leaf, which has -1.14% average return and -1.83% CAPM α, and N27 is the high-SUE, low-DOLVOL, high-BM_IA, and high-ROA leaf, which has 2.46% average returns and 1.97% CAPM α. By employing a simple monthly rebalanced strategy of buying N27 and selling short N21, one can expect an average return of 3.5%.

These leaf basis portfolios are generated via interactive and nonlinear splits on 61 characteristics. We expect them to be very hard to price by the prevalent linear factor models. Nine leaves have significant CAPM alphas among the 10 leaf basis portfolios. At a confidence level of 10%, the number of statistically significant non-zero alphas for FF5, Q5, RP-PCA 5, and IPCA 5 factor models are 10, 9, 9, and 7, respectively.

P-Tree is a goal-oriented clustering algorithm that maximizes the Sharpe ratio. Therefore, we anticipate these test assets having economically large and statistically significant alphas against those "inadequate" benchmark factor models. With such distinct basis portfolios, we expect the efficient frontier they span cannot be spanned

表3B、C给出前后20年的子样本分析。虽然重新拟合使树结构和叶身份发生变化，但仍发现相似模式，例如基准因子模型仍难以为这些叶组合定价。

表1：叶基础组合评价

本表报告第一棵P-Tree生成的十个叶基础组合。A面板对应图4，B、C分别对应图A.1的(a)、(b)。报告股票数中位数（叶规模）、平均收益、CAPM alpha（%）、beta、R^2，以及相对FF5、Q5、RP-PCA、IPCA五因子模型的alpha（%）。*、**、***分别代表10%、5%、1%的显著性水平。

表头中译：ID=节点编号；# Median=月度股票数中位数；AVG=平均收益；STD=标准差；alpha下标为比较模型；beta_CAPM=市场因子载荷；R^2_CAPM=CAPM解释度。

A：1981—2020年40年；B：1981—2000年前20年；C：2001—2020年后20年。完整表格数据及显著性星号保留于随后原文页，避免重排改变数值与列对应关系。

by benchmark factor models, possibly achieving higher mean-variance efficiency. Panels B and C of Table 3 provide subsample analysis for the first and the latter 20 years. Although the tree structure and leaf identity change because of refitting, we still find similar patterns. For example, pricing these portfolios based on leaf factors remains difficult using benchmark factor models.

Table 1: **Evaluation for Leaf Basis Portfolios**

This table reports the performance of 10 leaf basis portfolios generated by the first P-Tree. Panel A corresponds to Figure 4, whereas Panels B and C correspond respectively to (a) and (b) in Figure A.1. We report the median number of stocks (leaf size), average returns, CAPM α (%), β, R^2, and the alphas (%) with respect to FF5, Q5, RP-PCA, and IPCA five-factor models. *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

ID	# Median	AVG	STD	α_{CAPM}	β_{CAPM}	R^2_{CAPM}	α_{FF5}	α_{Q5}	α_{RP5}	α_{IP5}
					Panel A: 40 Years (1981-2020)					
N8	134	-0.40	5.76	-0.97***	0.84	0.42	-0.69***	-0.66***	-1.51***	-0.58*
N9	210	-0.92***	4.14	-1.35***	0.63	0.47	-1.02***	-1.25***	-1.84***	-0.36*
N20	179	-0.34	8.89	-1.34***	1.45	0.53	-0.36	-0.49*	-1.55***	-1.78***
N21	227	-1.14***	5.90	-1.83***	1.00	0.58	-1.06***	-1.56***	-2.34***	-1.19***
N11	900	0.36*	4.81	-0.35***	1.04	0.93	-0.24***	-0.15**	-0.29***	0.16
N12	210	0.94***	4.63	0.42**	0.76	0.54	0.64***	0.48***	-0.22	1.17***
N26	157	1.77***	6.04	1.16***	0.88	0.43	1.74***	1.37***	0.40**	1.40***
N27	112	2.46***	4.55	1.97***	0.71	0.49	2.07***	2.00***	1.18***	2.18***
N14	658	1.03***	6.88	0.21	1.18	0.59	1.15***	0.64***	-0.33**	0.06
N15	1768	0.81***	4.41	0.14***	0.98	0.98	0.08**	0.03	-0.26***	0.29
					Panel B: 20 Years (1981-2000)					
N16	97	0.25	5.99	-0.34	0.81	0.36	0.39	-0.08	-0.27	0.22
N17	256	-0.74***	4.40	-1.16***	0.57	0.33	-0.95*	-1.17***	-1.56***	-0.34
N18	95	1.72***	5.72	1.17***	0.75	0.34	2.07***	1.27***	0.84**	1.56**
N19	48	2.91***	4.97	2.44***	0.64	0.33	3.75***	2.50***	2.36***	2.26***
N5	3443	0.76***	4.39	0.03*	0.99	1.00	-0.03	-0.04***	-0.49***	0.41
N24	160	-3.53***	7.75	-4.36***	1.13	0.42	-1.74***	-3.29***	-3.58***	-2.54**
N25	250	-2.82***	6.85	-3.58***	1.04	0.45	-2.17***	-2.54***	-1.89***	-0.45
N13	286	-1.75***	9.94	-2.86***	1.52	0.46	-0.91	-1.19***	-1.29***	0.74
N14	217	-1.21*	9.60	-2.21***	1.37	0.40	0.00	-0.63	-0.09	1.61
N15	116	1.23*	10.07	0.16	1.46	0.41	2.35***	1.52***	1.39**	2.29
					Panel C: 20 Years (2001-2020)					
N8	153	-0.20	4.44	-0.63**	0.67	0.45	-0.54**	-0.46**	-0.93***	-0.10
N18	58	-0.31	6.82	-0.96***	1.01	0.44	-1.17***	-0.53*	-1.36***	-0.88**
N19	112	-0.90***	4.72	-1.39***	0.76	0.53	-1.41***	-1.31***	-1.89***	-0.93***
N5	1206	0.35	5.23	-0.38***	1.13	0.95	-0.27***	-0.25***	-0.21***	0.37*
N24	171	0.94**	6.47	0.26	1.05	0.53	0.49	0.57**	-0.13	0.59**
N25	104	2.44***	6.80	1.79***	1.02	0.45	2.17***	2.17***	1.16***	1.43***
N52	47	1.68***	5.28	1.17***	0.79	0.46	0.90***	1.45***	0.52**	1.56***
N53	76	3.53***	5.27	2.97***	0.86	0.54	2.68***	3.10***	2.24***	3.19***
N27	76	1.79***	4.05	1.36***	0.66	0.53	1.50***	1.37***	0.92***	1.6***
N7	1896	0.76***	4.27	0.16***	0.94	0.98	0.09**	0.09**	-0.05	0.63***

图6：使P-Tree测试资产更加多样化

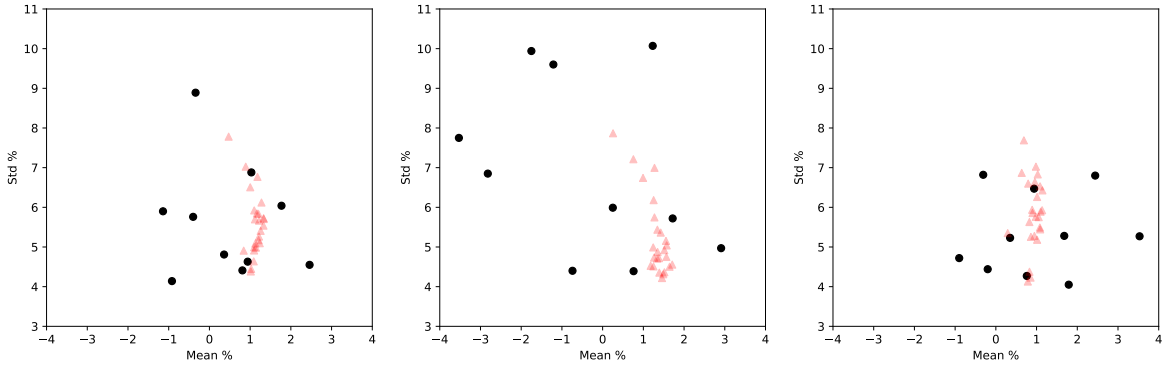
本图报告表1中P-Tree测试资产与5×5 ME-BM组合的表现。10个黑圆点表示P-Tree测试资产，25个浅红三角表示ME-BM组合。A面板为组合收益均值与标准差散点图，均以百分比计；B面板为CAPM alpha与beta散点图，alpha以百分比计。A1、B1对应1981—2020年；A2、B2对应1981—2000年；A3、B3对应2001—2020年。Mean=均值，Std=标准差。

多样化的测试资产。图6A描绘P-Tree测试资产的表现统计量。A1用黑圆点显示1981—2020年40年样本中第一棵P-Tree叶基础组合的均值—标准差散点图。预期收益介于-2%与3%之间，标准差介于4%与9%之间。这些组合的预期收益和风险差异较大，表明它们在MVE框架下具有多样性。子样本分析A2、A3中，组合之间也存在明显差异；较早样本更加分散，较晚样本更集中。相比之下，浅红三角所示的5×5 ME-BM组合聚集在相近的平均收益水平，其收益远小于P-Tree测试资产。

Figure 6: **Diversifying P-Tree Test Assets**

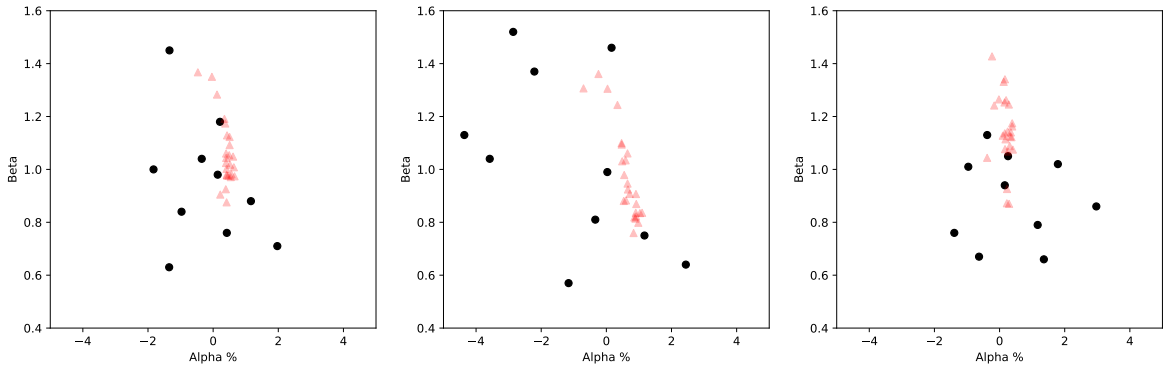
This figure reports the performance of P-Tree test assets (presented in Table 1) and the 5×5 ME-BM portfolios. The 10 black circles represent P-Tree test assets, and the 25 light-red triangles represent ME-BM portfolios. Panel A shows the scatter plots of the mean and standard deviation for portfolio returns in percentage. Panel B shows the scatter plots of CAPM alpha and beta, with alphas in percentage.

Panel A: P-Tree Test Asset: Mean v.s. Standard Deviation



(A1) 1981-2020 (A2) 1981-2000 (A3) 2001-2020

Panel B: P-Tree Test Asset: CAPM Alpha-Beta



(B1) 1981-2020 (B2) 1981-2000 (B3) 2001-2020

Diversified test assets. Figure 6, Panel A, depicts the performance statistics of P-Tree test assets. Subfigure A1 shows the mean-standard-deviation scatter plot for leaf basis portfolios of the first P-Tree on the 40-year sample from 1981 to 2020 in black circles. The expected returns are in the range of -2% to 3%, and the standard deviations are in the range of 4% to 9%. We observe large variations in these portfolios' expected returns and risks, which means they are diversified under the MVE framework. Significant variations exist among the portfolios in Panels A2 and A3 for subsample analysis, with the earlier sample showing more diversification and the latter sample being more concentrated. By contrast, the 5×5 ME-BM portfolios in light-red triangles cluster around the same level of average returns, which are much smaller than those of P-Tree

这一发现表明，在MVE框架下，P-Tree测试资产比ME-BM组合更加多样化。

图6B展示CAPM隐含的alpha—beta图。在B1的40年样本中，P-Tree测试资产的CAPM beta分布在1附近，范围为0.5—1.4；CAPM alpha分布在0附近，范围为-2%—2%。这些基础组合的CAPM alpha与beta存在很大差异。前20年的alpha和beta范围较宽，后20年的差异较小。5×5 ME-BM组合比P-Tree测试资产更容易由CAPM解释，因为它们紧密围绕alpha为0的竖线，各图中的分散程度均更小。总之，这为P-Tree测试资产相较传统资产更具多样性提供了积极证据，即使采用CAPM定价也是如此。

4 通过提升P-Tree生成多个因子与测试资产

4.1 扩展有效前沿

我们研究由P-Tree及提升P-Tree生成的测试资产所张成的有效前沿。许多P-Tree测试资产相对于多个知名基准因子模型具有显著alpha，说明这些模型不能有效张成该前沿。提升技术增强了生成更有效测试资产的能力，使有效前沿进一步扩展，即“在P-Tree上生长有效前沿”。

尚未被张成的有效前沿。图7以从蓝到红的渐变色展示P-Tree及提升P-Tree的MVE前沿，基准因子模型用黑色、基准测试资产用紫色表示。值得注意的是，第一棵树的测试资产生成的P-Tree1前沿，已经优于所有基准因子模型和测试资产所张成的前沿。基准测试资产中，285个双变量排序组合形成的前沿非常接近P-Tree1；但P-Tree1仅包含10个组合，远少于285个。因子模型中，IPCA五因子模型是一个有力的候选者，但效率仍低于P-Tree1前沿。总体上，这10个叶基础组合是用于张成有效前沿的有潜力的测试资产。

test assets. This finding indicates that P-Tree test assets are more diversified under the MVE framework than the ME-BM portfolios.

Panel B of Figure 6 shows the CAPM model-implied alpha-beta plots. In subfigure B1's 40-year sample, we find CAPM βs of P-Tree test assets are scattered around one in the range of 0.5 to 1.4. CAPM alphas are around zero, ranging from -2% to 2%. These basis portfolios have a lot of variations in CAPM alphas and βs. In the first 20 years in B1, alpha and beta ranges were larger, whereas the variation was smaller in the second 20 years in B2. The 5×5 ME-BM portfolios are more easily explained for the CAPM model than P-Tree test assets because they align closely with the vertical line of zero alpha with more negligible diversification in all figures. In summary, this is highly positive evidence for P-Tree test assets' better-diversified patterns than conventional ones, even when priced by CAPM.

4 Boosted P-Trees for Multiple Factors and Test Assets

4.1 Growing the Efficient Frontier

Our investigation focuses on the efficient frontiers spanned by test assets generated using (boosted) P-Tree. Many P-Tree test assets have significant alphas over multiple well-known benchmark factor models, indicating those factor models cannot effectively span the efficient frontier. The boosting technique enhances our ability to generate more effective test assets that expand the efficient frontier better, that is, "growing the efficient frontiers on P-Trees."

Unspanned efficient frontier. Figure 7 shows the MVE frontiers of P-Trees and boosted P-Trees in gradient color from blue to red. Benchmark factor models are printed in black, and benchmark test assets in purple. Notably, the P-Tree1 frontier (generated by the first P-Tree test assets) is already more efficient than any frontier spanned by the benchmark factor models and test assets. Among the benchmark test assets, 285 bivariate-sorted portfolios span a frontier very close to our P-Tree1 frontier. However, P-Tree1 contains only 10 portfolios, much smaller than 285. Among the factor models, the IPCA five-factor model is a strong candidate but still less efficient than the P-Tree1 frontier. Overall, these 10 leaf basis portfolios consist of promising test assets

在MVE框架下，它们张成的前沿比基准更加有效。换言之，所列基准因子模型与测试资产不足以张成真正的有效前沿。

提升有效前沿。在第一棵P-Tree之外，提升P-Tree将因子数从1增至20，从而提高组合效率。图7中，张成的前沿朝均值—方差图的左上方移动。我们的方法在20棵P-Tree的整个序列中持续推进有效前沿。将200项资产直接合并为一个MVE组合，会遇到维度灾难带来的困难。

图7：利用P-Tree刻画有效前沿

本图展示第一棵P-Tree、依次生成的提升P-Tree因子模型、基准因子模型和基准测试资产的均值—方差有效前沿。前沿上的圆点表示切点组合。均值与标准差均采用月度尺度。

图例中译：Mean（%）=均值（%）；Std（%）=标准差（%）；Uni-Sort=单变量排序组合；Bi-Sort=双变量排序组合；ME/BM=市值—账面市值比排序组合；Ind49=49个行业组合。1—20表示依次加入P-Tree因子后的前沿；FF5、Q5、RP5、IP5为对应基准五因子模型。

to span the efficient frontier, which is more efficient than the benchmarks under the MVE framework. In other words, the listed benchmark factor models and test assets are insufficient to span the real efficient frontier.

Boosting efficient frontier. Beyond the first P-Tree, boosted P-Trees improve portfolio efficiency from one factor to 20 factors. In Figure 7, the spanned frontiers move toward the top left corner of the mean-variance diagram. Our approach consistently advances the efficient frontier throughout the 20 P-Trees. Combining 200 assets in one MVE

Figure 7: **Characterizing the Efficient Frontier with P-Trees**

This figure shows the mean-variance efficient frontiers of the first P-Tree, the sequence of boosted P-Tree factor models, benchmark factor models, and benchmark test assets. The dots on the frontiers represent the tangency portfolios. The mean and standard deviation are on a monthly scale.

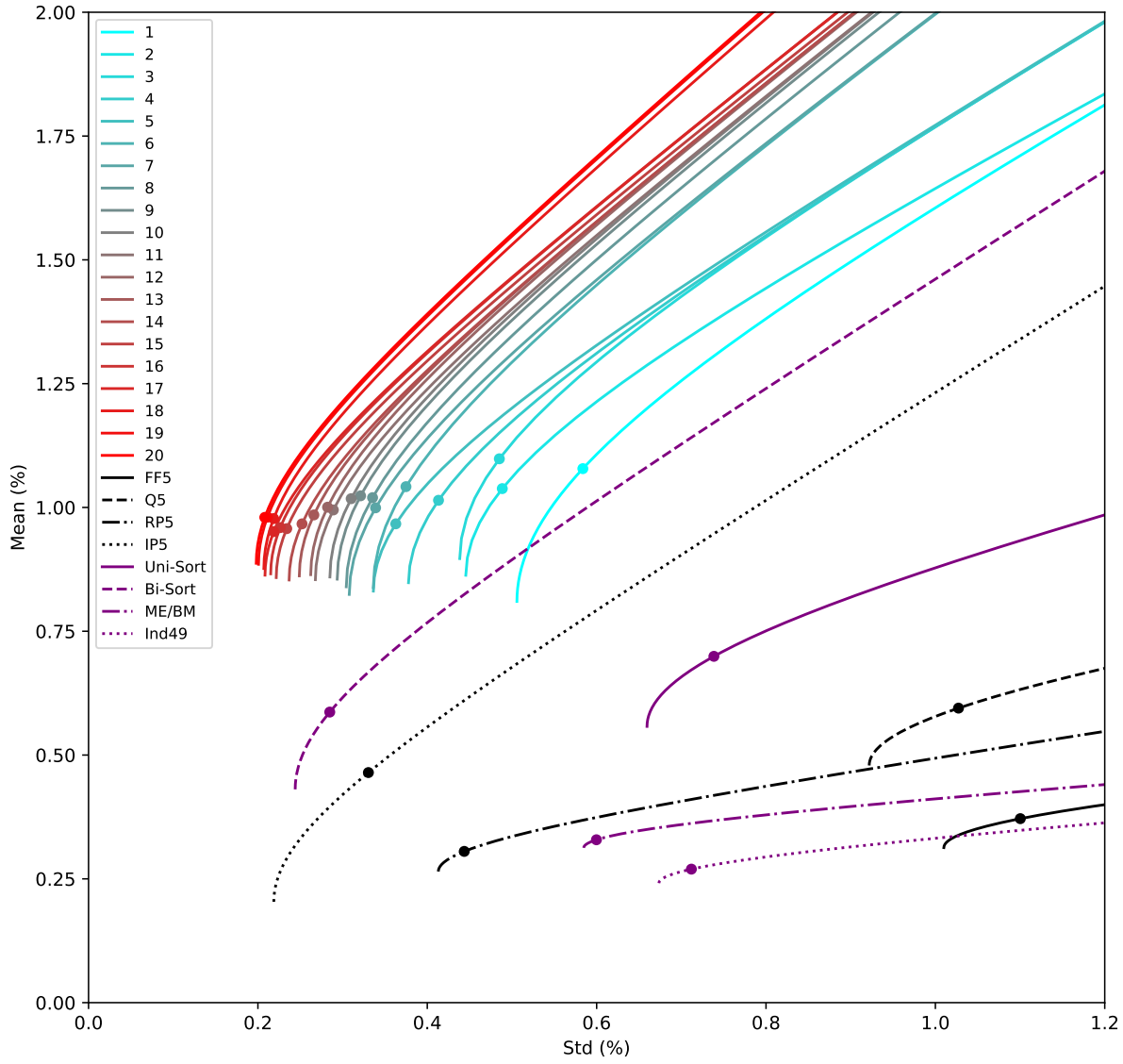


表2：检验提升P-Tree带来的扩展

本表报告顺序生成的各提升P-Tree因子的表现，包括夏普比率、CAPM检验、FF5检验、扩展因子检验及Barillas和Shanken（2017）检验。“Single”列为单个因子的夏普比率；“Cumu.”列为从第一个到当前因子共同张成的MVE组合的累计夏普比率。CAPM与FF5张成回归检验报告alpha（%）和t统计量。扩展因子检验将各P-Tree因子对此前已生成的提升P-Tree因子回归，报告alpha（%）、t统计量和R^2。表A.3提供子样本结果。BS Test=BS检验；p-value=p值。完整数值见随后原文表格。

不同于Ait-Sahalia和Xiu（2017）利用外生行业分类构造块对角结构，我们的提升P-Tree允许收益协方差呈现内生块对角结构。我们将每棵P-Tree的10个叶基础组合合并为一个P-Tree因子，再估计这些提升P-Tree因子的切点组合。这样便能在MVE框架下生成多个因子，并将多组测试资产合并到同一个前沿中。

从一棵P-Tree增加至20棵，会产生更加有效的前沿，使投资者在承担相同风险的情况下获得更高预期收益。P-Tree1与P-Tree10的前沿差异很大；P-Tree10之后，前沿仍继续抬升，但每次提升所带来的均值—方差效率改善更小。

Table 2: **Testing the Boosted P-Tree Growth**

This table presents the performance for each boosted P-Tree factor generated sequentially, including the Sharpe ratio, CAPM test, FF5 test, expanding factors test, and the Barillas and Shanken (2017) test. The "Single" column shows the Sharpe ratio of the single factor, and the "Cumu." (cumulative) column displays the Sharpe ratio of the MVE spanned by multiple factors from the first to the current P-Tree factors. The columns reporting on CAPM and FF5 spanning regression tests show the α (%) and t-statistic. For the expanding factor tests, we regress each P-Tree factor on all the boosted P-Tree factors generated and report the α (%), t-statistic, and R^2. Table A.3 shows the results for subsample analysis.

	Sharpe Ratio		CAPM Test		FF5 Test		Expanding Factors Test			BS Test
	Single	Cumu.	α (%)	t-stat	α (%)	t-stat	α (%)	t-stat	R^2	p-value
1	6.37	6.37	1.39	39.63	1.37	37.48	-	-	-	-
2	3.20	7.35	0.52	19.74	0.48	18.01	0.62	9.55	0.01	0.00
3	1.18	7.80	0.34	6.59	0.18	3.82	-0.86	-5.43	0.26	0.00
4	2.06	8.46	0.44	12.46	0.38	10.77	0.69	7.62	0.16	0.00
5	1.99	9.18	0.48	11.96	0.41	10.41	0.84	5.61	0.21	0.00
6	1.01	9.57	0.18	5.38	0.08	2.88	-0.50	-5.47	0.45	0.00
7	1.42	10.11	0.28	8.27	0.22	6.62	0.63	7.03	0.36	0.00
8	1.32	10.40	0.28	7.56	0.20	5.47	-0.50	-5.33	0.41	0.00
9	1.83	10.88	0.53	10.91	0.43	9.05	0.85	6.88	0.34	0.00
10	1.48	11.20	0.44	8.64	0.30	6.45	-0.68	-5.49	0.46	0.00
11	1.78	11.72	0.38	10.65	0.31	8.85	0.72	6.70	0.30	0.00
12	1.02	12.06	0.20	5.44	0.10	3.15	-0.55	-5.76	0.55	0.00
13	1.37	12.57	0.29	7.85	0.22	6.21	0.76	6.01	0.33	0.00
14	1.37	13.01	0.48	7.90	0.32	5.90	-0.91	-5.68	0.60	0.00
15	1.37	13.81	0.31	7.89	0.21	5.59	0.97	7.11	0.48	0.00
16	1.24	14.28	0.28	7.02	0.17	4.49	-0.74	-6.24	0.54	0.00
17	1.54	14.60	0.46	9.04	0.40	7.70	-0.89	-4.80	0.40	0.00
18	1.64	14.92	0.32	9.69	0.27	8.12	-0.65	-5.42	0.34	0.00
19	1.48	15.43	0.43	8.63	0.36	7.46	1.18	5.86	0.34	0.00
20	1.35	15.63	0.33	7.72	0.23	6.47	-0.59	-4.19	0.44	0.00

portfolio is challenging because of the curse of dimensionality. Unlike Ait-Sahalia and Xiu (2017), who use exogenous industry classifications to construct block-diagonal patterns, our boosted P-Trees entertain an endogenous block-diagonal structure in the return covariance. We combine the 10 leaf basis portfolios of each P-Tree into a single P-Tree factor and then estimate the tangency portfolio of the boosted P-Tree factors. As such, we can generate multiple factors and combine multiple sets of test assets in one frontier under the MVE framework.

Moving from a single P-Tree to 20 P-Trees results in more efficient frontiers, allowing investors to achieve higher expected returns while bearing the same level of risk. Although a significant difference exists between the frontiers of P-Tree1 and P-Tree10, the frontier lines increase after P-Tree10, indicating a more negligible improvement in mean-variance efficiency with each subsequent boosting.

我们采用资产定价检验展示提升P-Tree的增量效率。首先，报告每个提升P-Tree因子及多因子模型的夏普比率。其次，用CAPM和FF5评价每个提升因子，检查其alpha是否显著。由表1可知，P-Tree测试资产难以定价，因此预期P-Tree因子能够通过CAPM和FF5检验。第三，进行扩展因子检验，将每个P-Tree因子对此前生成的所有提升P-Tree因子回归。如果先前因子无法张成新增因子，则有效前沿的效率得到提高。

检验结果见表2。所有单个提升因子的夏普比率都高于1，包括第20个因子。与此同时，多因子累计夏普比率从6单调增加至15，意味着提升P-Tree测试资产为已有测试资产增加了新的定价信息。此外，全部CAPM和FF5 alpha均为正且高度显著，说明CAPM或FF5无法解释这些提升P-Tree因子。扩展因子检验中，我们发现显著的alpha及t统计量，以及中等水平的时间序列R^2。应用嵌套资产定价检验（Barillas和Shanken，2017）后，我们未发现证据表明已有因子模型足以解释提升P-Tree检验中的新增资产。[15]

总之，表2表明提升P-Tree生成了无法被解释的alpha，其叶组合因而能更好地张成有效前沿。相比之下，通常用于检验资产定价模型的单变量或双变量排序测试资产，未能完全覆盖有效前沿。

4.2 通过提升P-Tree生成测试资产

首先，我们展示叶基础组合的定价表现，说明它们是具有多样性的测试资产，并对FF5模型构成挑战。随后，使用提升P-Tree生成多组基础组合。在实证中，我们生长至20棵P-Tree，共产生200个叶基础组合。

[15] 第一行的“-”表示扩展因子检验和BS检验不适用于单因子情形。

We employ asset pricing tests to demonstrate the incremental efficiency of boosted P-Trees. First, we show the Sharpe ratios of each boosted P-Tree factor and the multi-factor model. Second, we evaluate each boosted factor with CAPM and FF5 by checking for a significant alpha. As we know from Table 1, P-Tree test assets are hard to price, and we expect the P-Tree factors to pass the CAPM test and FF5 test. Third, we regress each P-Tree factor on all the boosted P-Tree factors generated for the expanding factor tests. The efficiency of the frontier increases if the previous P-Tree factors do not span the additional P-Tree factor.

The test results are presented in Table 2. The Sharpe ratios of each boosted factor are above 1, even for the 20-th factor. Meanwhile, the cumulative multi-factor Sharpe ratio increases monotonically from 6 to 15, which means that the test assets of the boosted P-Tree add incremental pricing information to the existing test assets. Furthermore, all CAPM and FF5 alphas are positive and highly significant, indicating that all the boosted P-Tree factors cannot be explained by CAPM or FF5. In the expanding factor test, we find the significant α and t-statistic and the medium values for the time series R^2. After applying the nested asset pricing test (Barillas and Shanken, 2017), we do not find any evidence that supports a sufficient factor model to explain the assets in the boosted P-Tree test.[15] In summary, Table 2 shows that boosted P-Trees generate unexplained alphas, and their leaf portfolios thus better span the efficient frontier. By contrast, conventional univariate- or bivariate-sorted test assets used for testing asset pricing models do not fully cover the efficient frontier.

4.2 Generating Test Assets via Boosted P-Trees

First, we show the pricing performance of leaf basis portfolios, implying they serve as diversified test assets and are challenging against the FF5 model. Then, we utilize boosted P-Trees that generate multiple sets of basis portfolios. Empirically, we grow up to 20 P-Trees, yielding 200 leaf basis portfolios.

[15]The first row contains "-" signs, which indicate that the expanding factor and BS tests are not applicable for the single factor case.

P-Tree测试资产。表3A列出组合数量、GRS检验统计量及其p值、PY检验p值[16]、平均绝对alpha、alpha均方根、平均时间序列回归R^2，以及在10%、5%、1%水平上具有显著alpha的资产占比。行名P-Tree1代表第一棵树的10个叶基础组合；P-Tree1-5合并前五棵树的全部叶基础组合，P-Tree6-10包含第6至第10棵树。P-Tree11-15、P-Tree16-20与P-Tree1-20以同样方式定义。

对于所有P-Tree测试资产设置，GRS和PY检验均拒绝测试资产alpha联合为零的原假设，意味着FF5很难解释P-Tree测试资产的预期收益。最后三列也显示许多资产具有统计显著的alpha。作为对照，我们纳入文献中常用的四组测试资产：150个10×1单变量排序组合、285个双变量排序组合、5×5 ME-BM组合及49个行业组合，均可从Ken French网站下载。P-Tree测试资产的GRS统计量与alpha均大于这些基准测试资产。

此外，如预期所示，从第一棵树到后续提升树，alpha逐渐下降，表明接近有效前沿极限时的边际改善递减。第一棵树的测试资产比后续树具有更大的GRS统计量、更高的平均alpha和更高的显著alpha比例。从P-Tree1-5到P-Tree6-10、P-Tree11-15及P-Tree16-20，GRS统计量和alpha也呈现明显下降。第一棵P-Tree生成信息量最大的测试资产，而后续提升树的信息起补充作用且逐渐减少。

尽管alpha随提升下降，P-Tree16-20生成的测试资产仍无法由FF5解释：GRS统计量为4.21，平均绝对alpha为0.31%，52%的测试资产在10%水平上具有显著alpha。它们依然比单变量和双变量排序组合更难定价。总体而言，20棵P-Tree基于高维特征创建的这200个测试资产，对FF5模型提出了挑战。

[16] Pesaran和Yamagata（2023）的检验适用于被检验资产数量大于时间期数的情形，而Gibbons等（1989）的检验无法处理该情形。

P-Tree test assets. Table 3, Panel A, lists the number of portfolios, the GRS test statistic and its p-value, the p-value of PY test,[16] the average absolute α, the root mean squared α, the average time series regression R^2, and the percentage of significant alphas with respect to 10%, 5%, and 1% confidence levels. In the rows, P-Tree1 represents a set of test assets comprising 10 leaf basis portfolios in the first P-Tree. P-Tree1-5 combines all the leaf basis portfolios generated by the first five P-Trees, and P-Tree6-10 contains the 6-th to the 10-th P-Trees. Similarly, we define P-Tree11-15, P-Tree16-20, and P-Tree1-20.

We find the GRS test and PY test always reject the null hypothesis that the alphas of test assets are jointly zero for all specifications of P-Tree test assets, which means the expected returns of P-Tree test assets can hardly be explained by the FF5 model. Furthermore, the last three columns show many assets have statistically significant alphas. For comparison, we include four sets of test assets that are commonly used in literature: 150 10×1 univariate-sorted portfolios, 285 bivariate-sorted portfolios, 5×5 ME-BM portfolios, and 49 industry portfolios, all of which can be downloaded from Ken French's website. P-Tree test assets have larger GRS test statistics and alphas than these benchmark test assets.

Furthermore, we observe the decline of alphas from the first P-Tree to the subsequently boosted P-Trees as anticipated, indicating diminishing marginal convergence to the limit of the efficient frontier. The test assets of the first P-Tree have larger GRS statistics, higher average alphas, and higher proportions of significant alphas than the test assets generated by the follow-up P-Trees. In addition, we see an evident decline in GRS statistics and alphas from P-Tree1-5 to P-Tree6-10, P-Tree11-15, and P-Tree16-20. P-Tree generates the most informative test assets in the first P-Tree, whereas the incremental information in the boosted P-Trees is complementary and declining. Although the alphas decline along boosting, the test assets generated by P-Tree16-20 cannot be explained by FF5, with a 4.21 GRS test statistic, 0.31% mean absolute alpha, and 52% test assets having significant alphas at the 10% level. The test assets of P-Tree16-20 are still harder to price than univariate- and bivariate-sorted portfolios.

Overall, these 200 test assets created by 20 P-Trees on high-dimensional charac-

[16] Pesaran and Yamagata (2023) adapts to cases where the number of assets being tested is larger than the number of periods, which Gibbons et al. (1989) cannot address.

表3：测试资产比较

本表展示P-Tree测试资产与其他比较资产的表现统计量。P-Tree1为第一棵树中的10个组合；P-Tree1-5、P-Tree6-10、P-Tree11-15、P-Tree16-20及P-Tree1-20，分别合并对应树序号的叶基础组合。“Uni-Sort”含150个单变量排序组合，“Bi-Sort”含285个双变量排序组合，ME/BE含25个组合，“Ind49”含49个行业组合。

报告的统计量包括测试资产数N、GRS检验统计量与p值（Gibbons等，1989）、PY检验p值（Pesaran和Yamagata，2023）、平均绝对alpha（%）、alpha均方根（%）、组合对Fama-French五因子回归的平均R^2（%），以及在10%、5%、1%水平上具有无法解释的显著alpha的资产比例。三个面板报告在不同样本期重新训练模型的结果。

A面板：1981—2020年40年；B面板：1981—2000年20年；C面板：2001—2020年20年。表头p-GRS与p-PY分别为两类检验的p值；最后三列为三个显著性水平下的资产占比。原表数值完整保留在随后原文页。

Table 3: **Comparing Test Assets**

The table displays performance statistics of P-Tree test assets and others for comparison, including P-Tree1, which consists of 10 portfolios in the first P-Tree. P-Tree1-5, P-Tree6-10, P-Tree11-15, P-Tree16-20, and P-Tree1-20 are sets of test assets that combine leaf basis portfolios generated by the first 5, 6-10, 11-15, 16-20, and 1-20 P-Trees, respectively. "Uni-Sort" has 150 univariate-sorted portfolios, "Bi-Sort" has 285 bivariate-sorted portfolios, ME/BE has 25 portfolios, and "Ind49" includes 49 industry portfolios. The reported statistics include the number of test assets, GRS test statistics and p-values (Gibbons et al., 1989), p-values of PY test (Pesaran and Yamagata, 2023), average absolute α (%), root mean squared α (%), average R^2 (%) of regressing the portfolios on Fama-French five factors, and the proportion of test assets with unexplained significant alphas under 10%, 5%, and 1%. The three panels report on the re-trained models for different sample periods.

	N	GRS	p-GRS	p-PY	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$	$\overline{R^2}$	$\%\alpha_{10\%}$	$\%\alpha_{5\%}$	$\%\alpha_{1\%}$
Panel A: 40 Years (1981-2020)										
P-Tree1	10	141.27	0.00	0.00	0.92	1.11	75	100	90	80
P-Tree1-5	50	60.32	0.00	0.00	0.44	0.61	80	70	62	44
P-Tree6-10	50	4.60	0.00	0.00	0.29	0.37	79	56	50	34
P-Tree11-15	50	4.74	0.00	0.00	0.20	0.26	80	38	36	24
P-Tree16-20	50	4.21	0.00	0.00	0.31	0.42	77	52	44	30
P-Tree1-20	200	41.31	0.00	0.00	0.31	0.43	79	54	48	33
Uni-Sort	150	1.62	0.00	0.00	0.10	0.14	88	25	18	7
Bi-Sort	285	2.50	0.00	0.00	0.12	0.17	89	30	23	15
ME-BM	25	5.01	0.00	0.00	0.12	0.16	92	36	28	20
Ind49	49	1.99	0.00	0.00	0.28	0.35	60	39	31	18
Panel B: 20 Years (1981-2000)										
P-Tree1	10	84.36	0.00	0.00	1.58	1.95	70	90	80	80
P-Tree1-5	50	50.84	0.00	0.00	0.79	1.26	76	62	60	52
P-Tree6-10	50	7.27	0.00	0.00	0.58	0.87	75	56	44	38
P-Tree11-15	50	6.39	0.00	0.00	0.55	0.82	76	66	60	42
P-Tree16-20	50	8.42	0.00	0.00	0.52	0.74	76	62	54	50
P-Tree1-20	200	112.90	0.00	0.00	0.61	0.95	76	62	55	46
Uni-Sort	150	1.94	0.00	0.00	0.17	0.23	88	35	27	19
Bi-Sort	285	-	-	0.00	0.21	0.30	89	42	33	21
ME-BM	25	4.75	0.00	0.00	0.21	0.25	91	56	48	32
Ind49	49	2.44	0.00	0.00	0.52	0.61	61	59	49	27
Panel C: 20 Years (2001-2020)										
P-Tree1	10	56.76	0.00	0.00	1.09	1.35	68	90	90	90
P-Tree1-5	50	30.35	0.00	0.00	0.43	0.68	76	52	38	24
P-Tree6-10	50	5.17	0.00	0.00	0.29	0.37	75	34	28	14
P-Tree11-15	50	2.20	0.00	0.00	0.27	0.35	75	30	22	10
P-Tree16-20	50	2.52	0.00	0.00	0.31	0.40	76	42	28	10
P-Tree1-20	200	83.91	0.00	0.00	0.33	0.47	76	40	29	14
Uni-Sort	150	1.46	0.03	0.94	0.09	0.12	89	12	7	0
Bi-Sort	285	-	-	0.01	0.11	0.15	91	21	15	6
ME-BM	25	2.58	0.00	0.10	0.11	0.14	93	24	8	8
Ind49	49	1.29	0.11	0.36	0.25	0.32	62	18	8	2

这些测试资产比基准测试资产更难定价，为因子模型检验提出更高要求，回应了实证研究中提出的担忧（Lewellen等，2010；Ang等，2020）。

作为稳健性分析，表3B、C报告前后两个20年子样本。在最近20年中，FF5对基准测试资产表现良好；原文称，在10%水平上GRS与PY检验不能拒绝原假设，49个行业组合和单变量排序组合中显著alpha也更少。然而，P-Tree测试资产在GRS和PY检验中始终拒绝原假设，在两个子样本中均持续表现为难以定价。[17]

〔译注：上述“不能拒绝”的概括与表3C并非完全一致，例如Uni-Sort的GRS p值为0.03。此处保留作者论述，具体数值以原表为准。〕

P-Tree聚类的优势。P-Tree有两项优势使这些测试资产具有效率。第一，它是一种面向目标的聚类算法，专为迭代张成有效前沿而设计。每个叶根据相近特征值聚集个股收益，再汇总为组合收益时间序列；目标是最大化所有叶基础组合共同张成的MVE组合的夏普比率。它的经济目标与CART和其他现成机器学习模型大不相同：后者侧重统计意义上的收益预测，而不考虑经济效用，不能直接完成扩展有效前沿的任务。

第二，P-Tree利用高维特征之间复杂、非对称的交互和非线性。常见测试资产是按特征排序的组合，例如单变量和双变量排序，只考虑十分位或五分位分组，最多使用三个由研究者临时选择或凭领域知识选定的特征。相比之下，P-Tree更灵活，允许在统一框架下对更多特征进行一般性的顺序及同时排序，并允许非对称交互。如图4所示，它可以保留继续切分不能提高夏普比率的大叶N15，也允许存在N8这样的小叶。非对称切分使P-Tree在MVE框架下更关注信息丰富资产的精细簇，而较少关注信息不足的资产。

[17] 当测试资产数量超过观测期数时，GRS检验不适用。B、C面板的Bi-Sort即属于这种情况，用“-”表示。

teristics challenge the FF5 model. They are more difficult to price than benchmark test assets, setting a high standard for testing factor models, which respond to the concerns in empirical studies (Lewellen et al., 2010; Ang et al., 2020).

For robustness, Table 3, Panels B and C, report the first and the latter 20-year subsamples. In the recent 20 years, FF5 has performed well in pricing the benchmark test asset; that is, the GRS and PY tests cannot be rejected at the 10% level, and less significant alphas testing on 49 industry portfolios and univariate-sorted portfolios. However, the P-Tree test assets always reject the null hypothesis for GRS and PY tests and are consistently challenging to price in both subsamples.[17]

Advantages of P-Tree clustering. P-Tree has two advantages that make these test assets efficient. First, P-Tree is a goal-oriented clustering algorithm tailored for spanning the efficient frontier iteratively. Each leaf clusters individual asset returns based on similar characteristic values, aggregating as a time series of portfolio returns. The goal is to maximize the collective Sharpe ratio of the MVE portfolio spanned by the leaf basis portfolios. The economic objective of P-Tree is vastly different from the CART and other off-the-shelf ML models, which focus on statistical return prediction without economic utilities, and cannot ever tackle the task of growing efficient frontier.

Second, P-Tree exploits the complex asymmetric interactions and nonlinearities on high-dimensional characteristics. The commonly used test assets are characteristics-sorted portfolios, such as univariate- and bivariate-sorted portfolios, which only consider decile or quintile sorting, along with up to three characteristics chosen ad hoc or based on researchers' domain expertise. In contrast, P-Tree is more flexible, allowing for general sequential and simultaneous sorting with more characteristics and asymmetric interactions in a unified framework. P-Trees enable the inclusion of large leaves, like N15, as shown in Figure 4, where further splitting does not result in a higher Sharpe ratio, as well as small leaves, such as N8. The asymmetric splitting allows P-Trees to focus on the finer clusters of informative assets while paying less attention to non-informative assets under the MVE framework.

[17]The GRS test is not applicable in the presence of more test assets than observations, as is the case for Bi-Sort in Panels B and C, marked with "-".

4.3 P-Tree因子与资产定价表现

前面的分析已说明P-Tree测试资产不同于传统测试资产。接下来，我们用包含P-Tree叶基础组合在内的不同测试资产集合，交叉检验P-Tree因子与各类基准因子模型。采用横截面R^2衡量定价表现，公式见随后原文式（7）。

式（7）的含义为：横截面R^2等于1减去“平均收益预测误差的平方和”与“实际平均收益平方和”之比。其中，资产i的模型预测平均收益为估计因子载荷与估计风险溢价的内积。风险溢价通过横截面回归估计，因子载荷通过时间序列回归估计。横截面R^2表示模型隐含预期收益所解释的资产平均收益比例。

我们使用P-Tree测试资产、传统排序组合和行业组合等检验因子模型。表4的列为因子模型，行为被解释预期收益的测试资产。A面板为40年全样本，B、C为20年子样本。[18]

首先，P-Tree1因子解释了P-Tree1叶基础组合（10项资产）横截面收益变动的98.3%，基准模型表现则低得多。这并不意外，因为第一棵树的因子与测试资产是共同构造的。但FF5、Q5和RP-PCA五因子模型的R^2远小于P-Tree1，说明它们解释高度多样化的P-Tree1测试资产预期收益时存在较大定价误差。例外是IPCA五因子模型，定价P-Tree1测试资产的R^2达到93.4%。

其次，由前五棵树的五个因子组成的P-Tree五因子模型（P-Tree1-5，表示第一至第五因子组成的模型）表现出很强的资产定价能力。

[18] 第一行有两个“–”，表示资产数N=10小于或等于因子数10或20，因而不适合进行该横截面回归。

4.3 P-Tree Factors and Asset Pricing Performance

Using the findings from the previous analysis, we have established that P-Tree test assets are distinct from conventional test assets. Moving forward, we proceed with cross-examining P-Tree factors and various benchmark factor models on different sets of test assets, including P-Tree leaf basis portfolios. We adopt the cross-sectional R^2 to evaluate the asset pricing performance:

$$\text{Cross-sectional } R^2 = 1 - \frac{\sum_{i=1}^{N}\left(\bar{R}_i - \widehat{\bar{R}_i}\right)^2}{\sum_{i=1}^{N}\left(\bar{R}_i\right)^2}, \tag{7}$$

where $\widehat{\bar{R}_i} = \widehat{\boldsymbol{\beta}}^{\intercal}\widetilde{\boldsymbol{\lambda}}$. The risk premium estimation adopts the cross-sectional regression estimates of factors, and factor loadings are estimated from time series regressions. The cross-sectional R^2 represents the faction of assets' average returns explained by the model-implied expected returns. Various test assets, including P-Tree test assets, conventional sorted portfolios, and industry portfolios, are used to test the factor models. In Table 4, we display the factor models in the columns to explain the expected return of test assets in the rows. Panel A provides the asset pricing performances for the 40-year sample, and Panels B and C report for the 20-year subsamples.[18]

First, the P-Tree1 factor (from the first P-Tree) explains 98.3% of the cross-sectional return variation of the P-Tree1 leaf basis portfolios (10 assets), whereas the benchmark factor models provide much lower performance. This finding is not surprising, since the first P-Tree factor is constructed jointly with these test assets. However, the FF5, Q5, and RP-PCA five-factor models have much smaller R^2s than the P-Tree1 factor, indicating large pricing errors in explaining the expected returns of the highly diversified P-Tree1 test assets. One exception is that the IPCA five-factor model performs well in pricing the P-Tree1 test assets, with a 93.4% R^2.

Second, the P-Tree five-factor model (P-Tree1-5, read as model containing from the first to the fifth factors), which includes five P-Tree factors from the first five P-Trees, demonstrates tremendous asset pricing performance. It outperforms the benchmark

[18]The first row contains two "–," which indicate the number of assets $N = 10$ is equal to or smaller than the number of factors (10 or 20). Thus, the cross-sectional regression is not applicable.

在定价前10棵P-Tree的测试资产（P-Tree1-10）以及5×5 ME-BM组合时，该模型优于基准五因子模型。对其他测试资产集合，为保持同为五因子的公平比较，P-Tree1-5的横截面R^2也处于可比水平。其较好的定价表现，在全样本和子样本的A至C面板中保持一致。

表4：资产定价表现——横截面R^2

本表报告P-Tree因子模型和基准因子模型为不同测试资产定价时的横截面R^2（%）。列为因子模型，行为测试资产设置。列中包含1、5、10、20个因子的P-Tree模型，以及FF5、Q5、RP-PCA、IPCA五因子模型；行中包括前1、5、10、20棵树的测试资产，单变量排序组合、双变量排序组合、ME-BM 5×5组合，以及49个行业组合。

A面板：1981—2020年40年；B面板：1981—2000年20年；C面板：2001—2020年20年。P-Tree1F/5F/10F/20F中的F表示因子数。完整表格及数值见随后原文页。

five-factor models in pricing the first 10 P-Trees' (P-Tree1-10) test assets and 5×5 ME-BM portfolios. For other specifications of test assets, P-Tree1-5 shows comparable levels of cross-sectional R^2 among all the five-factor models for a fair comparison. The superior pricing performance remains consistent across Panels A to C for the entire sample and subsample analysis.

Table 4: **Asset Pricing Performance: Cross-Sectional R^2**

This table displays the cross-sectional R^2 (%) of pricing different sets of test assets by the P-Tree factor models and benchmark factor models. The factor models are listed in the columns, and the different specifications of test assets are listed in the rows. Specifically, the table includes 1, 5, 10, and 20 P-Tree factor models and FF5, Q5, RP-PCA, and IPCA five-factor models in the columns. In the rows, the test assets are from the top 1, 5, 10, and 20 P-Trees, univariate-sorted portfolios, bivariate-sorted portfolios, ME-BM 5×5 portfolios, and 49 industry portfolios.

	P-Tree1F	P-Tree5F	P-Tree10F	P-Tree20F	FF5	Q5	RP5	IP5
				Panel A: 40 Years (1981-2020)				
P-Tree1	98.3	99.2	–	–	36.8	33.9	51.1	93.4
P-Tree1-5	65.0	82.6	90.2	95.9	54.3	55.9	56.8	66.1
P-Tree1-10	68.5	74.0	87.9	92.1	66.9	70.0	69.1	73.2
P-Tree1-20	72.1	77.9	84.3	88.9	73.8	76.8	77.8	80.3
Uni-Sort	92.7	97.3	98.5	98.7	97.0	97.9	98.0	97.6
Bi-Sort	88.7	97.2	98.0	98.5	96.1	97.5	97.3	97.5
ME-BM	88.4	98.6	98.8	99.4	96.8	97.4	97.0	97.3
Ind49	82.0	92.9	95.3	97.8	96.1	95.9	95.7	92.8
				Panel B: 20 Years (1981-2000)				
P-Tree1	99.3	99.3	–	–	54.9	59.7	37.6	76.0
P-Tree1-5	49.2	88.5	92.6	96.0	25.8	31.8	41.1	32.2
P-Tree1-10	47.7	59.7	77.8	82.3	37.7	40.5	47.8	28.9
P-Tree1-20	42.8	54.0	70.2	74.5	42.2	45.6	52.5	30.4
Uni-Sort	80.4	96.5	96.9	97.6	94.6	95.9	97.0	87.6
Bi-Sort	66.4	92.1	93.9	94.9	89.1	91.5	92.7	78.8
ME-BM	66.6	94.0	96.2	99.0	93.4	95.3	96.4	88.3
Ind49	60.9	86.5	87.9	95.0	91.0	89.7	92.1	73.5
				Panel C: 20 Year (2001-2020)				
P-Tree1	99.2	99.7	–	–	52.2	43.6	69.7	53.9
P-Tree1-5	71.4	87.6	92.3	94.3	60.0	60.7	64.9	61.3
P-Tree1-10	70.4	78.1	86.4	88.4	66.4	65.7	69.8	65.4
P-Tree1-20	70.5	76.4	79.8	82.8	71.7	69.9	75.4	65.2
Uni-Sort	87.5	96.6	97.6	97.9	97.3	97.2	97.4	95.7
Bi-Sort	93.0	97.2	97.9	98.4	97.9	97.5	98.0	96.9
ME-BM	92.9	95.7	98.6	99.3	97.5	97.5	97.4	96.1
Ind49	75.3	92.6	94.8	97.7	90.2	90.8	92.8	86.0

第三，这些内生生成的P-Tree测试资产更好地扩展MVE前沿，提高了资产定价模型的检验门槛。常用测试资产对因子模型的检验门槛较低，因此基准模型似乎足以对其定价。如表4A所示，列出的基准五因子模型对基准测试资产的横截面R^2均超过90%，B、C子样本中也持续较高。然而，以P-Tree测试资产检验时，它们的R^2大幅下降。P-Tree测试资产利用大量特征间的非线性、非对称交互，基准测试资产最多只涉及两个特征。[19]

总之，P-Tree因子模型能够准确为P-Tree测试资产定价，而基准因子做不到；对基准测试资产，P-Tree模型的表现与基准模型相近或更好。P-Tree测试资产为资产定价模型设置了高于常用排序组合和行业组合的检验标准。

4.4 投资P-Tree因子

P-Tree框架将数千项资产聚为一定数量的叶基础组合，降低组合优化的复杂性。这些测试资产可以直接作为因子，也可以作为构建投资用MVE组合的组成部分。此外，可以将P-Tree投资策略的再平衡频率降低至季度或年度，以减少交易成本。

全样本评价。我们通过将P-Tree因子合并为一个组合——即这些因子的切点组合——来评价投资表现。表5A报告40年样本中的年化夏普比率和月度alpha。从单因子策略到5、10、15、20因子策略，夏普比率由6.37依次增加至9.20、11.19、13.80和15.59。为评估其增量投资信息，我们进一步相对于基准因子模型评价这些策略；模型调整后的alpha均超过0.80%，且统计上高度显著。

与已有研究相比，P-Tree投资策略具有很有竞争力的夏普比率。

[19] 行业组合是一个例外，因为它们并非根据公司特征排序。

Third, these endogenously generated P-Tree test assets better expand the MVE frontier and raise the bar for testing asset pricing models. The commonly used test assets set a low bar for testing factor models, so benchmark factor models seem sufficient to price them. As we see from Panel A of Table 4, the listed benchmark five-factor models have over 90% cross-sectional R^2s on the benchmark test assets, and these numbers are consistently large in Panels B and C for subsample analyses. However, their R^2s decline dramatically when tested against P-Tree test assets. P-Tree test assets use nonlinear and asymmetric interactions among a large set of characteristics, whereas benchmark test assets involve only up to two characteristics.[19]

In summary, P-Tree factor models accurately price P-Tree test assets, while benchmark factors do not, and perform similarly or better than benchmark factor models in pricing benchmark test assets. P-Tree test assets set a high standard for testing asset pricing models, surpassing common sorted and industry portfolios.

4.4 Investing P-Tree Factors

The P-Tree framework clusters thousands of assets into a number of leaf basis portfolios, reducing portfolio optimization complexity. These P-Tree test assets can be used as factors directly or as building blocks toward MVE portfolios for investment. Additionally, the frequency of rebalancing P-Tree investment strategies can be reduced to quarterly or annually to decrease transaction costs.

Full-sample evaluation. We evaluate the investment performance of P-Tree factors by combining them into one —– the tangency portfolio of P-Tree factors. Table 5, Panel A, reports the P-Tree investment strategy's annualized Sharpe ratios and monthly alphas in the 40-year sample. The Sharpe ratio increases from 6.37 to 9.20, 11.19, 13.80, and 15.59 for the one-factor strategy to 5, 10, 15, and 20 factors, respectively. To assess their additional investment information, we further evaluate these P-Tree investment strategies over benchmark factor models. These model-adjusted alphas are all greater than 0.80% and highly statistically significant.

The existing research finds that P-Tree investment strategies have highly com-

[19] Industry portfolios are an exception because they are not sorted based on characteristics.

表5：提升P-Tree的因子投资

本表展示P-Tree生成的初始因子，以及提升P-Tree多因子所构成的均值—方差有效组合的投资表现。报告统计量为年化夏普比率，以及相对于CAPM、FF5、Q5、RP-PCA和IPCA五因子模型的alpha。A面板为1981—2020年40年样本。B1、B2分别为1981—2000年样本内与2001—2020年样本外结果。C1、C2分别为2001—2020年样本内与1981—2000年样本外结果。*、**、***分别表示10%、5%、1%的显著性水平。

表头SR=年化夏普比率；alpha下标为相应比较模型；In-Sample=样本内；Out-of-Sample=样本外。各策略对应前1、5、10、15、20个P-Tree因子的组合。完整数值与显著性星号保留在随后原文页。

Ahn等（2009）基于相关性的聚类所生成的基础组合，表现不如按ME与BM进行双变量排序的组合。此外，Daniel等（2020）通过构建相应的特征有效组合，使特征排序组合的夏普比率接近翻倍。

Table 5: **Factor Investing by Boosted P-Trees**

This table presents the investment performance of the initial factors produced by P-Tree and the mean-variance efficient portfolio of multi-factors generated by boosted P-Trees. The reported statistics include the annualized Sharpe ratio and alphas with respect to CAPM, FF5, Q5, RP-PCA, and IPCA five-factor models. Panel A shows the 40-year sample from 1981 to 2020. Panels B1 and B2 present in-sample and out-of-sample results from 1981 to 2000 and 2001 to 2020, respectively. Panels C1 and C2 display in-sample and out-of-sample results from 2001 to 2020 and 1981 to 2000, respectively. *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

	SR	α_{CAPM}	α_{FF5}	α_{Q5}	α_{RP5}	α_{IP5}
			Panel A: 40 Years (1981-2020)			
P-Tree1	6.37	1.39***	1.30***	1.36***	1.30***	1.12***
P-Tree1-5	9.20	0.96***	0.90***	0.93***	0.89***	0.81***
P-Tree1-10	11.19	1.00***	0.95***	0.98***	0.95***	0.88***
P-Tree1-15	13.80	0.94***	0.89***	0.91***	0.90***	0.86***
P-Tree1-20	15.59	0.96***	0.92***	0.94***	0.92***	0.89***
			Panel B1: 20 Years In-Sample (1981-2000)			
P-Tree1	7.12	1.85***	1.88***	1.71***	1.65***	1.24***
P-Tree1-5	12.71	1.54***	1.59***	1.48***	1.35***	1.27***
P-Tree1-10	19.18	1.49***	1.49***	1.47***	1.39***	1.35***
P-Tree1-15	28.37	1.42***	1.39***	1.39***	1.36***	1.36***
P-Tree1-20	37.85	1.34***	1.34***	1.32***	1.29***	1.29***
			Panel B2: 20 Years Out-of-Sample (2001-2020)			
P-Tree1	3.24	1.34***	1.31***	1.23***	1.01***	0.91***
P-Tree1-5	3.41	1.02***	1.02***	0.95***	0.79***	0.63***
P-Tree1-10	3.21	0.94***	0.91***	0.87***	0.76***	0.57***
P-Tree1-15	3.12	0.89***	0.89***	0.83***	0.71***	0.49***
P-Tree1-20	3.13	0.83***	0.82***	0.77***	0.66***	0.49***
			Panel C1: 20 Years In-Sample (2001-2020)			
P-Tree1	5.82	1.51***	1.46***	1.50***	1.25***	1.36***
P-Tree1-5	9.32	1.32***	1.34***	1.30***	1.20***	1.24***
P-Tree1-10	14.33	1.13***	1.12***	1.11***	1.05***	1.08***
P-Tree1-15	20.45	1.04***	1.05***	1.03***	0.98***	1.00***
P-Tree1-20	26.11	1.05***	1.05***	1.04***	1.01***	1.02***
			Panel C2: 20 Years Out-of-Sample (1981-2000)			
P-Tree1	4.35	1.50***	1.59***	1.35***	1.43***	1.10***
P-Tree1-5	3.88	1.20***	1.24***	0.98***	1.11***	0.75***
P-Tree1-10	4.29	1.03***	1.10***	0.85***	0.90***	0.76***
P-Tree1-15	4.04	0.91***	0.91***	0.75***	0.78***	0.65***
P-Tree1-20	3.91	0.92***	0.91***	0.76***	0.78***	0.61***

petitive Sharpe ratios. The correlation-based clustering in Ahn et al. (2009) generates basis portfolios that underperform bivariate-sorted portfolios on ME and BM. Further-

具体而言，Daniel等（2020）将夏普比率从1.17提高至2.13。而单棵P-Tree在1981—2020年40年中取得6.37的夏普比率，相对于Ahn等（2009）和Daniel等（2020）有明显提高，尽管各研究的实证样本不同。

样本外评价。表5A基于全样本分析，可能引发样本内过拟合的担忧。因此，我们提供两项样本外投资策略实验。参照Kozak等（2020），将样本各分为一半构成训练集和测试集，以过去预测未来、未来预测过去的相互检验展示样本外表现。原文认为，使用未来样本训练模型预测过去，还可以进一步减轻有关前视偏差的担忧。

〔译注：上一句为作者原述。用未来训练、过去测试可以考察跨时期稳定性，但不是遵循实际时间顺序的可交易回测，不能据此认定消除了前视偏差。〕

一方面，过去预测未来的结果见B1、B2。前20年的样本内夏普比率甚至高于A面板的40年结果，样本外夏普比率超过3。尽管表现有所下降，许多经基准模型调整的样本外alpha仍接近1%，且高度显著。另一方面，在未来预测过去的结果中，样本外夏普比率高于3.88，模型调整后的alpha超过0.60%。这些发现为P-Tree测试资产和因子张成MVE前沿提供了较强的样本外证据。

这些样本外投资表现与近期研究报告的结果具有可比性。Cong等（2021）的深度强化学习策略在剔除小盘股后，于1990—2016年获得2.21的夏普比率。Bryzgalova等（2023）基于规模、营业盈利能力和投资排序组合的正则化组合，在2004—2016年取得0.69的样本外月度夏普比率，折合年化2.39。Feng等（2023）的深度神经网络方法在2002—2021年报告2.95—3.00的样本外夏普比率。P-Tree策略的突出之处在于透明、易实现、灵活，能够成为集测试资产生成、资产定价和投资管理于一体的框架。

more, Daniel et al. (2020) nearly doubles the Sharpe ratio of the characteristics-sorted portfolio from 1.17 to 2.13 by constructing the corresponding characteristic-efficient portfolios. However, a single P-Tree delivers a 6.37 Sharpe ratio over 40 years from 1981 to 2020, representing a significant improvement over Ahn et al. (2009) and Daniel et al. (2020), although the empirical samples differ.

Out-of-sample evaluation. The results in Panel A of Table 5 are based on a full-sample analysis, which may raise concerns about in-sample overfitting. Therefore, we provide two exercises on out-of-sample investment strategies. We follow Kozak et al. (2020) and perform a half-half split to construct the training and test samples. It suffices to demonstrate the out-of-sample performance when the past and future samples predict each other. This approach further mitigates concerns regarding the look-ahead bias when models trained by the future sample predict the past.

On the one hand, the past-predicting-future results are reported in Panels B1 and B2. The first twenty-year in-sample Sharpe ratios are even higher than the forty-year ones in Panel A, and the out-of-sample Sharpe ratios are over 3. Despite decreasing performance, many out-of-sample alphas, adjusted by benchmark models, are still close to 1% and highly significant. On the other hand, for the future-predicting-past result, the out-of-sample Sharpe ratios are larger than 3.88, and the model-adjusted alphas are over 0.60%. These findings demonstrate strong out-of-sample evidence for P-Tree test assets and factors for spanning the MVE frontier.

These out-of-sample investment performance results are comparable to those reported in recent studies. The deep reinforcement learning strategy in Cong et al. (2021) achieves a Sharpe ratio of 2.21 after excluding small stocks from 1990 to 2016. The regularized portfolio, based on sorted portfolios on size, operating profitability, and investment in Bryzgalova et al. (2023), achieves an out-of-sample monthly Sharpe ratio of 0.69 from 2004 to 2016, equivalent to a 2.39 annualized Sharpe ratio. The deep neural network approach in Feng et al. (2023) reports out-of-sample Sharpe ratios of 2.95-3.00 for 2002 to 2021. P-Tree strategies stand out for its transparency, ease of implementation, and flexibility as an all-round framework for test asset generation, asset pricing, and investment management.

5 可解释性、公司特征与宏观状态

5.1 利用随机P森林衡量特征重要性

首先，附录I的模拟证据展示了遗漏真实特征所造成的效率损失，并说明即使存在大量冗余或无用特征，能够处理非线性与交互的P-Tree仍可恢复真实特征集合。其次，传统顺序或独立资产排序方案受高维问题限制，实践中通常只关注不超过三个特征。无论是否使用树模型，穷举多个特征的所有可能顺序排序情形都是NP难问题；P-Tree的迭代生长提供了一种有效算法，使计算可行。因此，P-Tree算法选出的特征能够更好地代理SDF中内含的真实基本风险，而这些风险可能被依赖临时选定因子的线性因子模型忽略。

随机P森林。我们采用机器学习中另一种著名集成方案，评价特征重要性并考虑变量选择风险。随机森林拟合多棵树，以避免过拟合并量化模型不确定性。[20] 我们将这一方法用于评估P-Tree框架中变量的重要程度，以下称之为随机P森林（Random P-Forest）。

构造随机P森林的集成步骤如下：（i）沿时间维度进行有放回自助抽样，保留被选时期的完整面板横截面，利用资产收益序列相关性较低这一特点。（ii）随机抽取20项特征，在每个自助样本中独立生长P-Tree；重复1000次，形成由1000棵P-Tree组成的森林。（iii）统计各特征在P森林中的出现情况，评价其重要性。

我们的实证实验关注某个变量被选作决策树切分变量的频率。

[20] 标准随机森林（Breiman，2001）在自助训练样本上生长多棵树：对观测有放回随机抽样，并从完整训练集的变量中抽取一个子集。袋装法或随机森林的思想是，从经验分布抽样近似于从真实总体分布抽样，从而能够量化模型估计的不确定性。

5 Interpretability, Characteristics, and Macro Regimes

5.1 Characteristic Importance via Random P-Forest

First, our simulation evidence in Appendix I demonstrates the efficiency loss resulting from missing true characteristics and illustrates how the nonlinear and interactive P-Tree can recover the true set of characteristics, despite a large number of redundant or useless characteristics. Second, the conventional sequential or independent asset sorting scheme primarily focuses on no more than three characteristics in practice, due to the high-dimensional challenge. Because enumerating all possible sequential sorting cases with multiple characteristics is NP-hard, whether using tree models or not, P-Tree's iterative growth is an efficient algorithm for overcoming this problem and make computation feasible. Therefore, the characteristics selected by the P-Tree algorithm are better proxies for the true fundamental risk inherent in the SDF, which may be overlooked in a linear factor model with ad hoc selected factors.

Random P-Forest. We adopt another famous ensembling scheme in ML to evaluate characteristic importance and account for variable selection risk. Random Forest fits multiple trees to avoid overfitting and quantifying model uncertainty.[20] We utilize this approach to assess the significance of variables within the P-Tree framework. In the subsequent text, we refer to our approach as Random P-Forest.

The ensemble steps for constructing the Random P-Forest are as follows: (i) bootstrap the data on the time-series horizon with replacement, preserving the complete cross section of the panel data for the selected time periods to exploit the low serial correlations among asset returns. (ii) randomly draw 20 characteristics, and P-Trees grow on each bootstrap sample independently. We repeat the procedure 1,000 times to form a forest of 1,000 P-Trees. (iii) evaluate the importance of each characteristic by counting its appearance in the P-Forest.

Our empirical exercises focus on the frequency of selecting one variable as a split

[20] Standard Random Forest (Breiman, 2001) grows many trees on bootstrap training samples, which come from random samples of the observations with replacement and sample a subset of variables from the complete training set. The idea behind bagging or Random Forest is that sampling data from the empirical distribution approximates sampling from the true underlying distribution, thus enabling quantification of the uncertainty of model estimation.

直观而言，一个特征越频繁地被选作切分规则，它对于划分横截面、最大化均值—方差分散化就越重要。对每个自助样本，我们随机抽取特征子集来生长树，其中只有一部分最终被选为切分规则。统计第l项特征z_l在前J次切分中被使用的次数，以及它出现在自助抽取特征子集中的总次数。

特征重要性定义见式（8）：选择概率（z_l）=“z_l在前J次切分中被选中的次数”÷“z_l出现在自助子样本中的次数”。

表6汇总特征的选择概率。三行分别对应J=1、2、3。选择概率会随J从1增至3而提高，因此必须在同一行内比较。每行按选择概率降序列出前五项特征。

表6：以选择概率衡量特征重要性

本表报告1000棵树组成的随机P森林中最常被选中的特征。“Top 1”只统计每棵树的第一次切分；“Top 2”“Top 3”分别统计前两次、前三次切分。数值为这些特征在1000次集成中的选择频率，特征说明见表A.1。Top1：SUE、SVAR、CHPM、RVAR_CAPM、BASPREAD；Top2：SUE、SVAR、DOLVOL、CHPM、BM_IA；Top3：SUE、DOLVOL、BM_IA、SVAR、ME_IA。对应数值见原表。

首先，盈余意外（SUE）这一重要基本面指标，在自助样本中有约50%的概率成为首次切分特征，有66%的概率进入前两次切分。尽管FF5、Q5等线性因子模型未考虑它，SUE似乎是非线性交互建模中最有助于呈现收益横截面变动的特征。其他高频入选特征包括收益波动率（SVAR）、成交金额（DOLVOL）、特质波动率（RVAR_CAPM）、利润率变化（CHPM）和行业调整账面市值比（BM_IA）。

in a decision tree. Intuitively, if a characteristic is chosen more frequently as a split rule, it becomes more important for splitting the cross section to maximize the mean-variance diversification. For each bootstrap sample, a random subset of characteristics is drawn to grow the tree, and only a fraction of them are selected as split rules. We count the number of times a specific l-th characteristic z_l is used in the first J splits and the total number of appearances in bootstrap subsamples. We define the measure of characteristic importance as follows:

$$\text{Selection Probability } (z_l) = \frac{\#\left(z_l \text{ is selected at first } J \text{ splits }\right)}{\#\left(z_l \text{ appears in the bootstrap subsamples }\right)}. \tag{8}$$

Table 6 summarizes the selection probabilities (8) of characteristics. Three rows report the selection probabilities for $J = 1, 2, 3$, respectively. The selection probability increases as J grows from 1 to 3, so we must compare within the row. For each row, we list the top five characteristics in descending order of selection probability.

Table 6: **Characteristics Importance by Selection Probability**

This table reports the most frequently selected characteristics from the random P-Forest of 1,000 trees. The "Top 1" rows only count the first split for 1,000 trees. The "Top 2" or "Top 3" rows only count the first two or three splits. The numbers reported are the selection frequency for these top characteristics selected out of the 1,000 ensembles. The descriptions of characteristics are listed in Table A.1.

	1	2	3	4	5
Top1	SUE 0.51	SVAR 0.34	CHPM 0.25	RVAR_CAPM 0.23	BASPREAD 0.21
Top2	SUE 0.66	SVAR 0.38	DOLVOL 0.34	CHPM 0.31	BM_IA 0.31
Top3	SUE 0.71	DOLVOL 0.53	BM_IA 0.48	SVAR 0.41	ME_IA 0.41

First, we find the earnings surprise (SUE), one of the most important fundamentals, has a 50% chance of being the first splitting characteristic and a 66% chance of being one of the top two splitting characteristics in the bootstrap sample. Though not considered in linear factor models like FF5 and Q5, SUE seems most useful in nonlinear interactive modeling to display the cross-sectional variation of asset returns. Other frequently selected characteristics include return volatility (SVAR), dollar trading volume (DOLVOL), idiosyncratic volatility (RVAR_CAPM), change in profit margin

表6中靠前切分所选特征几乎涵盖所有主要类别——动量、规模、价值、摩擦/波动率和盈利能力。需要注意，某些特征（例如投资）没有入选，并不意味着它们无用，因为这里只展示最靠前的三次重要切分。

可解释性与过拟合。随机森林或提升树等树集成方法能够降低过拟合风险，但也使理解单棵树变得困难。P-Tree以图形形式展示特征的非线性和非对称交互，因而容易解释；但如果不能表明它没有过拟合，模型的有效性可能受到质疑。

除表5所示样本外实证评价外，我们还比较随机P森林与单棵P-Tree的重要特征，利用重要性指标检测过拟合。作者认为，特征相似性意味着P-Tree的全局切分准则能够像机器学习集成方法一样避免过拟合，同时提供更强的可解释性，以及对非线性和非对称交互的透明认识。随机P森林识别的重要特征，例如SUE、DOLVOL和BM_IA，也进入了图4中40年样本P-Tree的靠前切分。表6中的重要特征，同样出现在子样本P-Tree图的靠前切分中，见图A.1。

〔译注：“意味着……避免过拟合”为作者的解释。特征选择相似是稳定性证据，并不构成不会过拟合的数学保证。〕

5.2 利用P-Tree评价单项特征

作为另一种定价核，P-Tree能够直接评价新特征相对于基准的增量贡献，包括非线性贡献，从而补充已有文献。实证研究常依据某项特征排序构造新因子；一个重要任务是，通过张成回归控制基准模型，检验新因子的独立信息（Fama和French，1996）。然而，公司特征具有三维数据结构，实证研究很少直接检验相关特征是否提供新信息，因而可能损失信息（Lettau，2023）。

(CHPM), and industry-adjusted book-to-market ratio (BM_IA). The characteristics in Table 6 cover almost all major categories—momentum, size, value, frictions/volatility, and profitability—in the top splits. Notice that characteristics (e.g., investment) not being selected does not imply they are useless, because this table only shows the top three important characteristics.

Interpretability v.s. overfitting. Using tree ensembles such as random forests or boosted trees reduces the risk of overfitting but also makes understanding a single tree more difficult. P-Trees are easily interpretable because they graphically display characteristics for non-linearities and asymmetric interactions, but the model's effectiveness might be constrained if we cannot show it does not overfit.

In addition to the out-of-sample empirical evaluation presented in Table 5, we compare important characteristics between a random P-Forest and a single P-Tree, using importance metrics to detect overfitting. The similarity of characteristics implies that the global split criteria of P-Tree avoid overfitting, just like ML ensembling methods, but achieve greater interpretability and provide transparent insights into non-linearities and asymmetric interactions. The significant characteristics obtained from the random P-Forest, such as earning surprise (SUE), dollar trading volume (DOLVOL), and industry-adjusted book-to-market ratio (BM_IA), are also chosen in the top splits of P-Tree for the 40 years, as illustrated in Figure 4. In addition, the important characteristics listed in Table 6 are also selected in the top splits displayed in the P-Tree diagram for the subsample analysis (refer to Figure A.1).

5.2 Evaluating a Characteristic with P-Tree.

As an alternative pricing kernel, P-Tree can directly evaluate a new characteristic's incremental (and nonlinear) contribution over a benchmark to complement the literature. Researchers create a new factor by sorting based on a specific characteristic in empirical studies. A crucial research task is to test the independent information of new factors by controlling benchmark models through spanning regression (Fama and French, 1996). However, due to the three-dimensional data structure of firm characteristics, empirical studies rarely examine whether the associated characteristics provide

针对这一问题，P-Tree提供了新的评价视角：给定ME和BM等基准特征，检验其他特征对生成测试资产与潜在因子模型是否有用。

图8展示两棵基准P-Tree：一棵为三层ME基准树，所有切分都基于ME；另一棵为三层ME-BM基准树，第一层按ME切分、第二层按BM切分。为了评估某特征相对于ME在估计有效前沿方面的增量效用，我们在ME基准树上增加一次切分，切分变量为目标特征或ME。浅树上新增一次切分本来就可能提高MVE组合的夏普比率。因此，只有当加入该特征带来的夏普比率改善超过再按ME切分的改善时，才认为它提供新信息。这样便可衡量目标特征的边际贡献。

图8：ME基准P-Tree与ME-BM基准P-Tree示意

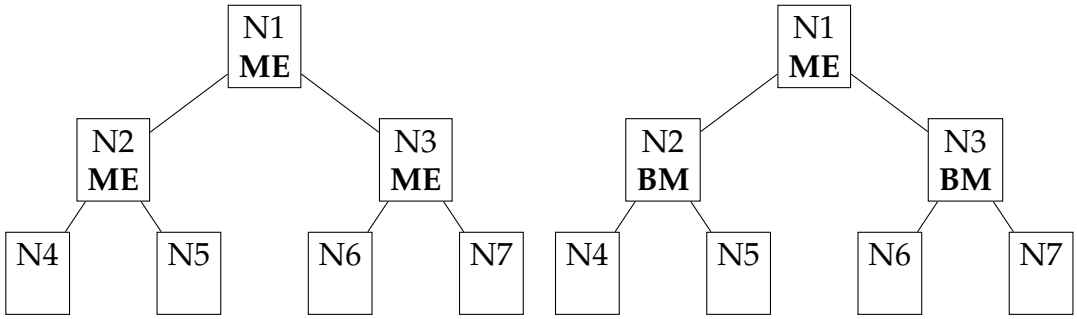
（a）ME基准树：根节点和下一层两个内部节点都按ME切分。（b）ME-BM基准树：根节点按ME切分，下一层两个内部节点按BM切分。N1—N7为节点编号。

图9（a）汇总各特征在ME基准树上的增量夏普比率。优于ME的特征用黑色表示，弱于ME的用灰色表示，并按夏普比率升序排列。即使对于浅层P-Tree，仍有24项特征相对于ME基准树不能提供增量信息。另外，规模之外的动量特征（SUE、ABR、NINCR）、价值特征（BM_IA、BM、CFP、EP）、盈利能力特征（CHPM、ROA、ROE）以及摩擦/波动率特征（MAXRET、SVAR），只需在ME基准树上再切分一次，就能明显改善投资表现。

new information and may suffer information loss (Lettau, 2023). To address this issue, P-Tree provides a new perspective for evaluating the usefulness of characteristics in generating test assets and latent factor models, given a set of benchmark characteristics, such as ME and BM.

In Figure 8, we demonstrate two baseline P-Trees: a three-layer ME baseline P-Tree where all splits are based on ME and a 3-layer ME-BM baseline P-Tree where the first layer is split on ME and the second layer is split on BM. To evaluate the incremental utility for estimating the efficient frontier of a characteristic against ME, we expand the ME baseline P-Tree with one more split on the characteristic of interest or ME. One might expect a split to increase the Sharpe ratio of an MVE portfolio for a shallow P-Tree. Therefore, we only determine if a characteristic provides new information, by comparing the Sharpe ratio improvement of adding this characteristic to the baseline P-Tree over a split on ME. This approach allows us to measure the marginal contribution of the characteristic of interest.

Figure 8: **Demonstration for ME baseline and ME-BM baseline P-Trees**



(a) ME Baseline P-Tree (b) ME-BM Baseline P-Tree

Figure 9, subfigure (a), summarizes the incremental Sharpe ratio of each characteristic on the ME baseline P-Tree. We color those characteristics outperforming ME in black, and those underperforming ME in grey. Also, we sort the characteristics by the Sharpe ratios in ascending order. Even for a shallow P-Tree, we find that 24 characteristics do not provide incremental information against the ME baseline P-Tree. Further, we identify categories other than size, including momentum (SUE, ABR, NINCR), value (BM_IA, BM, CFP, EP), profitability (CHPM, ROA, ROE), and frictions/volatility (MAXRET, SVAR), that vastly improve investment performance by a simple further split on the ME baseline P-Tree.

图9：利用P-Tree评价单项特征

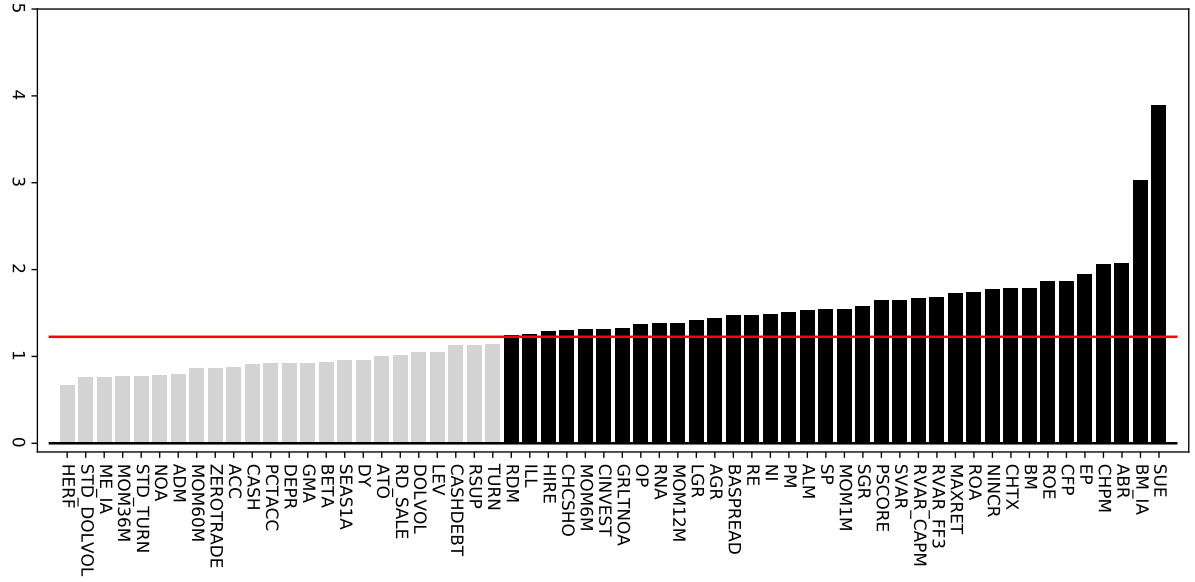
本图显示，在ME或ME-BM基准P-Tree上，按某特征进一步切分后基础组合的夏普比率。（a）、（b）分别为ME和ME-BM基准树的结果；特征按夏普比率升序排列。红线表示继续按控制特征切分一步后的夏普比率。柱高超过红线表示相对于基准特征提供增量信息，以黑色显示；否则以灰色显示，表示未提供增量信息。特征定义见表A.1。

图内Controlling ME=控制ME；Controlling ME-BM=控制ME与BM；纵向列出的特征缩写保留不变，其中文名称见表A.1译文；横轴为夏普比率。

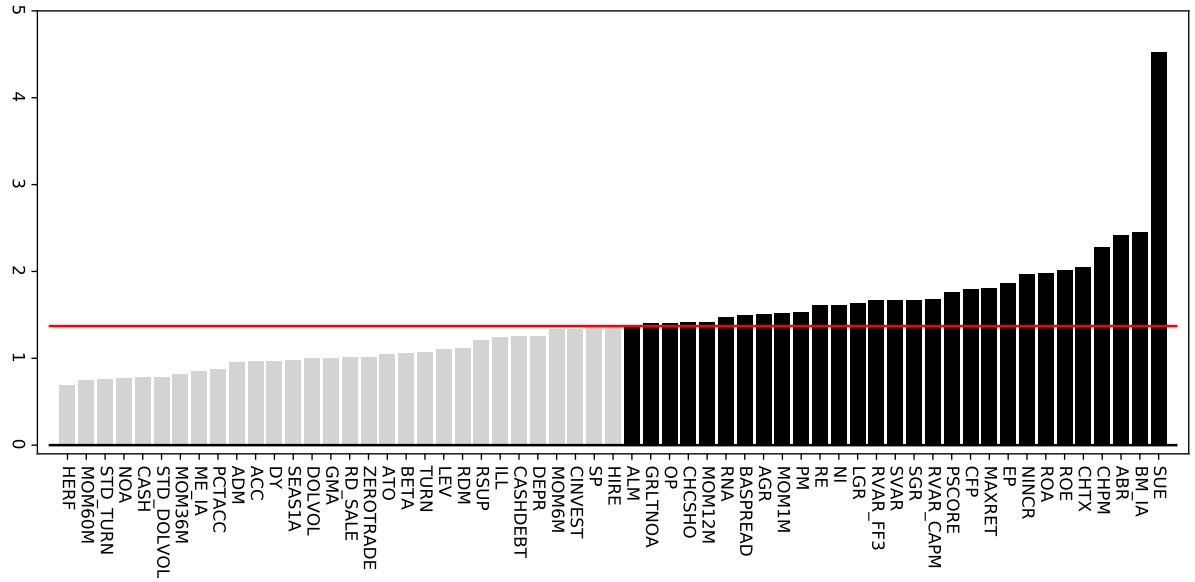
总之，我们通过在基准P-Tree上控制ME和/或BM，评价每项特征对投资目标的边际贡献。可以预期，当基准树更深时，这些边际贡献会更小。

Figure 9: **Evaluating a Characteristic with P-Tree**

This figure shows the Sharpe ratio of P-Tree basis portfolios after splitting on a characteristic, based on either ME or ME-BM baseline P- Trees. Sub-figures (a) and (b) display the results for ME and ME-BM baseline P-Trees, respectively. We sort the characteristics by their Sharpe ratios in ascending order. The red lines show the Sharpe ratios of P-Tree basis portfolios after splitting one step further on the controlling characteristics. A bar above the red line indicates incremental information against the benchmark characteristics and is colored black. Otherwise, the grey bar indicates the characteristic does not provide incremental information. The characteristics are listed in Table A.1.



(a) Controlling ME



(b) Controlling ME-BM

In summary, we assess each characteristic on a benchmark P-Tree by controlling `ME` and/or `BM` for its marginal contribution to the investment objective. We can expect these marginal contributions to be smaller when considering a deeper tree as the

改变基准P-Tree或控制特征时，单项特征的相对重要性也可能变化，这体现了P-Tree评价的适应性。该评价可补充传统因子张成回归，用于评价公司特征。

类似地，我们使用图8（b）的ME-BM基准树评价各项特征。这是一棵基于ME与BM的三层树。面对更强基准，RDM、ILL、HIRE、MOM6M、CINVEST、SP等特征不能再提供额外信息。相对于ME-BM基准树，共有30项特征不能为估计有效前沿提供增量信息。与ME基准结果一致，动量、价值和盈利能力特征在非线性交互模型中是有用的。

5.3 宏观经济状态下的P-Tree

越来越多的实证证据表明，因子风险溢价/风险价格和因子载荷beta可能随宏观经济条件显著变化。大量文献研究资本市场的状态变化和市场择时，包括牛市与熊市（Fabozzi和Francis，1977）、状态变化（Ang和Timmermann，2012）、时变通胀风险溢价（Boons等，2020）、时变因子模型（Cui等，2023）等。鉴于横截面驱动力可能随时间变化，考察P-Tree在不同状态下的表现，以及哪些特征被用于均值—方差分散化，具有价值。本节通过不同宏观状态下的子样本分析，展示P-Tree的稳健表现。

宏观状态与特征评价。为进行稳健性检查，我们使用表A.2的10个宏观变量，分析不同状态的子样本。结果见表7。P-Tree可以通过时间序列切分识别宏观状态变化，类似Cong等（2022）的方法。作为展示，附录III.D在替代切分准则下，通过数据驱动的时间序列切分识别状态转换。[21]

benchmark. The relative importance of a characteristic may change when the benchmark P-Tree or controlling characteristics are altered, highlighting the adaptable nature of P-Tree evaluation. This evaluation can complement the conventional factor-spanning regression in evaluating firm characteristics.

Similarly, we assess each characteristic using the ME-BM baseline P-Tree shown in subfigure (b) of Figure 8. This P-Tree has three layers and is based on `ME` and `BM`. With a stronger benchmark, some characteristics, such as `RDM`, `ILL`, `HIRE`, `MOM6M`, `CINVEST`, `SP`, do not provide additional information on this upgraded ME-BM baseline P-Tree. Against the ME-BM baseline P-Tree, we find that 30 characteristics do not offer incremental information for estimating the efficient frontier. Consistent with the ME baseline P-Tree results, we find momentum, value, and profitability characteristics are useful in the nonlinear interactive model.

5.3 P-Tree with Macroeconomic Regimes

Increasing empirical evidence indicates factor risk premia/price and factor loading betas can vary significantly under various macroeconomic conditions. Extensive literature studies regime changes and market timing in capital markets, including bull versus bear markets (Fabozzi and Francis, 1977), regime changes (Ang and Timmermann, 2012), time-varying inflation risk premia (Boons et al., 2020), time-varying factor models (Cui et al., 2023), and so on. Given that the cross-sectional driving forces may change over time, examining how P-Tree performs in different states and which characteristics are selected for the mean-variance diversification is valuable. This subsection presents the subsample analysis for implementing P-Trees under various macro regimes and demonstrates their robust performance.

Macroeconomic regimes and characteristics evaluation. For robustness check, We analyze ten macro variables from Table A.2 for subsample analysis in various regimes. The empirical outcomes of P-Tree models under different regimes are demonstrated in Table 7. P-Tree can identify macro regime shifts through time-series splits, similar to the approach of Cong et al. (2022). For demonstration, we detect regime switches

但在基准P-Tree中，我们依据10年滚动窗口的50%分位数，将样本分为“Top”（较高）与“Btm”（较低）两种状态，对每种状态分别拟合模型，从而评价P-Tree在各类正向和负向宏观环境中的作用。

表7：状态切换下的P-Tree表现

本表报告不同宏观状态下P-Tree模型的表现。每个宏观变量的时间序列，依据10年滚动窗口的50%分位数分为Top和Btm状态。在每种状态下构建一棵简单P-Tree，报告与表3类似的统计量，以及前三次切分所选特征。Com行报告将Top与Btm状态P-Tree因子合并后的平均收益和夏普比率。

表头中译：Regime=状态；AVG=平均收益；SR=夏普比率；GRS=GRS统计量；p-value=p值；|alpha|上横线=平均绝对alpha；平方平均后开方=alpha均方根；R^2=回归解释度；Top Three Characteristics=前三次切分特征。宏观变量定义见表A.2。数值见随后原表。

首先，有用特征集合可能随宏观条件变化。P-Tree表现稳健，能够适应宏观状态，并在前三次切分选择不同特征。例如，期限利差较低或市场波动较高时，代表公司基本面的SUE和交易量的DOLVOL，对横截面均值—方差分散化更加重要。

[21] 附录III.D定义宏观变量上的时间序列切分准则。例如，图I.4展示按市场波动率（SMVAR）进行时间序列切分的P-Tree，识别出高波动与低波动状态，再分别在各状态下建立横截面P-Tree。这种用于资产定价的状态划分仍然由数据驱动，并以宏观预测变量为工具。

under alternative split criteria and data-driven time-series splits in Appendix III.D.[21] However, for our baseline P-Tree, we split the sample into "Top" and "Btm" (Bottom) regimes using the 50% 10-year rolling-window quantiles and fit a separate model for each regime. Therefore, we evaluate the usefulness of P-Tree in all positive and negative macroeconomic environments.

Table 7: **P-Tree Performance under Regime Switches**

This table shows the performance of P-Tree models across various macroeconomic states. The time series of each macro variable is split into "Top" and "Btm" regimes using the 50% 10-year rolling-window quantiles. A simple P-Tree is then built for each regime, and similar performance statistics are reported in Table 3, along with the top three splitting characteristics. The rows labeled "Com" report the average return and Sharpe ratio of the combined P-Tree factor of the "Top" and "Btm" regimes.

	Regime	AVG	SR	GRS	p-value	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$	$\overline{R^2}$	Top Three Characteristics		
	Top	1.43	8.97	73.21	0.00	1.12	1.43	56.22	SUE	DOLVOL	CFP
DFY	Btm	1.3	8.16	63.08	0.00	1.58	1.88	43.70	BASPREAD	SUE	ZEROTRADE
	Com	1.37	8.52								
	Top	1.35	9.43	55.21	0.00	1.10	1.35	65.88	CHPM	BM_IA	ME
DY	Btm	1.31	7.7	72.39	0.00	1.09	1.33	47.08	ABR	ZEROTRADE	DOLVOL
	Com	1.32	8.21								
	Top	1.29	11.39	68.73	0.00	1.15	1.41	68.00	RVAR_CAPM	ME_IA	ME
EP	Btm	1.4	7.87	82.93	0.00	1.23	1.52	47.03	SVAR	SUE	ME
	Com	1.37	8.43								
	Top	1.24	8.48	89.34	0.00	1.17	1.45	48.88	RVAR_CAPM	SUE	ZEROTRADE
ILL	Btm	1.26	11.01	74.48	0.00	0.91	1.18	56.27	SUE	DOLVOL	SP
	Com	1.24	9.18								
	Top	1.15	7.3	31.89	0.00	1.07	1.33	66.79	CFP	ATO	ME
INFL	Btm	1.44	7.73	70.41	0.00	1.27	1.57	44.41	RVAR_CAPM	DOLVOL	SUE
	Com	1.34	7.41								
	Top	1.24	7.88	44.34	0.00	0.97	1.32	57.73	CFP	ALM	STD_DOLVOL
LEV	Btm	1.45	7.03	55.18	0.00	1.23	1.52	46.24	NI	MOM12M	BM_IA
	Com	1.37	7.18								
	Top	1.69	7.91	54.29	0.00	1.73	2.23	45.54	RVAR_CAPM	ME_IA	RD_SALE
NI	Btm	1.58	8.25	66.19	0.00	1.17	1.45	52.90	ROA	DOLVOL	SUE
	Com	1.63	8.05								
	Top	1.44	9.45	79.11	0.00	1.09	1.49	55.79	SUE	DOLVOL	CFP
SVAR	Btm	1.41	9.54	85.86	0.00	1.16	1.40	49.93	ROA	ZEROTRADE	DOLVOL
	Com	1.42	9.51								
	Top	1.08	10.5	61.48	0.00	0.97	1.25	54.85	CHCSHO	SUE	DOLVOL
TBL	Btm	1.43	8.98	108.59	0.00	1.19	1.49	50.87	SUE	DOLVOL	CFP
	Com	1.32	8.74								
	Top	1.27	9.33	75.74	0.00	1.12	1.28	62.06	OP	STD_DOLVOL	SUE
TMS	Btm	1.21	8.12	62.59	0.00	0.82	1.13	52.94	SUE	DOLVOL	BM_IA
	Com	1.24	8.65								

First, the set of useful characteristics might change across different macroeconomic conditions. P-Tree has robust performance, adapts to macro regimes, and selects different sets of characteristics in the top three splits. For example, when the term spread is low, or market volatility is high, the roles of SUE and DOVOL for firm fundamentals and trading volume are more important for cross-sectional mean-variance

[21] Appendix III.D defines the time-series split criteria on the macro variables. For example, Figure I.4 shows a time-series split P-Tree on Market Volatility (SMVAR), which identifies a high-volatility regime and a low-volatility regime. Then, two separate cross-sectional P-Trees are developed on each regime. This asset-pricing regime determination remains data-driven and instrumented by macro predictors.

不过，代表流动性和波动摩擦的BASPREAD、RVAR_CAPM等特征，常在净股票发行高、违约收益率低、市场流动性低或通胀低的情景中被选中。

其次，P-Tree测试资产始终为资产定价提供有信息的结果：在每个状态下，对Fama-French五因子的GRS检验均拒绝原假设，P-Tree测试资产张成的MVE组合夏普比率均高于7。值得注意的是，低通胀或低国库券利率情景的GRS统计量，即加权定价误差，明显更大，意味着常用因子在这些时期捕捉横截面信号的能力最弱。平均alpha大于表5A的P-Tree1，说明极端宏观状态下的测试资产更难被常用因子定价。最后，将两种状态的P-Tree因子合并后，Com因子的夏普比率高于表5A使用无条件模型的第一棵P-Tree因子。

6 结论

利用个体资产收益估计均值—方差有效（MVE）前沿，并生成具有多样性的测试资产，是实证资产定价长期面临的挑战（例如Markowitz，1952；Lewellen等，2010；Daniel等，2020）。本文提出新的可解释人工智能模型——面板树（P-Tree），有效应对广义证券排序这一传统实证难题，并在分析个体资产收益的非平衡面板方面取得优越表现。

P-Tree利用包含资产收益联合分布丰富信息的高维特征，生成基础组合，使其成为具有多样性的测试资产和定价核的组成部分。更一般地，P-Tree将树模型扩展到纯预测之外，以经济目标引导树生长，有效分析面板数据，同时保留可解释性，并处理噪声环境中的非对称与非线性交互。

对美国股票的实证研究表明，P-Tree测试资产张成的有效前沿，大幅优于常见可观测因子模型、机器学习潜在因子模型以及传统测试资产所形成的前沿。

diversification. However, characteristics such as BASPREAD and RVAR_CAPM, representing frictions for liquidity and volatility, are often chosen for scenarios involving high net equity issuance, low default yield, low market liquidity, or low inflation.

Second, P-Tree test assets consistently provide informative results for asset pricing considerations: the GRS tests against Fama-French five factors are rejected for each regime, and the Sharpe ratios of the MVE portfolios spanned by P-Tree test assets are all greater than 7. Interestingly, the GRS statistics (weighted pricing errors) are significantly larger for scenarios that involve low inflation or a low t-bill rate, which implies the weakest moments of commonly used factors capturing the cross-sectional signals. The average alphas are larger than those of "P-Tree1" in Table 5, Panel A, implying the test assets under extreme macro regimes are even more challenging to price by common factors. Finally, we combine the P-Tree factors in two regimes and find the "Com" (Combined) factor has larger Sharpe ratios than the first P-Tree factor in Table 5, Panel A, which implements an unconditional model.

6 Conclusion

Estimating the mean-variance efficient (MVE) frontier and generating diversified test assets using individual asset returns remain long-term challenges in empirical asset pricing (e.g., Markowitz, 1952; Lewellen et al., 2010; Daniel et al., 2020). Our paper presents a novel interpretable AI model, Panel Tree (P-Tree), that effectively addresses the traditional empirical challenge of generalized security sorting and delivers superior performance for analyzing the (unbalanced) panel of individual asset returns. P-Tree utilizes high-dimensional characteristics containing rich information on the joint distribution of asset returns, to generate basis portfolios as diversified test assets and building blocks of the pricing kernel. More generally, P-Trees broaden tree-based models beyond pure prediction and analyze panel data effectively with economics-guided tree growth, while retaining interpretability and handling asymmetric and nonlinear interactions in noisy environments.

Our empirical study of U.S. equities shows that the efficient frontiers spanned by P-Tree test assets vastly outperform those from commonly used observable and ma-

提升P-Tree生成一系列测试资产，对现有因子模型提出挑战，为按特征排序的组合提供具有多样性的替代测试资产。除了叶基础组合，MVE框架下的统一方法还提供可交易的P-Tree因子，其样本外年化夏普比率超过3。因子模型在解释横截面收益变动方面也优于基准模型。最后，P-Tree为评价高维特征的条件信息提供了新的非线性、交互式框架，补充了用于评价公司特征的传统因子张成回归。

参考文献

为便于检索与准确引用，参考文献的作者姓名、英文题名、期刊、卷期和页码均保留原文，不翻译或改写。参考文献自随后原文页开始，连续保留至原PDF第49页。

chine learning latent factor models, as well as conventional test assets. Boosted P-Trees generate sequences of test assets to challenge all existing factor models, providing a diversified test asset alternative to characteristic-sorted portfolios. In addition to leaf basis portfolios, our unified approach under the MVE framework provides tradable P-Tree factors and out-of-sample annualized Sharpe ratios over 3. The factor models also outperform benchmark models in explaining the cross-sectional return variation. Finally, P-Tree offers a new nonlinear and interactive framework to assess the conditional information of high-dimensional characteristics, complementing traditional factor-spanning regression in evaluating firm characteristics.

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附录

算法：生长单棵P-Tree

以下为原算法逐项中译，符号与原算法保持对应；精确数学排版见随后原文页。

1．过程：GROW_TREE（root），从根节点生长树。

2．输入：资产收益r_i,t与经过排序的公司特征z_i,t。

3．结果：从根节点生长P-Tree，形成叶基础组合。

4．令j从1循环至J，即遍历迭代次数。

5—6．如果当前深度达到或超过d_max，则返回。

7—8．否则，搜索整棵树，找出所有叶节点，记其集合为N。

9．遍历当前每个叶节点N。

10．遍历该节点候选切分集合C_N中的每条候选规则$\tilde{c}_{k,m,N}$。

11．依据候选规则对节点N的数据进行试切分。〔译注：原文写作“temporally”，依本算法上下文按暂时试切分理解；此处并非附录III.D的宏观时间切分。〕

12—13．若任一左/右子节点不满足最小叶规模，则将该规则的目标值L设为负无穷。

14—17．否则，计算叶基础组合；按式（2）使用所有叶基础组合估计SDF；按式（4）计算候选规则的切分准则L。

18—20．结束条件分支、候选切分循环和叶节点循环。

21．在所有叶节点及其候选切分规则中，寻找使切分准则最大化的节点与规则，作为第j次切分$\tilde{c}_j$。即对N与$\tilde{c}_{k,m,N}$联合求argmax L。

22．用第j条规则$\tilde{c}_j$切分选中的节点。

23—24．结束条件分支和迭代循环。

25．返回第J次切分后的叶组合收益向量$R_t^{(J)}$及因子收益$f_t^{(J)}$。

26．过程结束。

Appendices

Algorithm Growing a Single P-Tree

procedure GROWTREE(`root`)
Input Asset returns $\mathbf{r}_{i,t}$ and ranked characteristics $\mathbf{z}_{i,t}$
outcome Grow P-Tree from the `root` node, form leaf basis portfolios
  for j from 1 to J **do** ▷ Loop over number of iterations
    if current depth $\geq d_{\text{max}}$ **then**
      return.
    else
      Search the tree, find all leaf nodes $\mathcal{N}$
      for each leaf node N in $\mathcal{N}$ **do** ▷ Loop over all current leaf nodes
        for each candidate split $\tilde{\mathbf{c}}_{k,m,N}$ in $\mathcal{C}_N$ **do**
          Partition data temporally in N according to $\tilde{\mathbf{c}}_{k,m,N}$.
          if Either left or right child of N does not satisfy minimal leaf size **then**
            $\mathcal{L}(\tilde{\mathbf{c}}_{k,m,N}) = -\infty$.
          else
            Calculate leaf basis portfolios.
            Estimate SDF using *all* leaf basis portfolios as in (2).
            Calculate the split criteria $\mathcal{L}(\tilde{\mathbf{c}}_{k,m,N})$ in (4).
          end if
        end for
      end for
      Find the best leaf node and split rule that maximize split criteria

$$\tilde{\mathbf{c}}_j = \underset{N\in\mathcal{N},\tilde{\mathbf{c}}_{k,m,N}\in\mathcal{C}_N}{\arg\max} \{\mathcal{L}(\tilde{\mathbf{c}}_{k,m,N})\}$$

      Split the node selected with the j-th split rule of the tree $\tilde{\mathbf{c}}_j$.
    end if
  end for
  return $\left\{\mathbf{R}_t^{(J)}, f_t^{(J)}\right\}$
end procedure

图A.1：20年子样本的树结构图

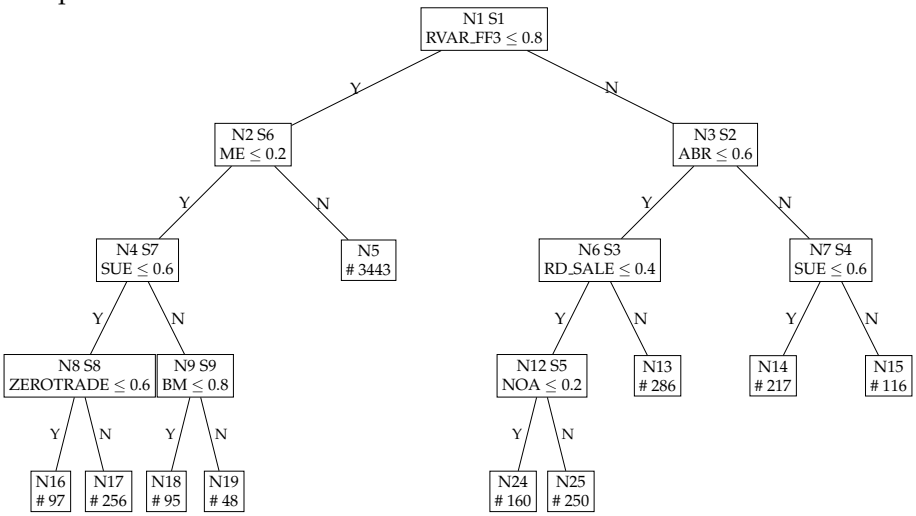
图示格式同图4。（a）为1981—2000年样本的P-Tree；（b）为2001—2020年样本的P-Tree。原图标题写1981—2000年，但两幅子图分别对应两个子样本。

图内N表示节点编号，S表示切分顺序；Y表示满足节点切分条件，N表示不满足；#后数字为叶节点股票数统计，沿用图4定义。各阈值为特征排序后所在分位。

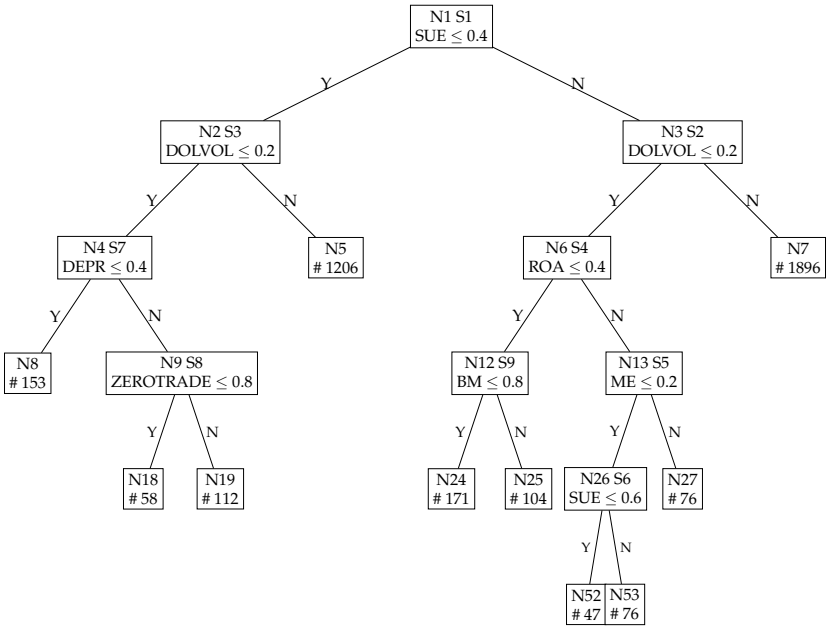
（a）根节点按RVAR_FF3≤0.8切分，后续涉及ME、SUE、ZEROTRADE、BM、ABR、RD_SALE、NOA。（b）根节点按SUE≤0.4切分，后续涉及DOLVOL、DEPR、ZEROTRADE、ROA、BM、ME与SUE。各路径、全部阈值和股票数保留于随后原图；变量中文名称见表A.1。

Figure A.1: **Tree Diagram, 20 Years, 1981-2000**

Format follows Figure 4. Subfigure (a) shows P-Tree in the sample from 1981-2000. Subfigure (b) shows P-Tree in the sample from 2001-2020.



(a) 20-year Sample (1981-2000)



(b) 20-year Sample (2001-2020)

表A.1：股票特征

本表列出实证研究使用的61项特征。以下依次为“编号｜特征缩写｜中文说明｜原文类别”；类别划分按原文保留。

1｜ABR｜盈余公告前后异常收益｜动量

2｜ACC｜经营性应计项目｜投资

3｜ADM｜广告费用与市值之比｜无形资产

4｜AGR｜资产增长｜投资

5｜ALM｜季度资产流动性｜无形资产

6｜ATO｜资产周转率｜盈利能力

7｜BASPREAD｜买卖价差（3个月）｜摩擦

8｜BETA｜贝塔（3个月）｜摩擦

9｜BM｜权益账面价值与市值之比｜价值相对成长

10｜BM_IA｜行业调整账面市值比｜价值相对成长

11｜CASH｜现金持有量｜价值相对成长

12｜CASHDEBT｜现金与债务之比｜价值相对成长

13｜CFP｜现金流与价格之比｜价值相对成长

14｜CHCSHO｜流通在外股份数量变化｜投资

15｜CHPM｜利润率变化｜盈利能力

16｜CHTX｜税费变化｜动量

17｜CINVEST｜公司投资｜投资

18｜DEPR｜折旧与物业、厂房及设备之比｜动量

19｜DOLVOL｜成交金额｜摩擦

20｜DY｜股息收益率｜价值相对成长

21｜EP｜盈利与价格之比｜价值相对成长

22｜GMA｜毛利润率/毛盈利能力｜投资

23｜GRLTNOA｜长期净经营资产增长｜投资

24｜HERF｜行业销售集中度｜无形资产

25｜HIRE｜员工增长率｜无形资产

26｜ILL｜滚动非流动性（3个月）｜摩擦

27 | LEV | 杠杆率 | 价值相对成长

28 | LGR | 长期债务增长 | 投资

29 | MAXRET | 最高日收益率（3个月） | 摩擦

30 | ME | 股票市值 | 摩擦

31 | ME_IA | 行业调整规模 | 摩擦

32 | MOM12M | 过去第2—12个月累计收益 | 动量

33 | MOM1M | 上月收益 | 动量

34 | MOM36M | 过去第13—35个月累计收益 | 动量

35 | MOM60M | 过去第13—60个月累计收益 | 动量

36 | MOM6M | 过去第2—6个月累计收益 | 动量

37 | NI | 净股票发行 | 投资

38 | NINCR | 盈余增长次数 | 动量

39 | NOA | 净经营资产 | 投资

40 | OP | 营业盈利能力 | 盈利能力

Table A.1: **Equity Characteristics**

This table lists the description of 61 characteristics used in the empirical study.

No.	Characteristics	Description	Category
1	ABR	Abnormal returns around earnings announcement	Momentum
2	ACC	Operating Accruals	Investment
3	ADM	Advertising Expense-to-market	Intangibles
4	AGR	Asset growth	Investment
5	ALM	Quarterly Asset Liquidity	Intangibles
6	ATO	Asset Turnover	Profitability
7	BASPREAD	Bid-ask spread (3 months)	Frictions
8	BETA	Beta (3 months)	Frictions
9	BM	Book-to-market equity	Value-versus-growth
10	BM_IA	Industry-adjusted book to market	Value-versus-growth
11	CASH	Cash holdings	Value-versus-growth
12	CASHDEBT	Cash to debt	Value-versus-growth
13	CFP	Cashflow-to-price	Value-versus-growth
14	CHCSHO	Change in shares outstanding	Investment
15	CHPM	Change in Profit margin	Profitability
16	CHTX	Change in tax expense	Momentum
17	CINVEST	Corporate investment	Investment
18	DEPR	Depreciation / PP&E	Momentum
19	DOLVOL	Dollar trading volume	Frictions
20	DY	Dividend yield	Value-versus-growth
21	EP	Earnings-to-price	Value-versus-growth
22	GMA	Gross profitability	Investment
23	GRLTNOA	Growth in long-term net operating assets	Investment
24	HERF	Industry sales concentration	Intangibles
25	HIRE	Employee growth rate	Intangibles
26	ILL	Illiquidity rolling (3 months)	Frictions
27	LEV	Leverage	Value-versus-growth
28	LGR	Growth in long-term debt	Investment
29	MAXRET	Maximum daily returns (3 months)	Frictions
30	ME	Market equity	Frictions
31	ME_IA	Industry-adjusted size	Frictions
32	MOM12M	Cumulative Returns in the past (2-12) months	Momentum
33	MOM1M	Previous month return	Momentum
34	MOM36M	Cumulative Returns in the past (13-35) months	Momentum
35	MOM60M	Cumulative Returns in the past (13-60) months	Momentum
36	MOM6M	Cumulative Returns in the past (2-6) months	Momentum
37	NI	Net Equity Issue	Investment
38	NINCR	Number of earnings increases	Momentum
39	NOA	Net Operating Assets	Investment
40	OP	Operating profitability	Profitability

表A.1（续）：股票特征

41 | PCTACC | 经营性应计项目比例 | 投资

42 | PM | 利润率 | 盈利能力

43 | PS | 业绩评分 | 盈利能力

44 | RD_SALE | 研发费用与销售额之比 | 无形资产

45 | RDM | 研发费用与市值之比 | 无形资产

46 | RE | 分析师盈余预测修正 | 无形资产

47 | RNA | 净经营资产收益率 | 盈利能力

48 | ROA | 资产收益率 | 盈利能力

49 | ROE | 权益收益率 | 盈利能力

50 | RSUP | 营业收入意外 | 动量

51 | RVAR_CAPM | CAPM特质波动率（3个月） | 摩擦

52 | RVAR_FF3 | Fama-French三因子残差方差（3个月） | 摩擦

53 | SVAR | 收益方差（3个月） | 摩擦

54 | SEAS1A | 一年季节性 | 无形资产

55 | SGR | 销售增长 | 价值相对成长

56 | SP | 销售额与价格之比 | 价值相对成长

57 | STD_DOLVOL | 成交金额标准差（3个月） | 摩擦

58 | STD_TURN | 股份换手率标准差（3个月） | 摩擦

59 | SUE | 未预期季度盈余 | 动量

60 | TURN | 股份换手率 | 摩擦

61 | ZEROTRADE | 零交易日数量（3个月） | 摩擦

表A.2：宏观经济变量

本表列出实证研究所用宏观变量。编号、缩写和中文说明如下。

1 | DFY | 违约收益率

2 | DY | 标普500股息收益率

3 | EP | 标普500盈利价格比

4 | ILL | Pastor-Stambaugh非流动性

5 | INFL | 通货膨胀

6 | LEV | 标普500杠杆率

7 | NI | 标普500净股票发行

8 | SMVAR | 股票市场（标普500）波动率

9 | TBL | 三个月国库券利率

10 | TMS | 期限利差

Continue: Equity Characteristics

No.	Characteristics	Description	Category
41	PCTACC	Percent operating accruals	Investment
42	PM	Profit margin	Profitability
43	PS	Performance Score	Profitability
44	RD_SALE	R&D to sales	Intangibles
45	RDM	R&D to market	Intangibles
46	RE	Revisions in analysts' earnings forecasts	Intangibles
47	RNA	Return on Net Operating Assets	Profitability
48	ROA	Return on Assets	Profitability
49	ROE	Return on Equity	Profitability
50	RSUP	Revenue surprise	Momentum
51	RVAR_CAPM	Idiosyncratic Volatility - CAPM (3 months)	Frictions
52	RVAR_FF3	Res. var. - Fama-French 3 factors (3 months)	Frictions
53	SVAR	Return variance (3 months)	Frictions
54	SEAS1A	1-Year Seasonality	Intangibles
55	SGR	Sales growth	Value-versus-growth
56	SP	Sales-to-price	Value-versus-growth
57	STD_DOLVOL	Std of dollar trading volume (3 months)	Frictions
58	STD_TURN	Std. of Share turnover (3 months)	Frictions
59	SUE	Unexpected quarterly earnings	Momentum
60	TURN	Shares turnover	Frictions
61	ZEROTRADE	Number of zero-trading days (3 months)	Frictions

Table A.2: **Macroeconomic Variables**

This table lists the description of macro variables used in the empirical study.

No.	Variable Name	Description
1	DFY	Default yield
2	DY	Dividend yield of S&P 500
3	EP	Earnings-to-price of S&P 500
4	ILL	Pastor-Stambaugh illiquidity
5	INFL	Inflation
6	LEV	Leverage of S&P 500
7	NI	Net equity issuance of S&P 500
8	SMVAR	Stock Market (S&P 500) Volatility
9	TBL	Three-month treasury bill rate
10	TMS	Term spread

表A.3：检验提升P-Tree扩展的子样本分析

本表报告表2结果的子样本分析。B面板对应1981—2000年前20年样本；C面板对应2001—2020年最近20年样本。

表头中译：Sharpe Ratio=夏普比率；Single=单个因子；Multiple=多个因子共同构成的组合；CAPM Test=CAPM检验；FF5 Test=FF5检验；Expanding Factors Test=扩展因子检验；BS Test=Barillas-Shanken检验；alpha（%）=回归截距；t-stat=t统计量；R^2=回归解释度；p-value=p值。1—20为依次生成的P-Tree因子。完整数值见随后原文页。

Table A.3: **Subsample Analysis for Testing the Boosted P-Tree Growth**

This table reports the subsample Analysis for results in Table 2. Panel B is for the first 20-year sample from 1981–2000, and panel C is for the recent 20-year sample from 2001-2020.

	Sharpe Ratio		CAPM Test		FF5 Test		Expanding Factors Test			BS Test
	Single	Multiple	α (%)	t-stat	α (%)	t-stat	α (%)	t-stat	R^2	p-value
					Panel B: 20 Years (1981-2000)					
1	7.12	7.12	1.86	31.23	1.78	28.53	-	-	-	-
2	1.09	7.75	0.25	4.09	0.11	1.91	-0.83	-4.80	0.28	0.00
3	2.26	9.45	0.39	9.66	0.34	8.30	0.92	8.19	0.17	0.00
4	6.48	11.36	1.80	28.41	1.76	26.07	1.75	10.16	0.03	0.00
5	1.59	12.71	0.33	6.54	0.28	5.37	1.11	6.85	0.36	0.00
6	1.65	13.67	0.37	6.83	0.26	4.74	-1.03	-5.34	0.40	0.00
7	1.72	14.94	0.33	7.20	0.32	6.44	1.25	5.67	0.20	0.00
8	1.52	15.78	0.39	6.19	0.31	4.75	-1.32	-5.12	0.35	0.00
9	3.02	17.89	1.03	13.04	0.93	11.56	2.68	8.42	0.26	0.00
10	1.93	19.18	0.83	8.14	0.67	7.18	-2.41	-6.27	0.52	0.00
11	1.34	20.72	0.38	5.36	0.19	2.85	-2.18	-7.33	0.45	0.00
12	2.22	22.89	0.36	9.49	0.31	8.24	1.46	7.36	0.39	0.00
13	1.51	24.14	0.35	6.15	0.23	4.07	1.82	6.06	0.43	0.00
14	2.11	25.71	0.58	8.97	0.48	7.27	-2.43	-5.93	0.40	0.00
15	2.39	28.38	0.87	10.21	0.76	9.10	3.69	7.64	0.52	0.00
16	2.53	30.37	0.53	10.84	0.44	8.85	2.40	7.12	0.31	0.00
17	1.50	32.42	0.30	6.10	0.31	5.78	2.65	7.17	0.35	0.00
18	1.63	33.89	0.37	6.76	0.28	4.99	-2.79	-5.64	0.37	0.00
19	1.84	35.78	0.47	7.71	0.37	5.95	3.54	7.34	0.30	0.00
20	2.70	37.93	0.89	11.62	0.79	10.21	-4.66	-6.90	0.44	0.00
					Panel C: 20 Years (2001-2020)					
1	5.82	5.82	1.51	25.60	1.47	24.01	-	-	-	-
2	2.14	6.64	0.58	9.23	0.52	8.04	0.91	6.44	0.03	0.00
3	0.91	7.53	0.18	3.37	0.09	1.85	-0.86	-5.82	0.36	0.00
4	1.35	8.60	0.29	5.58	0.31	5.72	1.00	8.65	0.17	0.00
5	1.08	9.30	0.31	4.24	0.26	3.46	-1.02	-6.38	0.42	0.00
6	2.63	10.36	0.60	11.41	0.56	10.31	1.04	7.61	0.12	0.00
7	1.39	11.31	0.29	5.75	0.26	4.90	1.00	7.23	0.28	0.00
8	1.60	11.93	0.40	6.71	0.30	5.26	-0.85	-5.62	0.44	0.00
9	1.04	12.73	0.21	4.08	0.18	3.64	-1.01	-7.45	0.36	0.00
10	1.64	14.32	0.30	6.92	0.25	5.67	1.11	7.80	0.44	0.00
11	1.69	15.73	0.36	7.15	0.30	5.94	1.43	6.79	0.33	0.00
12	1.58	16.82	0.60	6.65	0.51	5.64	2.49	5.72	0.24	0.00
13	1.48	18.07	0.41	6.16	0.29	4.61	-1.77	-5.52	0.47	0.00
14	1.39	19.46	0.32	5.73	0.26	4.61	1.68	6.05	0.40	0.00
15	1.48	20.59	0.35	6.15	0.21	4.19	-1.38	-6.25	0.58	0.00
16	1.40	21.91	0.34	5.78	0.27	4.61	1.84	6.29	0.46	0.00
17	1.27	23.40	0.39	5.18	0.25	3.39	-2.55	-6.99	0.51	0.00
18	1.44	24.36	0.34	5.99	0.28	4.99	-1.61	-4.97	0.48	0.00
19	1.58	25.45	0.33	6.62	0.28	5.56	-1.75	-5.21	0.42	0.00
20	1.88	26.52	0.58	8.02	0.50	6.90	-2.76	-4.96	0.39	0.00

《在面板树上扩展有效前沿》网络附录

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网络附录安排如下：第I节通过模拟展示P-Tree的优势；第II节说明P-Tree可灵活调整市场因子等基准因子；第III节采用最小化个股收益已实现定价误差的替代切分准则，进一步展示框架的灵活性，并在一个小节中将准则扩展到时间序列切分。

I 模拟

本节开展模拟研究，展示P-Tree选择特征的实际能力，以及P-Tree测试资产捕捉特征间非线性交互的效率。积极的模拟证据进一步支持P-Tree的建模能力。我们使用同一实证数据样本校准收益生成过程，并在不失一般性的情况下，考虑三个真实特征及其交互项、非线性项。此外，为模拟投资者的真实数据环境，还纳入大量与真实特征相关、但不参与收益生成的无用或噪声特征。

数据生成过程。具体而言，排序后的资产特征$z_{i,t}$是61×1向量，由VAR（1）模型模拟生成。这一设计保证冗余特征仍与真实特征相关。在信噪比较低或样本较小时，冗余特征可能被选中。真实收益生成过程假定只涉及61项中的三项特征：ME、BM与MOM12M。模型包含简单非线性和交互作用，具体见式（9）。

Internet Appendices for "Growing the Efficient Frontier on Panel Trees"

Lin William Cong Guanhao Feng Jingyu He Xin He

The online appendix is organized as follows. Section I uses simulation to demonstrate P-Tree's advantage. Section II illustrates the flexibility of P-Tree in adjusting for benchmark factors such as the market factor. Section III further reveals the framework's flexibility with an alternative split criterion that minimizes the realized pricing error of individual stock returns, with a subsection extending the criteria to split the time series.

I Simulation

In this section, simulation studies are conducted to demonstrate the practical capability of P-Tree in choosing characteristics and the efficiency of P-Tree test assets in capturing non-linear interactions among characteristics. The positive simulation evidence provides further support for the modeling ability of P-Tree. We calibrate the return-generating process using the same empirical data sample and consider three genuine characteristics and their interactions and nonlinear terms without loss of generality. Moreover, to simulate the true investor data environment, we consider a large set of useless or noisy characteristics that correlate with the true characteristics but do not impact generating returns.

Data-generating process. Specifically, the ranked asset characteristics $\mathbf{z}_{i,t}$ (a vector of size 61×1) are simulated from a VAR(1) model. This design ensures that even redundant characteristics still correlate with genuine ones. Redundant characteristics may be selected in scenarios of low signal-to-noise ratios or small sample sizes. The true return-generating process is assumed only to involve three characteristics among all 61 (`ME`, `BM`, and `MOM12M`). The model incorporates simple nonlinearity and interaction,

式（9）完整保留在随后原文页。其含义是：股票超额收益r_i,t等于市场超额收益mkt_t，加上由κ缩放的五项特征效应，再加误差ε_i,t。五项效应依次为c1·ME、c2·BM、c3·ME×BM、c4·MOM12M、c5·MOM12M²，各特征均使用t-1期值。

其中，MEBM为ME与BM的交互项，MOM12M²为二次项；误差独立同分布于均值0、方差σ²的正态分布。σ和系数c1至c5均用真实数据校准。κ取0.5、1、2，用于调整特征信号强度；κ=1对应实证研究中的实际数据。

每个模拟样本包含1000项资产，500期用于样本内分析，另500期用于样本外分析。表I.1报告10次重复的平均结果，包括P-Tree测试资产的最大夏普比率及市场alpha。前者展示有效前沿上的切点组合，后者展示市场因子对这些资产的定价难度。

传统特征排序组合。表I.1的DECILE AVG（30）报告按三个真实特征分别构造十分位组合所得的三组MVE组合与测试资产的平均表现。κ=1时，夏普比率为3.12，平均绝对alpha为0.39%，alpha均方根为0.47%。由于预先指定真实特征，样本外结果也相当稳定。在不知道真实特征的情况下，对全部61项特征使用DECILE AVG（610），夏普比率降至0.97，两项alpha指标降至0.10%与0.12%；样本外夏普比率更低。比较不同κ可见，较大κ对应较高的夏普比率与alpha。

式（9）的收益生成过程包含交互和二次项。因此，我们还考察基于真实特征的5×5双变量排序组合。这些组合持续取得高于十分位组合的夏普比率与alpha，显示出更优表现。结果说明，考虑潜在特征交互能够获得更多定价信息。

as described below:

$$r_{i,t} = \texttt{mkt}_t + \kappa\left[c_1\texttt{ME}_{i,t-1} + c_2\texttt{BM}_{i,t-1} + c_3\texttt{MEBM}_{i,t-1} + c_4\texttt{MOM12M}_{i,t-1} + c_5\texttt{MOM12M}^2_{i,t-1}\right] + \epsilon_{i,t}, \tag{9}$$

where $r_{i,t}$ represents the stock excess return, $\texttt{mkt}_t$ denotes the market excess return, $\texttt{MEBM}$ is the interaction of $\texttt{ME}$ and $\texttt{BM}$, $\texttt{MOM12M}^2$ is the quadratic term, and the error terms are simulated from $\epsilon_{i,t} \sim i.i.d.N(0, \sigma^2)$. The parameter σ is calibrated from real data, and so is $[c_1, \cdots, c_5]$. The parameter $\kappa = [0.5, 1, 2]$ allows us to adjust the strength of the characteristic signal, with the value of $\kappa = 1$ representing our actual data in the empirical study.

Each simulation sample consists of 1,000 assets, 500 periods for in-sample analysis, and another 500 periods for out-of-sample analysis. Table I.1 presents the average results of 10 repetitions, including the maximal Sharpe ratios and market alphas of P-Tree test assets. The former displays the tangency portfolio on the efficient frontier, whereas the latter demonstrates the pricing difficulty of the market factor.

Conventional characteristics-sorted portfolios. In Table I.1, "DECILE AVG(30)" presents the average performance of three MVE portfolios and test assets for decile portfolios sorted on three true characteristics. This configuration yields a Sharpe ratio of 3.12, an average absolute α of 0.39%, and a root mean squared α of 0.47% when $\kappa = 1$. Meanwhile, the performance of "DECILE AVG(30)" is quite consistent in out-of-sample results, because of the pre-specified true characteristics. Without knowing the true characteristics, "DECILE AVG(610)" on 61 characteristics reduces the Sharpe ratio to 0.97 and the alphas to 0.10% and 0.12%, whereas the out-of-sample results reveal lower Sharpe ratios. Across the different κ's, we find higher Sharpe ratios and alphas for large κ than for small κ cases.

The return-generating process described in (9) involves interaction and quadratic terms. We also examine bivariate-sorted portfolios (5×5) using the true characteristics. These portfolios consistently yield higher Sharpe ratios and alphas than decile portfolios, demonstrating their superior performance. Our findings indicate that accounting for potential characteristic interactions can lead to pricing information gains.

表I.1：模拟表现比较

本表报告不同测试资产集合的最大夏普比率、平均绝对市场alpha（%）及alpha均方根（%），数值为10次模拟的平均结果。最多检验20棵P-Tree，每棵包含10个叶基础组合。括号内数字为平均叶基础组合数。基准包括：（1）市场因子；（2）按三个真实特征构造的三组十分位组合的平均结果；（3）按61项特征构造的61组十分位组合的平均结果；（4—6）分别按ME-BM、ME-MOM12M、BM-MOM12M两项真实特征构造的5×5双变量排序组合。

A面板=样本内结果；B面板=样本外结果。κ=0.5、1.0、2.0为三种信号强度。SR=夏普比率；其后两列分别为平均绝对alpha与alpha均方根。全部数值见随后原表。

P-Tree测试资产。P-Tree为横截面切分、测试资产创建和有效前沿张成提供内生解决方法。表I.1中，κ=1时，P-Tree1取得较高的样本内夏普比率6.14、平均绝对alpha 0.73%、alpha均方根0.94%，以及样本外夏普比率5.61、平均绝对alpha 0.88%、alpha均方根1.06%。κ增加至2时，样本内与样本外的夏普比率和alpha都更大。同时，第一棵P-Tree较为简约，达到相近夏普比率所需资产少于替代方法。

Table I.1: **Simulation Performance Comparison**

The table presents the maximal Sharpe ratios, average absolute market α (%), and root mean squared α (%) for different sets of test assets. The average results are reported based on 10 simulations. We test up to 20 P-Trees, each containing 10 leaf basis portfolios. The numbers in parentheses represent the average number of leaf basis portfolios. Benchmark results include the following specifications: (1) market factor; (2) average results for 3 sets of decile portfolios of true char.; (3) average results for 61 sets of decile portfolios; and (4 - 6) bivariate-sort (5×5) portfolios on either two true char., including `ME` and `BM`, `ME` and `MOM12M`, and `BM` and `MOM12M`.

	$\kappa = 0.5$			$\kappa = 1.0$			$\kappa = 2.0$		
	SR	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$	SR	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$	SR	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$
Panel A: In-Sample Results									
P-Tree1	3.31	0.43	0.54	6.14	0.73	0.94	12.00	1.54	1.89
P-Tree1-5	4.47	0.24	0.33	7.58	0.34	0.52	14.23	0.63	1.00
P-Tree1-10	6.15	0.19	0.27	9.75	0.26	0.42	17.45	0.49	0.79
P-Tree1-15	7.86	0.17	0.24	12.04	0.23	0.37	21.11	0.41	0.68
P-Tree1-20	9.76	0.16	0.22	14.49	0.22	0.34	25.45	0.37	0.61
MKT(1)	0.55	0.00	0.00	0.55	0.00	0.00	0.55	0.00	0.00
DECILE AVG(30)	1.73	0.21	0.25	3.12	0.39	0.47	5.97	0.76	0.92
DECILE AVG(610)	0.81	0.08	0.09	0.97	0.10	0.12	1.36	0.15	0.19
ME-BM(25)	2.63	0.31	0.40	4.92	0.57	0.78	9.60	1.12	1.54
ME-MOM12M(25)	2.42	0.29	0.37	4.49	0.55	0.70	8.74	1.09	1.37
BM-MOM12M(25)	2.95	0.35	0.46	5.60	0.68	0.89	11.00	1.35	1.75
Panel B: Out-of-Sample Results									
P-Tree1	2.66	0.61	0.72	5.61	0.88	1.06	11.38	1.56	1.90
P-Tree1-5	2.35	0.58	0.63	5.27	0.68	0.77	10.80	0.89	1.14
P-Tree1-10	2.09	0.59	0.62	4.90	0.65	0.72	10.21	0.77	0.98
P-Tree1-15	1.95	0.59	0.62	4.69	0.63	0.70	9.90	0.72	0.91
P-Tree1-20	1.90	0.59	0.62	4.55	0.63	0.68	9.58	0.70	0.87
P-Tree16-20	0.80	0.59	0.61	1.60	0.61	0.65	3.12	0.64	0.73
MKT(1)	0.45	0.00	0.00	0.45	0.00	0.00	0.45	0.00	0.00
DECILE AVG(30)	1.55	0.20	0.25	3.03	0.39	0.48	6.00	0.77	0.94
DECILE AVG(610)	0.49	0.08	0.09	0.70	0.10	0.13	1.17	0.16	0.20
ME-BM(25)	2.23	0.29	0.41	4.58	0.56	0.78	9.20	1.12	1.54
ME-MOM12M(25)	2.20	0.31	0.39	4.40	0.58	0.72	8.70	1.12	1.41
BM-MOM12M(25)	2.72	0.36	0.46	5.41	0.69	0.89	10.73	1.35	1.76

P-Tree test assets. P-Trees are an endogenous solution for cross-sectional splitting, test asset creation, and efficient frontier spanning. In Table I.1, the P-Tree test assets ("P-Tree1") exhibit an impressive in-sample Sharpe ratio (6.14), absolute alpha (0.73%), squared alpha (0.94%), out-of-sample Sharpe ratio (5.61), absolute alpha (0.88%), and squared alpha (1.06%) when $\kappa = 1$. When the κ increases to 2, we observe larger Sharpe ratios and alphas for both in-sample and out-of-sample results. Meanwhile, the first P-Tree is parsimonious, requiring fewer assets than alternative approaches to

例如，第一棵P-Tree仅需10项资产即可取得5.61的样本外夏普比率，而使用真实特征的双变量排序组合需要25项资产才能取得相近水平。此外，第一棵P-Tree测试资产的样本外alpha也高于其他基准测试资产。

提升P-Tree表示为P-Tree1-5至P-Tree1-20。尽管加入提升因子使样本内夏普比率上升，但从第一棵到第20棵树，alpha下降。因此，P-Tree在较早提升步骤捕捉主要信息，随后提供的信息逐渐减少。所有P-Tree设置在样本外都表现出更低夏普比率和更高alpha；P-Tree1提供最大的样本外夏普比率和alpha。这确认：在式（9）这种仅含三个真实特征的简单收益生成过程中，较早的P-Tree捕捉了更多信息。

总之，在夏普比率和alpha等风险调整或基于模型的投资标准下，P-Tree测试资产优于大多数预先指定的排序组合设置。预测信号 κ 越大，优势越明显。在式（9）的交互、非线性收益生成情景中，P-Tree相对于所有传统排序组合均具有竞争力。

信息不完全的P-Tree。文献记载了数百项特征、因子和异象（Cochrane，2011；Green等，2017；Hou等，2020；Feng等，2020；Bryzgalova等，2023）。要完整张成有效前沿，必须包含所有有价值的特征；但我们不知道哪些特征确实有用，也不知道是否仍有尚未发现的有用特征。

因此，在式（9）的收益生成过程中，我们考察未观察到全部真实特征时的P-Tree表现。具体做法是忽略MOM12M并重新训练P-Tree，模拟结果见表I.2。遗漏一个真实特征后，样本内和样本外的夏普比率及alpha均明显下降。

achieve comparable Sharpe ratios; for example, the first P-Tree only needs 10 assets to achieve an out-of-sample Sharpe ratio of 5.61, whereas the alternative bivariate-sorted portfolios on true characteristics require 25 assets to get comparable Sharpe ratios. Further, the test assets of the first P-Tree have larger out-of-sample alphas than other benchmark test assets.

The boosted P-Trees are denoted as "P-Tree1-5" to "P-Tree1-20." Although the in-sample Sharpe ratio increases when adding boosted factors, the alphas are lower from the first to the 20-th P-Tree. Therefore, P-Trees capture the important information in early boosting steps, whereas the subsequent P-Trees provide declining information. We observe lower Sharpe ratios and higher alphas for out-of-sample results for all P-Tree specifications. Meanwhile, "P-Tree1" provides the largest out-of-sample Sharpe ratios and alphas. This finding confirms the early P-Trees capture more information, given such a simple return-generating process involving only three true characteristics as in (9).

In conclusion, the P-Tree test assets outperform most pre-specified sorted portfolio specifications in risk-adjusted or model-based investment criteria, such as Sharpe ratios and alphas. The outperformance is more pronounced when the predictive signal κ is larger. In this interactive and nonlinear return-generating scenario (9), P-Tree performances are competitive with all the conventional sorted portfolios.

P-Tree with incomplete information. The literature documents hundreds of characteristics, factors, and anomalies (Cochrane, 2011; Green et al., 2017; Hou et al., 2020; Feng et al., 2020; Bryzgalova et al., 2023). We must include all the valuable characteristics to span the efficient frontier fully. However, we still do not know whether all the characteristics are helpful. Also, we are agnostic if more valuable characteristics are still not found.

To this end, we investigate the performance of P-Tree without observing the full set of true characteristics, given the return-generating process (9). Specifically, we ignore `MOM12M` and re-train P-Trees. Table I.2 reports the simulation results. Once we ignore one of the true characteristics, we find a considerable decline in Sharpe ratios and alphas for both in-sample and out-of-sample analysis. Also, the alphas of P-Tree

表I.2：遗漏MOM12M特征时P-Tree的模拟表现

本表报告遗漏真实特征MOM12M后P-Tree测试资产的夏普比率和alpha。格式同表I.1。A面板为样本内，B面板为样本外；三组列对应 κ=0.5、1.0、2.0，组内依次为夏普比率、平均绝对alpha和alpha均方根。完整数值见随后原文页。

忽略MOM12M后，P-Tree测试资产的alpha更小。因此，缺少关键特征会使P-Tree效率下降。实证应用中，我们并不知道真实经济的收益生成过程，所以纳入许多特征，以减轻遗漏关键特征的担忧。

包含真实特征的P-Tree。即使不预先指定真实特征，P-Tree也能生成信息丰富的测试资产，获得较高夏普比率，并相对于市场因子产生较大定价误差。因此，我们推测P-Tree能够在高维空间中正确选择真实特征。表I.3列出各特征在P-Tree切分中被选中的概率。为简化展示，仅报告10次模拟中第一棵P-Tree的结果。真实特征ME、BM、MOM12M在前三次、前五次以及全部九次切分中均以较高概率入选。

κ 越大，选中真实特征的概率越高。在低信噪比 κ=0.5时，噪声特征也会进入靠前切分，例如NI和BASPREAD在前五次切分中的选择概率非零。而在高信噪比 κ=2时，10次重复中的前五次切分只选择真实特征。因此，我们确认，P-Tree的优良表现源自对式（9）收益生成过程中真实特征的正确识别。

Table I.2: **Simulation Performance for P-Tree Omitting a Characteristic MOM12M**

This table shows the Sharpe ratios and alphas of P-Tree test assets omitting a true characteristic MOM12M. Table format follows Table I.1.

	$\kappa = 0.5$			$\kappa = 1.0$			$\kappa = 2.0$		
	SR	$\lvert\alpha\rvert$	$\sqrt{\alpha^2}$	SR	$\lvert\alpha\rvert$	$\sqrt{\alpha^2}$	SR	$\lvert\alpha\rvert$	$\sqrt{\alpha^2}$
Panel A: In-Sample Results									
P-Tree1	2.73	0.32	0.41	4.86	0.60	0.77	9.42	1.30	1.66
P-Tree1-5	4.00	0.18	0.26	6.20	0.26	0.40	11.52	0.47	0.81
P-Tree6-10	1.66	0.12	0.15	2.35	0.16	0.21	3.14	0.23	0.31
P-Tree11-15	1.71	0.12	0.16	2.19	0.16	0.21	3.45	0.26	0.35
P-Tree16-20	1.52	0.13	0.16	1.71	0.15	0.19	3.16	0.24	0.31
Panel B: Out-of-Sample Results									
P-Tree1	1.91	0.59	0.68	4.23	0.76	0.94	8.91	1.36	1.72
P-Tree1-5	1.54	0.59	0.63	3.83	0.63	0.70	8.28	0.77	0.99
P-Tree6-10	0.48	0.60	0.62	1.38	0.61	0.64	2.73	0.60	0.67
P-Tree11-15	0.82	0.60	0.62	1.45	0.61	0.64	2.95	0.63	0.71
P-Tree16-20	0.69	0.60	0.62	1.28	0.59	0.63	2.80	0.58	0.66

test assets are smaller, ignoring MOM12M. Therefore, P-Tree would be inefficient if it were missing some key characteristics. In empirical applications, we are agnostic regarding the return-generating process of the real economy. Therefore, we cover many characteristics to relieve the concern about missing key ones.

P-Tree with true characteristics. Even without pre-specifying the true characteristics, P-Tree can produce informative test assets to achieve high Sharpe ratios and large pricing errors against the market factor. Thus, we conjecture that P-Tree can correctly select the true characteristics in a high-dimensional space. Table I.3 tabulates the probability of selecting each characteristic in P-Tree splits. For simplicity, we only report for the first P-Tree over 10 simulation repetitions. We find the true characteristics, ME, BM, and MOM12M, are selected with high probabilities in the top three, top five, and all nine splits.

When κ is large, the probability of selecting the true characteristics is higher. For low signal-to-noise setting $\kappa = 0.5$, the noisy characteristics are selected in top splits; for example, NI and BASPREAD are selected with non-zero probabilities in the top five splits. However, for high signal-to-noise setting $\kappa = 2$, only the true characteristics are selected in the top five splits, over 10 simulation repetitions. Therefore, we confirm

表I.3：第一棵P-Tree的特征选择概率

本表展示10次重复中第一棵P-Tree所选特征及其概率。每棵树包含九次切分，因而有十个叶。“Top3”只统计前三次切分，Top5与Top9类似。每行仅展示选择最频繁的五项特征。例如，A面板前三次切分中，BM以40%的概率最常被选择。A、B、C面板分别对应 $\kappa = 0.5$、1、2。左侧为表I.1中完整特征信息的P-Tree，右侧为表I.2中遗漏MOM12M的P-Tree。

表头Complete Characteristics Set=完整特征集合；Remove MOM12M=移除MOM12M。数值和排名见随后原表。

进一步地，我们考察遗漏重要特征MOM12M时的选择结果，与表I.2的处理相同。移除MOM12M后，BM和ME入选概率更高；更重要的是，噪声特征也更频繁地被选中。但如表I.2所示，噪声特征不能帮助P-Tree完整张成有效前沿。这一现象提醒我们，必须向P-Tree提供全部真实特征。

Table I.3: **Characteristics Selection Probability for the First P-Tree**

This table shows the selected characteristics of the first P-Tree with its selection probability over 10 repetitions. Each P-Tree contains nine splits and, thus, ten leaves. The "Top3" columns specify that we only count the first three splits of the P-Tree model, and similarly, we defined "Top5" and "Top9." The rows only show the five most frequently selected characteristics. For example, in Panel A, the characteristic BM is the most frequently selected, with a 40% probability, in the top three splits. Panels A, B, and C report for $\kappa = 0.5, 1, 2$. The left panel reports for P-Trees with complete characteristics information as in Table I.1, whereas the right panel is for P-Trees omitting MOM12M as in Table I.2.

	Complete Characteristics Set					Remove MOM12M				
	1	2	3	4	5	1	2	3	4	5
					Panel A: $\kappa = 0.5$					
Top3	BM	MOM12M	ME			ME	BM			
	0.40	0.33	0.27			0.63	0.37			
Top5	BM	ME	MOM12M	NI	BASPREAD	ME	BM	BASPREAD	HIRE	SEAS1A
	0.40	0.34	0.22	0.02	0.02	0.46	0.38	0.02	0.02	0.02
Top9	BM	ME	MOM12M	ALM	CFP	ME	BM	ILL	RNA	CASHDEBT
	0.33	0.26	0.12	0.02	0.02	0.29	0.26	0.03	0.02	0.02
					Panel B: $\kappa = 1$					
Top3	BM	MOM12M	ME			ME	BM			
	0.37	0.33	0.30			0.60	0.40			
Top5	BM	ME	MOM12M			BM	ME	DEPR	TURN	SVAR
	0.44	0.32	0.24			0.48	0.44	0.02	0.02	0.02
Top9	BM	ME	MOM12M	CHCSHO	RE	ME	BM	TURN	DEPR	ILL
	0.36	0.29	0.16	0.03	0.02	0.40	0.34	0.03	0.02	0.02
					Panel C: $\kappa = 2$					
Top3	BM	MOM12M	ME			ME	BM			
	0.37	0.33	0.30			0.67	0.33			
Top5	BM	ME	MOM12M			BM	ME			
	0.48	0.28	0.24			0.60	0.40			
Top9	BM	ME	MOM12M	MOM36M	CFP	ME	BM	CFP	SP	TURN
	0.43	0.27	0.19	0.02	0.01	0.47	0.42	0.02	0.01	0.01

that the superb performance of P-Trees originates from correctly selecting the true characteristics in the return-generating process (9).

Further, we investigate the characteristics selection results when omitting an important characteristic MOM12M, which is the same as the treatment in Table I.2. We find BM and ME are selected with an even higher probability, once we omit MOM12M. More importantly, we find noisy characteristics are selected more frequently. However, the noisy characteristics cannot help P-Tree fully span the efficient frontier, as shown in Table I.2. This phenomenon alerts us that we must feed all true characteristics to P-Tree.

II 经基准调整的P-Tree

如第2.3节所述，P-Tree提升方法可以灵活纳入任意基准因子，例如CAPM、Fama-French三因子和PCA因子。这里展示以市场因子为基准模型拟合提升P-Tree的例子。具体而言，在提升切分准则式（6）中，定义F=[mkt_t，f_1,t，⋯，f_K,t]。

实证结果中，即使将市场因子作为基准，初始P-Tree仍保持不变，说明横截面特征在市场因子之外具有信息价值。因此，树图仍与图4相同。但后续P-Tree变化很大，实证表现也不同于第3节。以下图表报告包含市场因子基准的P-Tree测试资产与因子的实证结果。

图I.1描绘P-Tree因子和市场因子共同张成的有效前沿。其模式与图7中不含市场基准的主要结果相似：随着提升树增多，P-Tree前沿向均值—方差图左上角移动。不过，市场因子在切点组合中仅占很小比例，第一棵P-Tree与市场因子共同形成的前沿较平。

表I.4展示P-Tree测试资产的定价表现。与表3主要结果一致，第一棵树测试资产的GRS统计量、平均alpha和显著alpha比例均高于后续树生成的资产。此外，除2001—2020年子样本中的P-Tree11-15外，GRS和PY检验始终拒绝原假设。总体而言，包含市场基准的P-Tree测试资产难以被FF5定价。

表I.5考察每个P-Tree因子的表现。与表2一致，较低的p值表明，提升P-Tree因子无法由此前P-Tree因子解释。与表2中不能检验第一个因子的情况不同，这里可以相对于市场基准检验第一个P-Tree因子。

II Benchmark-Adjusted P-Trees

As described in section 2.3, P-Tree boosting is a flexible approach to include any benchmark factor(w), such as CAPM, Fama-French three factors, and PCA factors. Here, we demonstrate an example of fitting boosted P-Trees with the market factor as a benchmark factor model. Specifically, we define $\mathbf{F} = [\text{mkt}_t, f_{1,t}, \cdots, f_{K,t}]$ in the boosting split criteria (6).

The initial P-Tree remains unchanged in the empirical findings, even when the market factor is included as a benchmark, suggesting the information usefulness of the cross-sectional characteristics beyond the market factor. Therefore, the P-Tree diagram remains identical to Figure 4. However, the subsequent P-Trees change significantly, and the empirical performances differ from those in section 3. The following tables and figures present the empirical results for the P-Tree test assets and factors with the market factor benchmark.

Figure I.1 depicts the efficient frontiers spanned by P-Tree factors and the market factor. We find similar patterns as our main results without a market benchmark in Figure 7. The P-Tree frontiers move to the top-left corner of the mean-variance diagram, as more boosted P-Trees are generated. However, the market factor only takes a small proportion of the tangency portfolio, and the frontier is flat for the first P-Tree and the market factors.

Table I.4 shows the asset pricing performance of P-Tree test assets. Consistent with our main results in Table 3, we find the test assets of the first P-Tree have larger GRS statistics, higher average alphas, and higher proportions of significant alphas than the test assets generated by the follow-up P-Trees. Additionally, the GRS and PY tests always reject the null hypothesis, except for P-Tree11-15 in subsample 2001-2020. In general, the test assets of P-Trees with a market factor benchmark are challenging to be priced by FF5.

Table I.5 investigates the performance of each P-Tree factor. Consistent with our main results in Table 2, the boosted P-Tree factors cannot be explained by the previous P-Tree factors, as evidenced by the low p-values. In contrast to the results in Table

正如预期，第一个P-Tree因子无法由市场因子解释。

表I.6报告投资表现。加入市场因子略微改善了风险调整投资表现。原文称，在2001—2020年和1981—2020年的样本外期间，投资夏普比率超过3。所列可观测及潜在因子模型仍无法解释P-Tree因子，其无法解释的alpha较大。

〔译注：此处保留原文表述，但表I.6的另一组样本外期间标为1981—2000年，且B2若干多因子策略的夏普比率低于3。核对时应以原表具体行列为准。〕

总之，本节展示P-Tree提升过程的灵活性：构造P-Tree测试资产与因子时，可以将任意给定因子模型作为控制基准加入。

2, where the first factor could not be tested, we test the first P-Tree factor against the market benchmark. As expected, the first P-Tree factor cannot be explained by the market factor.

Table I.6 reports the investment performances. Adding the market factor slightly improves the risk-adjusted investment performance. The investment Sharpe ratios are over 3 for the out-of-sample periods 2001-2020 and 1981-2020. Again, those listed observable and latent factor models cannot explain our P-Tree factors with large unexplained alphas. In conclusion, this section demonstrates the flexibility of boosting in P-Trees: one can add any given factor model as a control when building P-Tree test assets and factors.

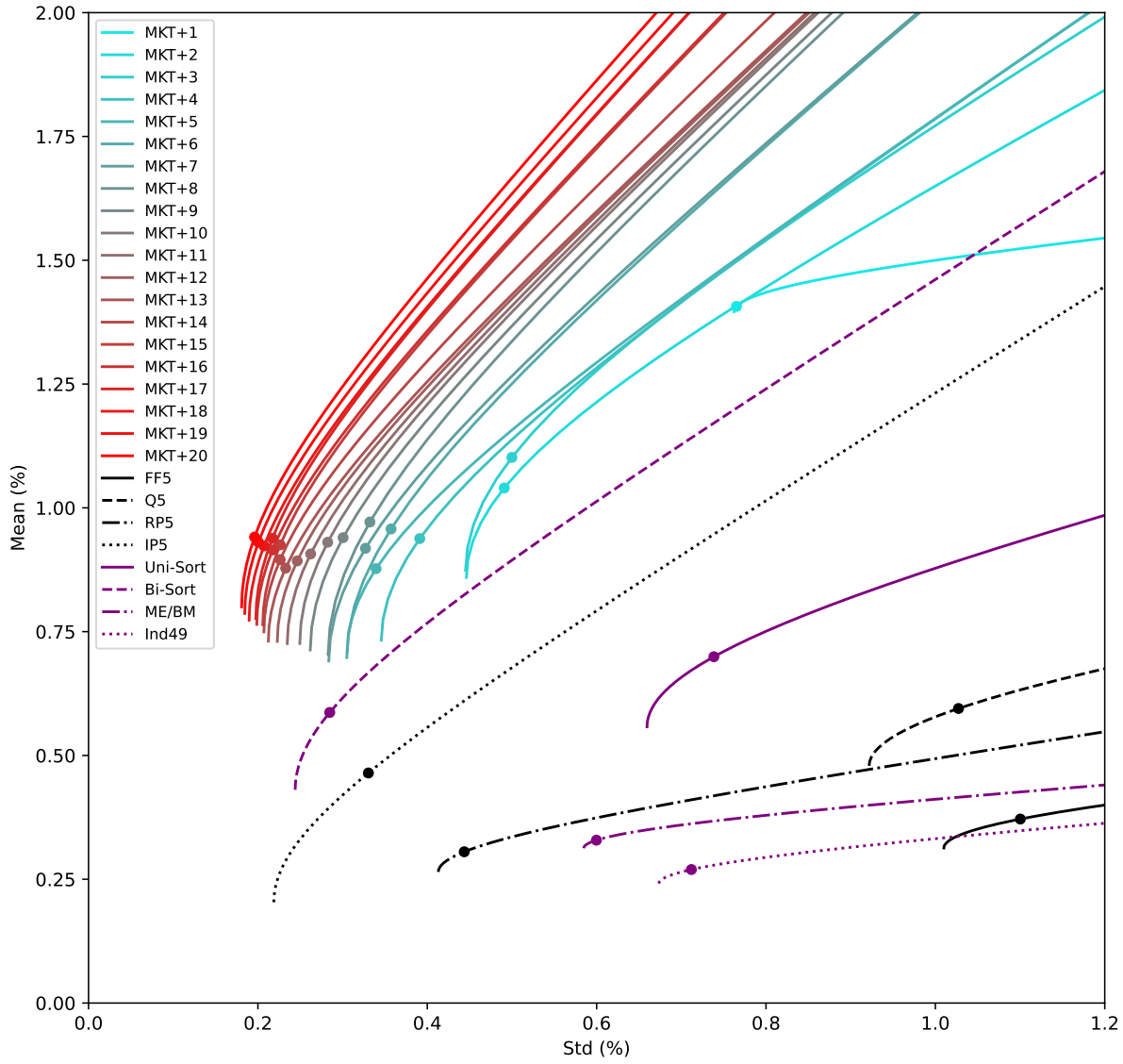
图I.1：市场因子基准——刻画有效前沿

本图补充图7，展示在P-Tree生长过程中考虑市场因子基准的结果。

纵轴Mean（%）=月度均值；横轴Std（%）=月度标准差。MKT+1至MKT+20表示市场因子与前1至20个P-Tree因子共同形成的前沿。FF5、Q5、RP5、IP5为基准五因子模型；Uni-Sort=单变量排序组合；Bi-Sort=双变量排序组合；ME/BM=市值—账面市值比排序组合；Ind49=49个行业组合。曲线、切点与坐标数值完整保留于随后原图。

Figure I.1: **Market Factor Benchmark: Characterizing the Efficient Frontier**

This figure presents complementary results to Figure 7 for considering the market factor benchmark while growing P-Trees.



表I.4：市场因子基准——P-Tree生成的测试资产

本表补充表3，展示在P-Tree生长过程中考虑市场因子基准的结果。

A面板：1981—2020年40年；B面板：1981—2000年20年；C面板：2001—2020年20年。P-Tree行名指对应树的叶组合集合。

表头依次为：测试资产数N、GRS统计量、GRS检验p值、PY检验p值、平均绝对alpha、alpha均方根、平均时间序列R^2、在10%/5%/1%水平上具有显著alpha的资产比例。单位及计算方式同表3。完整数值见随后原表。

Table I.4: **Market Factor Benchmark: Test Assets Generated by P-Tree**

This table presents complementary results to Table 3 for considering the market factor benchmark while growing P-Trees.

	N	GRS	p-GRS	p-PY	$\overline{\lvert\alpha\rvert}$	$\sqrt{\overline{\alpha^2}}$	$\overline{R^2}$	$\%\alpha_{10\%}$	$\%\alpha_{5\%}$	$\%\alpha_{1\%}$
Panel A: 40 Years (1981-2020)										
P-Tree1	10	141.27	0.00	0.00	0.92	1.11	75	100	90	80
P-Tree1-5	50	57.09	0.00	0.00	0.39	0.59	78	64	52	46
P-Tree6-10	50	4.99	0.00	0.00	0.24	0.31	79	48	38	24
P-Tree11-15	50	6.82	0.00	0.00	0.39	0.52	71	50	40	30
P-Tree16-20	50	3.74	0.00	0.00	0.27	0.36	72	48	40	32
P-Tree1-20	200	39.88	0.00	0.00	0.32	0.46	75	53	43	33
Panel B: 20 Years (1981-2000)										
P-Tree1	10	84.36	0.00	0.00	1.58	1.95	70	90	80	80
P-Tree1-5	50	50.61	0.00	0.00	0.65	1.04	77	54	50	42
P-Tree6-10	50	9.05	0.00	0.00	0.46	0.66	75	52	42	26
P-Tree11-15	50	3.55	0.00	0.00	0.29	0.38	76	46	36	28
P-Tree16-20	50	10.01	0.00	0.00	0.46	0.59	78	74	64	48
P-Tree1-20	200	118.72	0.00	0.00	0.47	0.71	77	56	48	36
Panel C: 20 Years (2001-2020)										
P-Tree1	10	56.76	0.00	0.00	1.09	1.35	68	90	90	90
P-Tree1-5	50	31.26	0.00	0.00	0.35	0.66	78	32	30	24
P-Tree6-10	50	3.55	0.00	0.00	0.28	0.43	80	38	28	18
P-Tree11-15	50	1.12	0.28	0.82	0.15	0.21	77	16	6	2
P-Tree16-20	50	1.76	0.00	0.02	0.25	0.36	76	24	12	6
P-Tree1-20	200	51.47	0.00	0.00	0.26	0.45	78	28	19	13

表I.5：市场因子基准——检验提升P-Tree带来的扩展

本表补充表2，展示在P-Tree生长过程中考虑市场因子基准的结果。A面板对应1981—2020年40年样本。

表头中译：Sharpe Ratio=夏普比率；Single=单个因子；Cumu.=累计多因子组合；CAPM Test=CAPM检验；FF5 Test=FF5检验；Expanding Factors Test=扩展因子检验；BS Test=Barillas-Shanken检验；alpha（%）=回归截距；t-stat=t统计量；R^2=回归解释度；p-value=p值。1—20为依次生成的因子，完整数值见随后原表。

Table I.5: **Market Factor Benchmark: Testing the Boosted P-Tree Growth**

This table presents complementary results to Table 2 for considering the market factor benchmark while growing P-Trees.

	Sharpe Ratio		CAPM Test		FF5 Test		Expanding Factors Test			BS Test
	Single	Cumu.	α (%)	t-stat	α (%)	t-stat	α (%)	t-stat	R^2	p-value
					Panel A: 40 Years (1981-2020)					
1	6.37	6.37	1.39	39.63	1.37	37.48	1.39	39.63	0.01	0.00
2	2.24	7.33	0.40	13.61	0.37	12.65	0.66	10.99	0.13	0.00
3	0.65	7.61	0.12	2.36	-0.03	-0.80	-0.62	-5.65	0.66	0.00
4	1.57	8.29	0.27	9.25	0.26	8.42	0.58	8.51	0.25	0.00
5	1.63	8.92	0.46	9.62	0.42	8.81	0.97	8.06	0.17	0.00
6	0.72	9.24	0.10	3.02	0.05	1.69	-0.47	-5.59	0.70	0.00
7	1.40	9.66	0.36	8.09	0.29	6.58	0.74	6.37	0.29	0.00
8	0.83	10.04	0.17	4.02	0.09	2.20	-0.66	-5.93	0.55	0.00
9	1.39	10.72	0.29	8.03	0.25	6.94	0.77	7.78	0.36	0.00
10	1.37	11.27	0.36	7.88	0.27	6.24	0.93	6.86	0.31	0.00
11	1.22	11.82	0.37	6.87	0.40	7.34	1.14	6.59	0.31	0.00
12	1.24	12.36	0.36	6.99	0.28	5.41	1.09	6.49	0.35	0.00
13	1.27	12.87	0.40	7.22	0.25	4.85	-1.02	-6.07	0.46	0.00
14	1.64	13.47	0.53	9.68	0.48	8.61	1.12	6.58	0.44	0.00
15	0.69	13.88	0.10	2.73	0.05	1.56	-0.68	-5.40	0.73	0.00
16	1.16	14.31	0.41	6.47	0.29	5.18	-1.00	-5.16	0.58	0.00
17	0.80	14.67	0.15	3.71	0.11	2.62	-0.79	-4.84	0.55	0.00
18	1.69	15.09	0.34	10.07	0.28	8.51	0.71	5.24	0.33	0.00
19	1.37	15.59	0.33	7.92	0.29	6.68	-1.00	-6.07	0.40	0.00
20	1.08	16.09	0.23	5.85	0.15	3.72	-0.87	-5.58	0.52	0.00

表I.5（续）：市场因子基准——检验提升P-Tree带来的扩展

B面板对应1981—2000年20年样本；C面板对应2001—2020年20年样本。

列设置与上一页一致：单因子和累计组合夏普比率；CAPM和FF5检验的alpha及t统计量；扩展因子检验的alpha、t统计量与R^2；BS检验p值。完整数值见随后原表。

Table Continued: Market Factor Benchmark: Testing the Boosted P-Tree Growth

	Sharpe Ratio		CAPM Test		FF5 Test		Expanding Factors Test			BS Test
	Single	Cumu.	α (%)	t-stat	α (%)	t-stat	α (%)	t-stat	R^2	p-value
					Panel B: 20 Years (1981-2000)					
1	7.12	7.12	1.86	31.23	1.78	28.53	1.86	31.23	0.01	0.00
2	0.71	8.02	0.12	1.89	-0.01	-0.08	-0.93	-7.26	0.79	0.00
3	3.17	9.43	0.74	13.75	0.74	12.78	1.14	8.72	0.09	0.00
4	0.85	10.49	0.19	2.76	0.13	1.75	-1.20	-7.12	0.58	0.00
5	1.83	12.14	0.39	7.65	0.30	5.71	1.26	8.55	0.29	0.00
6	1.48	13.61	0.33	6.02	0.34	5.89	1.38	7.52	0.32	0.00
7	0.96	14.61	0.20	3.43	0.05	0.87	-1.14	-5.91	0.62	0.00
8	2.14	16.17	0.70	9.08	0.62	7.80	2.24	7.09	0.20	0.00
9	1.30	17.37	0.32	5.12	0.18	3.41	-1.45	-6.00	0.47	0.00
10	3.50	19.16	0.92	15.19	0.82	13.42	2.03	7.10	0.21	0.00
11	1.02	20.95	0.21	3.73	0.21	3.55	1.95	6.76	0.52	0.00
12	1.36	22.02	0.29	5.54	0.21	4.00	-1.40	-5.07	0.52	0.00
13	1.23	24.02	0.22	4.81	0.19	3.76	1.89	6.71	0.39	0.00
14	1.17	25.66	0.26	4.50	0.15	2.60	-2.02	-5.83	0.51	0.00
15	1.00	27.74	0.14	3.65	0.11	2.74	1.73	6.36	0.57	0.00
16	0.91	29.86	0.14	3.12	0.02	0.45	-1.73	-6.24	0.74	0.00
17	4.13	32.95	0.73	18.02	0.65	16.36	2.28	7.48	0.38	0.00
18	1.57	35.18	0.27	6.43	0.24	6.07	2.22	6.00	0.41	0.00
19	1.53	38.25	0.36	6.23	0.29	5.15	-3.28	-6.91	0.56	0.00
20	1.79	41.12	0.39	7.49	0.28	5.36	3.30	7.16	0.57	0.00
					Panel C: 20 Years (2001-2020)					
1	5.82	5.82	1.51	25.60	1.47	24.01	1.51	25.60	0.01	0.00
2	1.39	7.13	0.34	5.74	0.34	5.86	0.97	9.42	0.33	0.00
3	0.98	8.23	0.22	3.75	0.20	3.23	0.97	8.00	0.43	0.00
4	0.63	9.03	0.11	1.73	0.03	0.52	-0.95	-6.55	0.76	0.00
5	0.78	9.61	0.12	2.72	0.06	1.57	-0.59	-5.47	0.63	0.00
6	1.51	10.58	0.44	6.30	0.35	5.04	1.18	6.88	0.39	0.00
7	1.07	11.46	0.25	4.22	0.21	3.66	1.08	6.12	0.37	0.00
8	1.02	12.24	0.20	4.00	0.13	2.80	-0.81	-5.62	0.57	0.00
9	2.24	13.22	0.54	9.68	0.50	8.76	1.15	6.24	0.30	0.00
10	0.74	14.24	0.16	2.44	0.10	1.53	-1.41	-6.23	0.68	0.00
11	1.09	15.86	0.25	4.31	0.20	3.28	1.68	7.68	0.51	0.00
12	0.93	16.97	0.23	3.47	0.13	2.03	-1.51	-5.83	0.59	0.00
13	1.46	18.29	0.34	6.09	0.23	4.61	1.49	6.45	0.47	0.00
14	0.91	19.37	0.17	3.40	0.10	2.05	-1.34	-5.66	0.63	0.00
15	0.58	20.52	0.08	1.38	0.05	0.83	-1.77	-5.62	0.83	0.00
16	1.30	21.91	0.27	5.34	0.22	4.36	-1.70	-6.54	0.54	0.00
17	1.05	22.94	0.47	4.09	0.45	3.87	3.93	5.40	0.44	0.00
18	0.65	24.43	0.08	1.92	0.08	1.89	1.66	6.32	0.79	0.00
19	1.33	25.80	0.29	5.48	0.20	3.87	2.10	5.92	0.51	0.00
20	0.84	26.95	0.16	3.01	0.18	3.17	2.19	5.33	0.62	0.00

表I.6：市场因子基准——提升P-Tree的因子投资

本表补充表5，展示在P-Tree生长过程中考虑市场因子基准的结果。

A面板：1981—2020年全样本。B1面板：1981—2000年样本内；B2：2001—2020年样本外。C1面板：2001—2020年样本内；C2：1981—2000年样本外。

表头SR=年化夏普比率；其余各列依次为CAPM、FF5、Q5、RP-PCA5、IPCA5调整的alpha。P-Tree1、1-5、1-10、1-15、1-20表示对应累计因子组合。显著性星号含义同表5。完整数值保留在随后原表。

Table I.6: **Market Factor Benchmark: Factor Investing by Boosted P-Trees**

This table presents complementary results to Table 5 for considering the market factor benchmark while growing P-Trees.

	SR	α_{CAPM}	α_{FF5}	α_{Q5}	α_{RP5}	α_{IP5}
			Panel A: 40 Years (1981-2020)			
P-Tree1	6.37	1.39***	1.30***	1.35***	1.29***	1.12***
P-Tree1-5	8.92	0.75***	0.68***	0.72***	0.70***	0.63***
P-Tree1-10	11.27	0.52***	0.49***	0.51***	0.49***	0.47***
P-Tree1-15	13.88	0.45***	0.43***	0.44***	0.43***	0.42***
P-Tree1-20	16.09	0.39***	0.38***	0.39***	0.38***	0.37***
			Panel B1: 20 Years In-Sample (1981-2000)			
P-Tree1	7.12	1.85***	1.88***	1.71***	1.64***	1.24***
P-Tree1-5	12.14	0.80***	0.79***	0.77***	0.75***	0.71***
P-Tree1-10	19.16	0.64***	0.65***	0.62***	0.61***	0.57***
P-Tree1-15	27.74	0.53***	0.53***	0.52***	0.52***	0.49***
P-Tree1-20	41.12	0.47***	0.47***	0.46***	0.46***	0.45***
			Panel B2: 20 Years Out-of-Sample (2001-2020)			
P-Tree1	3.24	1.34***	1.31***	1.23***	1.30***	0.91***
P-Tree1-5	2.98	0.52***	0.47***	0.49***	0.48***	0.32***
P-Tree1-10	2.57	0.36***	0.29***	0.33***	0.34***	0.19***
P-Tree1-15	2.48	0.30***	0.23***	0.27***	0.27***	0.16***
P-Tree1-20	2.57	0.26***	0.20***	0.24***	0.25***	0.15***
			Panel C1: 20 Years In-Sample (2001-2020)			
P-Tree1	5.82	1.50***	1.46***	1.50***	1.37***	1.36***
P-Tree1-5	9.61	0.66***	0.65***	0.66***	0.64***	0.62***
P-Tree1-10	14.24	0.46***	0.46***	0.46***	0.45***	0.45***
P-Tree1-15	20.52	0.38***	0.38***	0.38***	0.37***	0.37***
P-Tree1-20	26.95	0.34***	0.34***	0.34***	0.33***	0.33***
			Panel C2: 20 Years Out-of-Sample (1981-2000)			
P-Tree1	4.35	1.50***	1.59***	1.35***	1.43***	1.10***
P-Tree1-5	4.11	0.57***	0.71***	0.51***	0.60***	0.30***
P-Tree1-10	4.07	0.40***	0.49***	0.35***	0.44***	0.23***
P-Tree1-15	3.98	0.36***	0.45***	0.31***	0.40***	0.18***
P-Tree1-20	3.82	0.31***	0.40***	0.26***	0.35***	0.14***

III 采用其他经济目标切分准则的P-Tree

III.A 另一种全局切分准则：最小化已实现定价误差

基准模型生成覆盖有效前沿的测试资产，并构造相应潜在因子，以恢复定价核、最大化MVE组合的夏普比率，以及最小化HJ距离（Hansen和Jagannathan，1991）。现在，我们使用直接最小化个股已实现定价误差的准则，展示P-Tree框架的灵活性。我们运行回归模型，为叶参数提供条件期望值，目标函数见式（10）。

式（10）对所有时期t和各期股票i的残差平方求和：残差为r_i,t减去β_i,t-1(z_i,t-1)f_t。其中，r_i,t为股票i在时期t的超额收益；f_t为P×1可交易因子收益向量；β(z_i,t-1)=b_0+b转置乘z_i,t-1，表示依赖过去公司特征的条件beta。精确符号排版见随后原式。

以最小化定价误差为资产定价目标，与近期利用深度学习生成潜在因子、构造SDF的文献理念相近，例如Gu等（2021）、Chen等（2023）、Feng等（2023）。在该准则下，P-Tree联合解决潜在因子构建与beta估计问题。已实现定价误差就是所生成P-Tree因子模型的收益残差；P-Tree生成潜在可交易因子，以降低总的已实现定价误差平方和。

图I.2和I.3展示使用替代切分准则、分别包含和不包含市场基准的P-Tree。由于选择的公司特征不同，其树结构也不同于图4基于最大化夏普比率平方的主要结果。

III.B 提升

针对替代准则，提出如下提升程序。

1．第一个因子f_1,t可以预先指定，也可以按第2.3节由P-Tree生成。然后固定f_1,t及其相应因子载荷β_1（·）。

III P-Tree with Alternative Economically Guided Split Criteria

III.A An Alternative Global Split Criteria: Minimizing Realized Pricing Errors

Our baseline model generates test assets covering the efficient frontier and constructs the corresponding latent factor, recovering the pricing kernel, maximizing the Sharpe ratio of the MVE portfolio, and minimizing the HJ-distance (Hansen and Jagannathan, 1991). We now demonstrate the flexibility of the P-Tree framework using a criterion that directly minimizes the realized pricing errors of individual stocks. We run a regression model and provide a conditional expected value for the leaf parameter:

$$\mathcal{L}_{\mathrm{A}}(\tilde{\mathbf{c}}_{k,m}) = \sum_{t=1}^{T} \sum_{i=1}^{N_t} \left[\left(r_{i,t} - \beta_{i,t-1}(\mathbf{z}_{i,t-1}) f_t \right) \right]^2, \tag{10}$$

where $r_{i,t}$ is the excess return for stock i at period t, f_t is the $P \times 1$ traded factor return, and $\beta\left(z_{i,t-1}\right) = b_0 + b^\top z_{i,t-1}$ are conditional betas on the past firm characteristics $z_{i,t-1}$.

The asset pricing goal of minimizing objectives on pricing errors is similar in spirit to the recent literature about latent factor generation for constructing the SDF via deep learning (e.g., Gu et al., 2021; Chen et al., 2023; Feng et al., 2023). With this criterion, P-Tree solves a joint problem of constructing the latent factors and estimating betas. The realized pricing errors are the return residuals from the generated P-Tree factor model. P-Tree generates latent traded factors to reduce the aggregate quadratic realized pricing errors.

Figures I.2 and I.3 display P-Trees with and without a market benchmark, fitted on the alternative split criteria. The tree structure differs from our main results based on the maximum squared Sharpe ratio shown in Figure 4 for selecting different firm characteristics.

III.B Boosting

We propose the boosting procedure for the alternative criteria as follows:

1. The first factor $f_{1,t}$ can be pre-specified or generated by P-Tree as discussed in section 2.3. Then, we keep $f_{1,t}$ and its corresponding factor loadings $\beta_1(\cdot)$.

2．生成第二个因子以补充第一个因子。树生长步骤与第2.3节相同，但使用式（11）的提升切分准则：对r_i,t−β_1(z_i,t−1)f_1,t−β_2(z_i,t−1)f_2,t的平方，在所有股票和时期上求和。β_1（·）固定为第1步估计值，因此r_i,t−β_1(z_i,t−1)f_1,t代表此前嵌套因子模型未解释的收益部分，即已实现定价误差。生成第二个因子后，固定f_2,t及其对应载荷β_2（·）。

3．类似地生成第三个因子f_3,t以补充前两个因子，同时将前两个因子载荷固定为此前估计值。式（12）将第三项β_3(z_i,t−1)f_3,t加入拟合收益，再对剩余误差平方求和。

4．重复上述过程K次，生成K个因子f_t=[f_1,t，…，f_K,t]及相应载荷β=[β_1，…，β_K]。式（11）、（12）完整排版见随后原文页。

图I.2：包含市场因子的P-Tree

本图展示以替代切分准则在1981—2000年训练的P-Tree，共20个叶。第一个因子为市场因子，随后在资产定价准则下应用P-Tree生成一个提升树因子。图示格式同图4。

图中N为节点编号，S为切分顺序；Y/N表示条件满足/不满足；#为叶中股票数统计。变量缩写中文说明见表A.1。根切分为RVAR_FF3≤0.8；全部后续切分、阈值与股票数保留于原图。

2. The second factor is generated to augment the first one. The tree-growing steps are the same as those in section 2.3, except we use the boosting split criteria:

$$\mathcal{L}(\tilde{\mathbf{c}}_{k,m}) = \sum_{t=1}^{T}\sum_{i=1}^{N_t}\Big(r_{i,t} - \beta_1(\mathbf{z}_{i,t-1})f_{1,t} - \beta_2(\mathbf{z}_{i,t-1})f_{2,t}\Big)^2. \tag{11}$$

Note $\beta_1(\cdot)$ is fixed from step 1, and then, $r_{i,t} - \beta_1(\mathbf{z}_{i,t-1})f_{1,t}$ represents the unexplained part of returns or the realized pricing error from the previous nested factor model. The second factor is generated to augment the first one. Then, we keep $f_{2,t}$ and its corresponding factor loadings $\beta_2(\cdot)$.

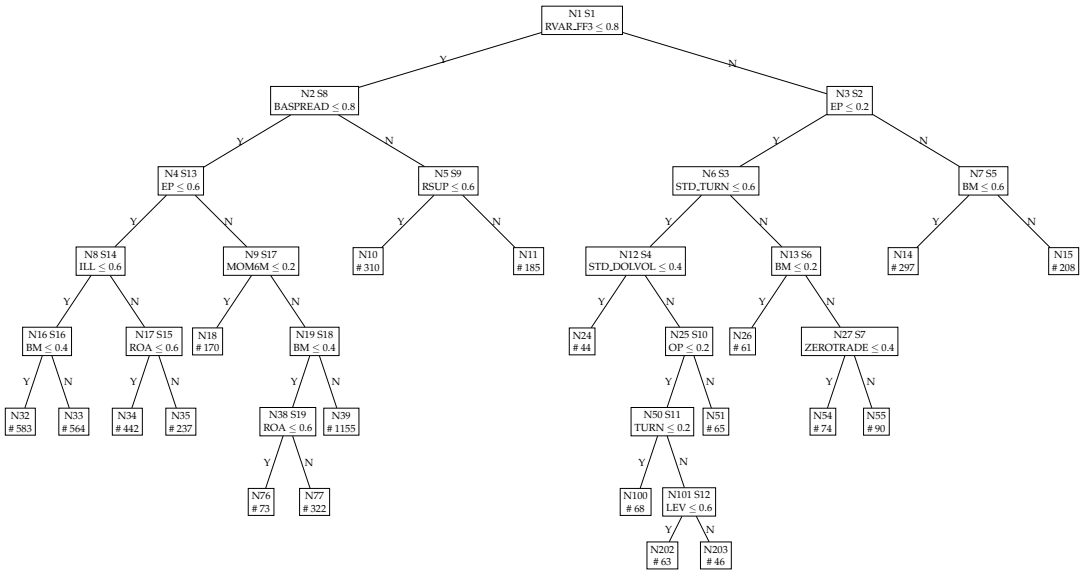
3. The third factor $\mathbf{f}_{3,t}$ is generated similarly to augment the first two factors, whereas factor loadings for the first two factors are fixed at previous estimations:

$$\mathcal{L}(\tilde{\mathbf{c}}_{k,m}) = \sum_{t=1}^{T}\sum_{i=1}^{N_t}\Big(r_{i,t} - \beta_1(\mathbf{z}_{i,t-1})f_{1,t} - \beta_2(\mathbf{z}_{i,t-1})f_{2,t} - \beta_3(\mathbf{z}_{i,t-1})f_{3,t}\Big)^2. \tag{12}$$

4. Repeat the above process K times to generate K factors $\mathbf{f}_t = [f_{1,t}, \cdots, f_{K,t}]$ and obtain corresponding factor loadings $\boldsymbol{\beta} = [\beta_1, \cdots, \beta_K]$.

Figure I.2: **P-Tree with Market**

This figure displays P-Tree with alternative split criteria trained from 1981 to 2000, with 20 leaves. The first factor represents the market factor, and then P-Tree is applied to generate a boosted tree factor under the asset pricing criterion. The figure format follows Figure 4.



图I.3：不包含市场因子的P-Tree

本图展示使用替代切分准则、未控制市场因子的P-Tree。训练样本为1981—2000年。图示格式同图I.2。根切分为STD_DOLVOL≤0.8，后续路径、变量缩写、阈值与股票数完整保留在原图。

提升P-Tree因子增强已有因子模型。假设拟合K=3的P-Tree模型，可以按重要程度由高到低将三个因子排列为[f_1,t，f_2,t，f_3,t]。实证结果提供两种设置：一种以市场因子为基准，另一种为普通P-Tree。生成额外P-Tree因子类似于增加主成分，每个后续因子都有助于减少剩余定价误差。例如，三因子提升模型见式（13）：T(z_i,t-1|Θ)等于三个条件载荷与因子收益乘积之和，即β_1f_1,t+β_2f_2,t+β_3f_3,t，各β均依赖滞后公司特征。

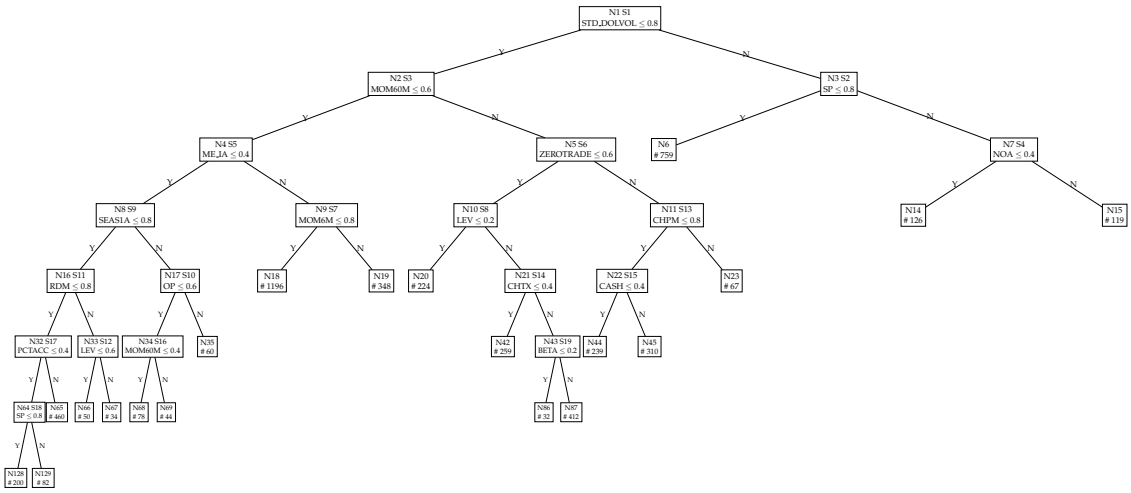
III.C 实证结果

表现指标。我们考虑多个指标。总R^2与横截面R^2分别评价时间序列变动与横截面变动的定价表现。具体使用式（14）—（16）。

总R^2见式（14）：1减去所有资产、所有时期的平均收益残差平方，与实际收益平方平均值之比。各平均都除以NT；预测收益为估计条件载荷与同期因子收益的乘积。数学排版见随后原文页。

Figure I.3: **P-Tree without Market**

This figure displays a P-Tree on alternative split criteria without controlling for market factors. The training sample spans from 1981 to 2000. The figure format follows Figure I.2.



The boosted P-Tree factors enhance an existing factor model. Assuming we fit the P-Tree model with $K = 3$ factors, we can arrange the three factors in descending order of significance as $[f_{1,t}, f_{2,t}, f_{3,t}]$. In the empirical results, we provide two specifications: one with the market factor as a benchmark and one with a vanilla P-Tree. Generating additional P-Tree factors is similar to generating additional principal components, and each subsequent factor helps reduce the remaining pricing errors. For example, a three-factor boosted P-Tree model follows:

$$\mathcal{T}(\mathbf{z}_{i,t-1}|\Theta) = \beta_1(\mathbf{z}_{i,t-1})f_{1,t} + \beta_2(\mathbf{z}_{i,t-1})f_{2,t} + \beta_3(\mathbf{z}_{i,t-1})f_{3,t}. \tag{13}$$

III.C Empirical Results

Performance metrics. We consider multiple performance metrics. Total R^2 and cross-sectional R^2 metrics evaluate asset pricing performance for time-series and cross-sectional variations, respectively. Specifically, we evaluate the asset pricing performance using these metrics:

$$\text{Total } R^2 = 1 - \frac{\frac{1}{NT}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(r_{i,t} - \widehat{r}_{i,t}\right)^2}{\frac{1}{NT}\sum_{t=1}^{T}\sum_{i=1}^{N} r_{i,t}^2}, \tag{14}$$

在式（14）中，预测值r̂_{i,t}=β̂(z_{i,t-1})f_t。总R²表示模型隐含同期收益所解释的已实现收益变动比例。

个股横截面R²见式（15）：先对每只股票的实际收益与预测收益之差取时间平均，再平方并对股票求平均；以该数值除以“每只股票时间平均实际收益的平方”的横截面平均值，最后用1减去该比值。预测值仍为条件载荷乘因子收益。该指标衡量公共因子模型对无条件平均收益平方的解释比例。

组合横截面R²见式（16）：1减去各组合平均收益预测误差平方和与实际平均收益平方和之比。其中，预测组合平均收益等于估计因子载荷β̂_i的转置乘风险溢价估计λ̃_f。风险溢价通过横截面回归估计，载荷通过时间序列回归估计。三项指标的精确公式见相邻原文页。

比较资产定价模型。表I.7汇总资产定价准则下P-Tree的表现。A面板为以市场为基准、分别增加1个和4个P-Tree因子的两种模型。通过Barillas和Shanken（2017）的夏普比率检验，选择提升过程中生成的额外因子数。B面板比较CAPM、FF5和Hou等（2021）的Q5等可观测因子模型，同时比较机器学习金融文献中的潜在因子模型，如Lettau和Pelger（2020b）的RP-PCA、Kelly等（2019）的IPCA。C面板报告按市场波动进行时间序列切分的P-Tree，详见III.D。

对个股收益定价时，P-Tree的总R²始终为正，样本内和样本外均优于大多数可观测因子模型。多因子P-Tree的总R²与两种强潜在因子模型相当。个股横截面R²衡量因子模型总体定价误差；多因子P-Tree在样本内和样本外均取得较高正值，体现模型隐含预期收益的表现。这些数值接近表现较强的IPCA基准。

where $\widehat{r}_{i,t} = \widehat{\beta}(\mathbf{z}_{i,t-1})f_t$. Total R^2 represents the fraction of realized return variation explained by the model-implied contemporaneous returns.

$$\text{Stock CS } R^2 = 1 - \frac{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{T}\sum_{t=1}^{T}(r_{i,t} - \widehat{r}_{i,t})\right)^2}{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{T}\sum_{t=1}^{T} r_{i,t}\right)^2}, \tag{15}$$

where $\widehat{r}_{i,t} = \widehat{\beta}(\mathbf{z}_{i,t-1})f_t$. Cross-sectional R^2 for individual stocks represents the fraction of the squared unconditional mean returns described by the common factor model:

$$\text{Portfolio CS } R^2 = 1 - \frac{\sum_{i=1}^{N}\left(\bar{R}_i - \widehat{\bar{R}}_i\right)^2}{\sum_{i=1}^{N}\left(\bar{R}_i\right)^2}, \tag{16}$$

where $\widehat{\bar{R}}_i = \widehat{\boldsymbol{\beta}}_i^{\intercal}\widetilde{\boldsymbol{\lambda}}_f$. The risk premium estimation adopts the cross-sectional regression estimates of factors, and factor loadings are estimated from time series regressions.

Comparing asset pricing models. Table I.7 summarizes the performance of the P-Tree model under the asset pricing criterion. Panel A shows the performance of two P-Tree models on the market benchmark, with additional one and four P-Tree factors. The Sharpe ratio test from Barillas and Shanken (2017) selects the number of additional P-Tree factors generated through boosting. Panel B of Table I.7 compares these results with observable factor models such as CAPM, FF5, and the recent Q5 model of Hou et al. (2021). We also compare the performance of P-Tree with latent factor models from the ML finance literature, such as RP-PCA of Lettau and Pelger (2020b) and IPCA of Kelly et al. (2019). Panel C reports the results of the P-Tree model with time-series splits on market volatility, which is discussed in section III.D.

The total R^2 values for P-Trees are consistently positive in pricing individual stock returns, outperforming most observable factor models both in-sample and out-of-sample. The multifactor P-Tree models show comparable total R^2 to two strong latent factor models. Cross-sectional R^2 for individual stocks quantifies the aggregate pricing errors by the factor model. For the cross-sectional R^2, the multifactor P-Tree models produce highly positive numbers for in-sample and out-of-sample analysis, which demonstrates the performance of the model-implied expected returns. These numbers are close to the strong IPCA benchmark.

表I.7：资产定价表现

本表报告资产定价表现。Tot（总R^2，%）和CS（横截面R^2，%）分别为式（14）、（15）的个股收益指标。样本内为1981—2000年，样本外为2001—2020年。还报告式（16）的组合横截面R^2（%），行中的因子模型为列中的测试资产组合定价。

Leaf20表示图I.2的20个基础组合，Leaf40表示图I.2和I.3合计40个基础组合。A面板为不同因子数的资产定价面板树：P-Tree2为市场因子加一个P-Tree因子的两因子模型；P-Tree5*为市场因子加四个P-Tree因子的五因子模型。*表示因子数通过Barillas和Shanken（2017）的夏普比率平方检验选定。B面板比较第4.3节介绍的基准模型。C面板为时间序列切分P-Tree，选择市场波动率SMVAR切分时间维度。

表头Individual Stocks=个股；Portfolios=组合；In-Sample=样本内；Out-of-Sample=样本外；Entire Sample=全样本；FF25=5×5 ME-BM组合；Ind49=49个行业组合。数值见原表。

我们还考察P-Tree为多组测试组合定价的表现，包括5×5 ME-BM组合与49个行业组合。此外，图I.2产生20个叶基础组合，图I.3在不含市场因子的情况下再产生20个。生成基础组合是P-Tree框架的另一项数据驱动输出，它解决高维特征挑战，并遵循选定的经济目标切分准则。式（16）的横截面R^2使用全样本计算，量化模型隐含预期收益未能解释的总体定价误差。

结果表明，包括CAPM、FF5和Q5在内的可观测因子模型可以解释5×5 ME-BM和49个行业组合的横截面平均超额收益。

Table I.7: **Asset Pricing Performance**

This table reports the asset pricing performances. "Tot" (total R^2 %) and "CS" (cross-sectional R^2 %) in equations (14) and (15) are measures for individual stock returns. The in-sample period is from 1981 to 2000, and the out-of-sample period is from 2001 to 2020. We also report cross-sectional R^2 % in (16), using the factor models in the rows to price the test asset portfolios in the columns. "Leaf20" indicates the 20 basis portfolios from Figure I.2, and "Leaf40" indicates the 40 basis portfolios from Figures I.2 and I.3. Panel A shows results for the asset pricing panel-tree models with #factors. Specifically, "P-Tree2" indicates a two-factor model of the market factor and a P-Tree factor. "P-Tree5*" indicates a five-factor model of the market factor and four P-Tree factors. "*" indicates the optimal number of factors selected by the squared Sharpe ratio test of Barillas and Shanken (2017). Panel B compares benchmark models introduced in section 4.3. Panel C reports for the asset pricing P-Tree applying the time-series split, which chooses market volatility (SMVAR) to split on the time-series dimension.

	Individual Stocks				Portfolios			
	In-Sample		Out-of-Sample		Entire Sample			
	Tot	CS	Tot	CS	FF25	Ind49	Leaf20	Leaf40
Panel A: P-Tree								
P-Tree2	11.1	25.5	11.1	10.4	93.1	93.1	98.5	68.6
P-Tree5*	13.0	22.7	13.7	16.5	94.2	99.2	94.0	85.0
Panel B: Other Benchmark Models								
CAPM	7.0	1.3	8.4	0.6	91.7	88.1	97.3	10.4
FF3	10.5	7.5	10.7	5.1	95.1	94.1	98.4	38.7
FF5	11.0	13.1	11.3	5.1	96.8	96.1	98.7	75.2
Q5	10.9	18.1	11.5	6.4	97.4	95.9	98.5	79.2
RP-PCA5	12.1	18.3	13.6	15.0	97.0	95.7	98.6	64.0
IPCA5	13.8	27.8	14.9	17.7	97.3	92.8	99.4	89.2
Panel C: P-Tree with Time-Series Split on Market Volatility								
TS-P-Tree2	11.7	24.7	11.3	9.3	87.6	81.8	92.1	26.6
TS-P-Tree5*	13.4	26.4	13.6	18.4	96.1	91.2	97.8	80.7

We also investigate P-Tree performance in pricing multiple sets of test portfolios, including 5×5 ME-BM and 49 industry portfolios. In addition, 20 leaf basis portfolios are generated in Figure I.2, and another 20 are generated without the market factor in Figure I.3. Generating basis portfolios is another data-driven output from our P-Tree framework, which resolves the high-dimensional characteristics challenge and follows a chosen economics-guided split criterion. Cross-sectional R^2 in (16) quantifies the aggregate unexplained pricing errors by the model-implied expected returns and is calculated using the entire sample of data.

The results suggest observable factor models, including CAPM, FF5, and Q5, explain the cross-sectional average excess returns for 5×5 ME-BM and 49 industry port-

但它们无法解释P-Tree生成的叶基础组合。相比之下，A、C面板的P-Tree因子模型能持续为所有测试资产集合定价。在其他潜在因子模型中，IPCA具有稳健的正向表现和竞争力，但P-Tree提供了更高透明度与可解释性。

表I.8：投资P-Tree因子

本表报告股票因子模型的投资表现，包括因子均值—方差有效（MVE）组合与等权（1/N）组合的月度平均收益（%）、CAPM alpha（%）和年化夏普比率。模型名后的*表示依据Barillas和Shanken（2017）的夏普比率平方检验选定的最优因子数；t统计量对应的*、**、***分别表示10%、5%、1%显著性。

样本内为1981—2000年，样本外为2001—2020年；AVG=月度平均收益，SR=年化夏普比率，alpha=CAPM截距。A面板为P-Tree，B面板为其他基准模型，C面板为按市场波动进行时间序列切分的P-Tree。完整数值见原表。

投资表现。为比较可观测或潜在因子的风险调整投资表现，我们考虑两种常见多因子投资策略：1/N等权组合与MVE组合。表I.8报告训练与测试样本的月度平均收益、CAPM alpha和年化夏普比率。

P-Tree因子模型显示良好的资产定价与投资表现。样本内分析中，A面板五因子模型（市场加四个提升P-Tree因子）的MVE策略取得3.47的年化夏普比率，月度CAPM alpha超过1%；这些指标在样本外也保持较高。若聚焦于作为强基准的仅做多1/N策略，数值降低但持续为正。总体上，P-Tree因子与IPCA因子表现最好；P-Tree还提供透明性和互操作性。

folios. However, they cannot explain the leaf basis portfolios generated by P-Trees. In contrast, the P-Tree factor models in Panels A and C consistently price all sets of test assets. IPCA competes with robust positive performance among other latent factor models, but P-Trees offer more transparency and interpretability than other latent factor models.

Table I.8: **Investing P-Tree Factors**

This table reports the investment performance of the equity factor models. The monthly average return (%) and CAPM α (%) are reported, along with the annualized Sharpe ratio for the factors' mean-variance efficient (MVE) and equal-weighted ($1/N$) portfolios. "*" indicates the optimal number of factors selected by the squared Sharpe ratio test of Barillas and Shanken (2017). For t-statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	In-Sample (1981-2000)						Out-of-Sample (2001-2020)					
	MVE			$1/N$			MVE			$1/N$		
	AVG	SR	α	AVG	SR	α	AVG	SR	α	AVG	SR	α
Panel A: P-Tree												
P-Tree2	1.75	1.58	1.51***	1.31	1.34	0.86***	0.30	0.31	0.29	0.45	0.56	0.17
P-Tree5*	1.26	3.47	1.20***	1.06	1.69	0.72***	0.80	1.93	0.76***	0.70	1.14	0.42***
Panel B: Other Benchmark Models												
FF5	0.45	1.48	0.38***	0.38	1.34	0.33***	0.27	0.64	0.13*	0.25	0.59	0.12
Q5	0.77	2.78	0.74***	0.63	2.10	0.53***	0.34	1.22	0.34***	0.31	1.10	0.25***
RP-PCA5	0.82	3.48	0.76***	1.07	1.77	0.75***	0.34	1.49	0.32***	0.50	1.00	0.27***
IPCA5	1.50	10.37	1.48***	0.90	3.15	0.80***	0.97	4.60	0.98***	0.73	2.14	0.61***
Panel C: Time-Series Split P-Tree - Market Volatility												
TS-P-Tree2	1.80	1.67	1.56***	1.32	1.36	0.86***	0.60	0.66	0.57***	0.60	0.77	0.31***
TS-P-Tree5*	1.20	3.65	1.14***	1.03	1.61	0.69***	0.78	1.69	0.73***	0.74	1.14	0.46***

Investment performance. To compare the risk-adjusted investment performances of observable or latent factors, we consider two common strategies for investing in multiple factors: the $1/N$ equal-weighted portfolio and the MVE portfolio.Table I.8 reports the monthly average return, CAPM alpha, and the annualized Sharpe ratio of the training and test samples.

P-Tree factor models show excellent asset pricing and investment performance. For the in-sample analysis, the MVE strategies of the five-factor model (market plus four boosted P-Tree factors) deliver a 3.47 annualized Sharpe ratio in Panel A. The monthly CAPM alphas are beyond 1%. These numbers are consistently high out-of-sample. If we focus on the long-only $1/N$ strategies (a strong benchmark), the numbers become lower but remain consistently positive. Overall, P-Tree factors and the IPCA

表I.9：因子张成alpha检验

本表报告因子张成检验的月度alpha（基点，bp）及显著性。A面板为表I.8中P-Tree模型各因子的结果：将各行因子对各列因子模型回归。P-Tree因子按交互树结构中前两次切分选定的特征命名。还展示投资这五个因子的MVE与1/N策略。FF25与INF49分别表示5×5 ME-BM组合和49个行业组合。t统计量的*、**、***分别表示10%、5%、1%显著性。

In-Sample=样本内；Out-of-Sample=样本外；A面板=P-Tree因子；B面板=其他测试资产；4 factors + mkt=四个因子加市场。数值及星号见随后原表。

我们将每个P-Tree因子对基准模型分解，以表I.9中的alpha评价风险调整投资表现。例如，A面板第一个投资因子RVAR_FF3-EP相对FF5与Q5具有经济与统计显著的alpha。原文称，IPCA五因子模型在样本内和样本外均无法解释这一首要因子；四个投资P-Tree因子加市场的MVE组合仍处处具有显著alpha。另外，A面板P-Tree因子的表现略低，但持续为正。

〔译注：上述显著性概括与表中部分数值不一致，例如RVAR_FF3-EP的样本外alpha未标显著性星号，MVE对IP5的alpha也未标星。保留作者论述，具体以原表为准。〕

P-Tree因子基于叶组合生成，而叶组合由P-Tree利用多个特征生成。B面板将P-Tree因子的投资表现与传统分组（5×5 ME-BM和49个行业组合）比较，后者表现更低。我们据此认为，P-Tree因子的正向表现既来自特征的交互与非线性，也来自识别这些关系的数据驱动方法，而不是临时设定的排序。

Table I.9: **Factor-Spanning Alpha Test**

This table reports the monthly alphas (bp) and significance for the factor-spanning test. Panel A shows the results for each factor of the P-Tree models in Table I.8. We regress each factor in the rows against a factor model in the columns. We name the P-Tree factors by the first two splitting characteristics selected in the interactive tree structure. We also show the mean-variance efficient (MVE) and $1/N$ strategies investing these five factors. Notably, "FF25" and "INF49" are the 5×5 ME-BM and 49 industry portfolios. For t-statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	In-Sample				Out-of-Sample			
	FF5	Q5	RP5	IP5	FF5	Q5	RP5	IP5
Panel A: P-Tree factors								
RVAR_FF3-EP	130***	101***	83***	107**	12	4	13	31
BM_IA-Ill	35**	33	19	-100***	107***	110***	81***	34
MOM12M-STD_DOLVOL	82***	53***	53	-24	25	22	22	-95***
ME-RDM	52***	48***	47***	109***	29**	27**	30**	13
MVE (4 factors + mkt)	58***	45***	58***	-21	36***	34***	36***	-11
1/N (4 factors + mkt)	60***	47***	60***	49*	35***	33***	35***	1
Panel B: Other Test Assets								
MVE-FF25	55***	42***	32***	27*	19***	15**	8	10
MVE-IND49	13*	20	-36**	14	10	8	20*	28*
1/N-FF25	-8***	-8	-68***	57***	3	8**	-23***	5
1/N-IND49	63*	30	56	62	-2	10	-13**	4

factors perform the best. P-Trees further offer transparency and interoperability.

We decompose each P-Tree factor on benchmark factor models to evaluate the risk-adjusted investment performances, alphas, in Table I.9. For example, the first investment P-Tree factor `RVAR_FF3-EP` in Panel A shows economically and statistically significant alphas against FF5 and Q5 models. A 5-factor IPCA model cannot explain this top P-Tree factor for either in-sample or out-of-sample evaluation. The MVE portfolio of four investment P-Tree factors plus the market factor still shows significant alphas everywhere. Moreover, the performances for P-Tree factors in Panel A are slightly lower but still consistently positive.

P-Tree factors are generated based on leaf portfolios, which are generated by P-Trees using multiple characteristics. In Panel B, we compare the investment performance of P-Tree factors with conventional clustering (5×5 ME-BM and 49 industry portfolios), which shows lower performance. We conclude that the positive performance of P-Tree factors comes from both the interaction and nonlinearity of characteristics and the data-driven approach used to identify them, as opposed to ad hoc

III.D 时间序列切分

资产收益面板同时具有横截面与时间序列维度。大多数文献与基准P-Tree一样，重点解释横截面收益变动，而不考虑潜在结构突变或状态切换。随着树从上向下生长，资产面板依据过去特征被划分为多个叶节点；每个叶保留从1到T的完整时间序列，但只包含一部分资产。

然而，实证发现表明，不同宏观状态下的因子选择、风险溢价与beta都可能随时间变化。例如，大量文献考察金融市场在高低波动、高低通胀、繁荣或衰退期间的状态变化（如Hamilton和Lin，1996；Buraschi和Jiltsov，2005）。

P-Tree能够沿时间序列维度切分节点，为构建与估计具有状态切换的因子—beta模型提供另一种方法。宏观变量提取的时间序列信息也有助于构造动态SDF。用宏观变量而非公司特征切分节点（例如根节点），就能构造时变因子模型，并估计宏观状态切换下相应的条件beta。每个子节点保持与父节点相同的资产横截面，但时间样本不重叠。这提高了P-Tree同时捕捉横截面和时间序列变动的灵活性，实现二维面板聚类。

第一次时间序列切分的资产定价准则见式（17）：分别计算A状态和B状态中，股票实际收益减去该状态条件beta乘因子收益后的残差平方，再将两部分加总。候选规则$\tilde{c}$把时间序列划为互不重叠的时期集合R_A与R_B，例如高通胀和低通胀状态。两组时期都不必连续。N_A,t表示子节点R_A在时期t的资产子样本；N_B,t表示R_B中的对应资产子样本。

sorting.

III.D Time-Series Split

The panel data of asset returns has cross-sectional and time-series dimensions. Most literature focuses on explaining the cross-sectional return variation, as do the baseline P-Tree models, without considering potential structural breaks or regime shifts. Because the tree grows from top to bottom, the panel of assets is partitioned into many leaf nodes based on past characteristics. Each leaf node maintains the entire time series from period 1 to T but a subset of assets. However, empirical findings suggest that factor choices, risk premia, and betas can be time-varying under different macroeconomic states. For example, extensive literature examines regime changes in financial markets involving periods of high and low volatility or inflation, booms, or recessions (e.g., Hamilton and Lin, 1996; Buraschi and Jiltsov, 2005).

With its ability to split nodes in the time-series dimension, P-Tree provides an alternative approach to constructing and estimating factor-beta with regime-switching. Moreover, the time-series information extracted from macro variables proves helpful in constructing the dynamic SDF. By splitting nodes (e.g., root node) by macro variables instead of firm characteristics, we can create time-varying factor models and estimate corresponding conditional betas under switching macroeconomic states. Each child node maintains the same cross section of assets as the parent node, but their time-series samples do not overlap. This approach improves P-Tree's flexibility in capturing cross-sectional and time-series variations, providing clustering over the two-dimensional panel.

The asset pricing split criterion for the first time-series split is

$$\mathcal{L}(\tilde{\mathbf{c}}) = \sum_{t\in\mathcal{R}_A}\sum_{i\in N_{A,t}} \left(r_{i,t} - \beta_A(\mathbf{z}_{i,t-1})f_{A,t}\right)^2 + \sum_{t\in\mathcal{R}_B}\sum_{i\in N_{B,t}} \left(r_{i,t} - \beta_B(\mathbf{z}_{i,t-1})f_{B,t}\right)^2, \tag{17}$$

where the candidate split $\tilde{c}$ partitions the time series of data into two non-overlapped sets of periods $\mathcal{R}_A$ and $\mathcal{R}_B$, for example, high- or low-inflation states. Note both sets of periods are not necessarily consecutive. We use $N_{A,t}$ to represent the subsample of assets in the child node $\mathcal{R}_A$ during time period t, and $N_{B,t}$ for the corresponding

f_A,t和f_B,t分别表示左右子节点的P-Tree因子收益。R_A与R_B之间没有重叠时期，且各自可包含宏观状态变化形成的多个时段。不同于式（4）的横截面准则，时间序列切分准则基于两个时期集合、各自对应因子f_A,t与f_B,t的总定价损失。

图I.4：按高/低市场波动切分的P-Tree

本图展示同时考虑横截面与时间序列变动的P-Tree。最重要的宏观预测变量是股票市场波动率SMVAR，首次切分判断当前SMVAR是否低于过去十年的60%分位数。高、低SMVAR时期分别对应两个子节点的树模型。图示格式同图I.2。

图内TS Split=时间序列切分；SMVAR≤60Q{10Y}=SMVAR不高于过去十年的60%分位数。Y/N表示条件满足/不满足，节点编号、切分顺序和省略号均按原图保留。

包含条件beta的两状态模型见式（18）：T(z_i,t-1|Θ)=β_A(z_i,t-1)f_A,t I(t属于R_A)+β_B(z_i,t-1)f_B,t I(t属于R_B)。其中I为指示函数，仅在时期t属于对应状态时取1，否则为0。精确符号见随后原式。

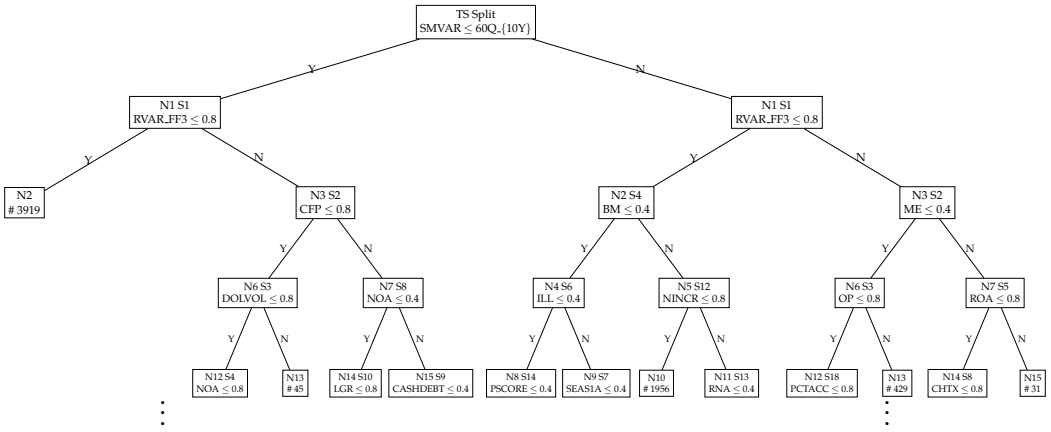
由于股票数量远大于时间期数，为作展示，在一开始找到最佳时间序列切分规则后，后续所有切分只根据横截面特征选择。根节点任一子节点上的后续切分，仅依赖该侧子样本。这一扩展不仅增强P-Tree捕捉横截面与时间序列变动的建模灵活性，还揭示宏观变量如何与公司特征交互来解释资产收益，因为二者共同定义空间划分。

图I.4展示按市场波动率SMVAR进行时间序列切分的P-Tree。

subsample in the child node $\mathcal{R}_B$. The factors $f_{A,t}$ and $f_{B,t}$ represent the P-Tree factor returns for the left and right child nodes, respectively. Note no overlapping periods exist between $\mathcal{R}_A$ and $\mathcal{R}_B$. Moreover, $\mathcal{R}_A$ and $\mathcal{R}_B$ may contain multiple periods for the changing macroeconomic states. In contrast to the cross-sectional split criteria in (4), the time-series split criterion is based on the *total* pricing loss of two time periods, with corresponding factors $f_{A,t}$ and $f_{B,t}$.

Figure I.4: **P-Tree Split on High/Low Market Volatility**

This figure depicts P-Tree, considering both cross-sectional and time-series variations. The most important macro predictor is stock market volatility `SMVAR`, and the first split is implemented when the current `SMVAR` level is lower than the 60% quantile of the past decade. Two tree models are provided as two child leaves for high- and low-`SMVAR` periods. The figure format follows Figure I.2.



Below is the two-period model with conditional betas:

$$\mathcal{T}(\mathbf{z}_{i,t-1}|\Theta) = \beta_A(\mathbf{z}_{i,t-1})f_{A,t}\mathbf{I}(t \in \mathcal{R}_A) + \beta_B(\mathbf{z}_{i,t-1})f_{B,t}\mathbf{I}(t \in \mathcal{R}_B). \tag{18}$$

Because the stock universe is much larger than the number of periods, after finding the best time-series split rule at the beginning, all subsequent split rules are selected based on cross-sectional characteristics only for demonstration purposes. Note any further split on either child of the root node depends only on the subsample on one side. This extension not only improves the modeling flexibility of P-Tree to capture cross-sectional and time-series variations, but also informs how macro variables interact with firm characteristics to explain asset returns, because they jointly define the space partition.

Figure I.4 shows a time-series split P-Tree on market volatility (`SMVAR`), which

该模型将市场波动水平识别为状态切换指标，并可能生成多个状态。随后，在每个状态下分别生长横截面P-Tree。不同状态中的P-Tree模型不同，但特质波动率仍是关键指标。表I.7的资产定价表现和表I.8的投资结果也显示，模型在不同市场波动状态下表现稳健。这些发现为不同时间段收益存在时变因子结构提供了支持性证据。

identifies the regime-switching indicator as the levels of market volatility and might generate multiple regimes. Then, two separate cross-sectional P-Trees are developed on each regime. The P-Tree model varies under different regimes, though the idiosyncratic volatility remains one key indicator. Also, we show the asset pricing performance in Table I.7 and investment results in Table I.8, demonstrating robust performances over the market volatility regimes. All of these discoveries provide supportive proof for the time-varying factor structure of returns across different time periods.